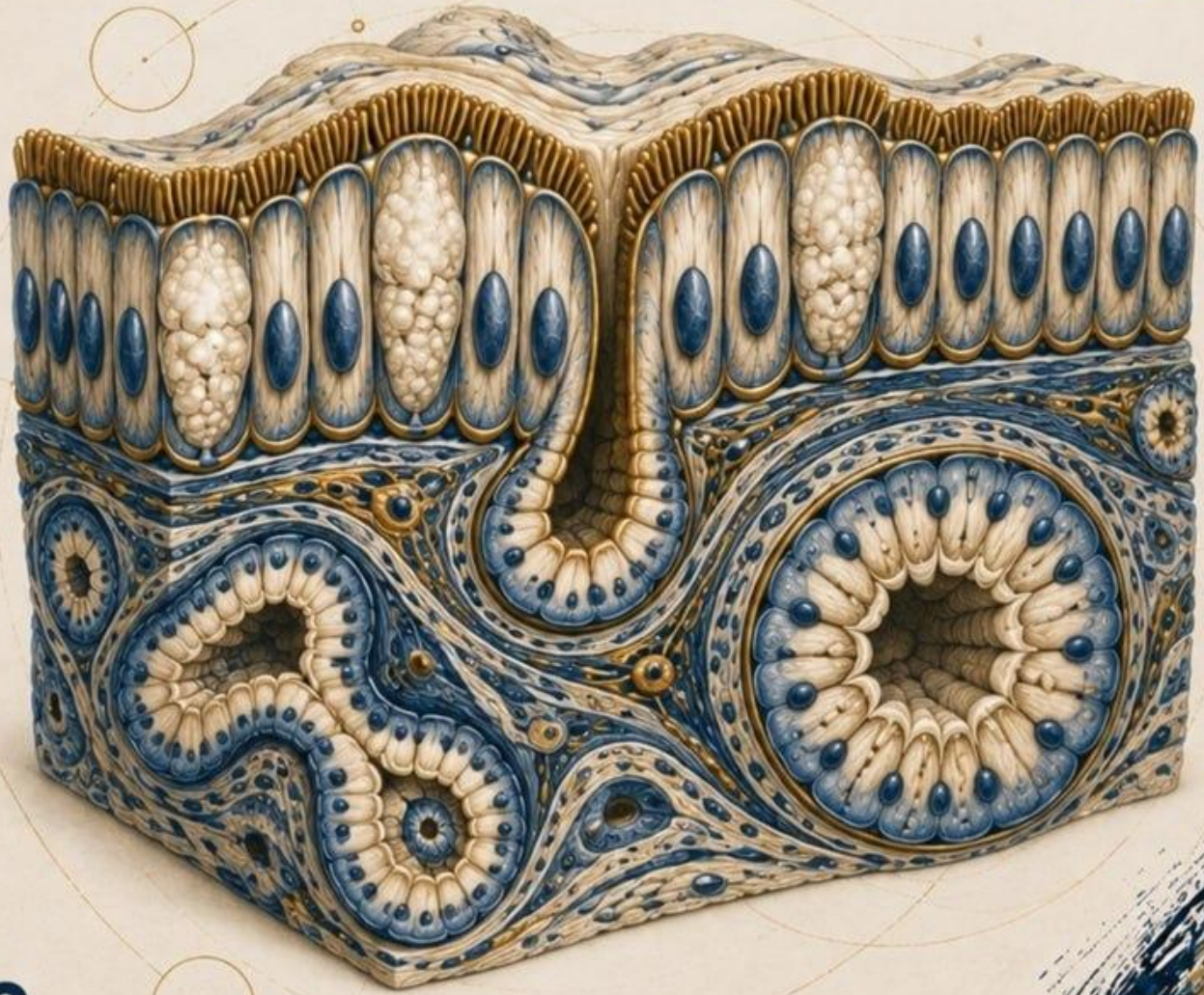
HISTOLOGY

MCQS



2026

بسم الله الرحمن الرحيم

والصلاة والسلام على خير المرسلين محمد بن عبد الله وعلى آله وصحبه وسلم أما بعد، نقدم بين يديك ملف

(General Histology MCQ)

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ابدأ مستعيناً بالله وكرر الحل ما استطعت واحرص على وقتك فهو رأس مالك وأغلى ما تملك

حسابات الفريق على منصات التواصل الاجتماعي

جميع الحقوق محفوظة لفريق مِعوان

ولا نحلل بيعها بعائد ربحي أو سرقتها بأي شكل من الأشكال

Lecture 1: Microscopy and Microtechnique

Pre Exams & Formatives

1-	What is the maximum resolving power seen with a compound microscope? (MID 30, Department Qs)				
A-	2 millimeters	B-	0.2 millimeters	C-	2 micrometers
D-	0.2 micrometers	E-	2 angstroms		

2-	Which microscope makes things appear three dimensional? (MID 30, FINAL 30, Department Qs)		
A-	Simple microscope	B-	Compound microscope
C-	Phase contrast microscope	D-	Dissection microscope
E-	Transmission electron microscope		

ملحوظة من د. هدي

3-	Which microscope would be particularly useful for looking at living cells? (FINAL 30, 2ND TRIAL 35, Department Qs)		
A-	Simple microscope	B-	Compound microscope
C-	Phase contrast microscope	D-	Dissection microscope
E-	Transmission electron microscope		

4-	Scanning electron microscope is useful in? (MID 31)
A-	studying the organoids inside the cells in thin sections
B-	studying living cells and objects
C-	studying details of surfaces of cells
D-	studying the organoids in thick sections
E-	studying nuclear structure

5-	Identification of glycogen in the cells occurs by using? (MID 31)				
A-	vital staining method	B-	general histological method	C-	specific histological method
D-	histochemical method	E-	supravital staining method		

6-	Vital staining technique is?(FINAL 31)		
A-	Staining of dead tissue outside the body	B-	Staining of a living cell inside the body
C-	Staining of a living cell outside the body	D-	Staining of a dead tissue inside the body
E-	Staining of inclusions inside the cell		

7-	During preparation of paraffin section which order is correct? (MID 32)
A-	Fixation, dehydration, clearing, embedding, sectioning and staining
B-	Clearing, fixation, embedding, staining and sectioning

C-	Staining, washing, fixation, dehydration and embedding
D-	Cutting, fixation, washing, dehydration and staining
E-	None of the above

8-	**Hematoxyline is? (MID 33)**		
A-	Acidic dye	B-	Stain acidic structures
C-	Basic dye	D-	Both:b&c

9-	**Which of the following microscopes can be used for examination of living cells? (MID 33)**		
A-	Fluorescent microscope	B-	Phase contrast
C-	High voltage EM	D-	Both:b&c

10-	**What is the limit resolution? (MID 34)**
A-	the largest distance between two things below it it can't be seen
B-	the smallest distance between two things below it they appear one thing
C-	The power of microscope to distinguish fine details

11-	**About acidophilic?(FINAL 34)**		
A-	Have great affinity to basic dye	B-	is a basic structure
C-	have no affinity to acidic dye	D-	is an acidic structure

12-	**The plasma membrane? (2ND TRIAL 35)**		
A-	Is visible with the light microscope	B-	Is visible with the electron microscope

13-	**What is the suitable method for preparation of soft tissues ? (MID 37)**						
A-	Sectioning	B-	Grinding	C-	Filming	D-	smearing

14-	**To discover living cells we use....?(FINAL 37)**		
A-	Phase contrast Microscope	B-	Compound microscope

15-	**In histological examination, the light microscope is commonly used to detect and evaluate which of the following cellular features? (FINAL SNU-M 1)**
A-	The detailed ultrastructure of mitochondria
B-	The shape of the cell's nucleus
C-	The internal components of ribosomes
D-	The molecular structure of the plasma membrane

16-	Limit of resolution of Light microscope is:- (Formative 36)?						
A-	0.24 mm	B-	0.24 nm	C-	240 nm	D-	200 um

17-	What is the power of magnification of the ocular lens on a compound microscope? (Formative 36, Formative 32)?						
A-	1X	B-	100X	C-	10X	D-	1000X

18-	Which of the following is an acid stain? (Formative 32)						
A-	Azure II	B-	Toluidine blue	C-	Methylene blue	D-	Eosin

19-	The functional unite of the kidney is (2ND TRIAL 32)?						
A-	Nephrons	B-	collecting tubule	C-	renal artery	D-	renal veins

20-	Which of the following microscopes permits direct observation of living cells? (2ND TRIAL 37)
A-	Phase contrast microscope
B-	Transmission electron microscope
C-	Scanning electron microscope
D-	High voltage electron microscope

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
D	D	C	C	D	B	A	D	D	B
11	12	13	14	15	16	17	18	19	20
B	B	A	A	B	C	C	D	A	A

Lecture Qs

1-	What color does hematoxylin stain structures? (Lecture Qs, Department Qs)								
A-	Orange	B-	Blue	C-	Pink	D-	Red	E-	Green

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
B	-	-	-	-	-	-	-	-	-

Department Qs

1-	What is "compound microscope"? (Department Qs)
A-	Microscope with the capability to view oil immersion
B-	Microscope with the capability to view compounds
C-	Microscope with a single lens
D-	Microscope with two lenses
E-	Microscope with three lenses

2-	What is the total magnification achieved with a compound microscope? (Department Qs)
A-	Magnification of objective lens
B-	Magnification of ocular lens
C-	Magnification of ocular lens added to the magnification of the objective lens
D-	Magnification of ocular lens multiplied by the magnification of the objective lens
E-	Magnification of condenser lens multiplied by the magnification of the objective lens

3-	On a microscope, what structure connects the eyepiece to the objective lens? (Department Qs)								
A-	Base	B-	Nosepiece	C-	Stage	D-	Tube	E-	Diaphragm

4-	What is another name for the light microscope? (Department Qs)		
A-	Simple microscope	B-	Compound microscope
C-	Phase contrast microscope	D-	Dissection microscope
E-	Transmission electron microscope		

5-	Which microscope does not rely on visible light? (Department Qs)		
A-	Simple microscope	B-	Compound microscope
C-	Phase contrast microscope	D-	Dissection microscope
E-	Transmission electron microscope		

6-	When using a compound microscope, what is the magnification of the oil immersion lens? (Department Qs)								
A-	4X	B-	10X	C-	40X	D-	100X	E-	1000X

7-	What is the usual magnification of the ocular lens on a compound microscope? (Department Qs)								
A-	1X	B-	10X	C-	100X	D-	1000X	E-	10,000X

8-	What is the role of the condenser lens? (Department Qs)		
A-	Control the aperture of light	B-	Increase the magnification
C-	Focus the light on the specimen	D-	Initial magnification of 10X

E-	Provide light

9-	On a microscope, what structure varies the diameter of the cone of light? (Department Qs)								
A-	Base	B-	Nosepiece	C-	Stage	D-	Tube	E-	Diaphragm

10-	Where do you place the slide when using a microscope? (Department Qs)								
A-	Base	B-	Nosepiece	C-	Stage	D-	Tube	E-	Diaphragm

11-	What is the bottom of a microscope called? (Department Qs)								
A-	Base	B-	Nosepiece	C-	Stage	D-	Tube	E-	Diaphragm

12-	What is another name for the bright field microscope? (Department Qs)		
A-	Simple microscope	B-	Compound microscope
C-	Phase contrast microscope	D-	Dissection microscope
E-	Transmission electron microscope		

13-	Which of the following stain blue with H&E stain? (Department Qs)				
A-	Cytoplasm	B-	Collagen fibers	C-	Nucleus
D-	Elastic fibers	E-	Decalcified bone matrix		

14-	During the preparation of a routine H&E slide, how is the tissue preserved? (Department Qs)				
A-	Fixation	B-	Embedding in paraffin	C-	Staining
D-	Slicing	E-	Dehydration		

15-	During the preparation of a routine H&E slide, what allows the tissue to be visualized? (Department Qs)				
A-	Fixation	B-	Embedding in paraffin	C-	Staining
D-	Slicing	E-	Dehydration		

16-	During the preparation of a routine H&E slide, what step occurs after the tissue is embedded in paraffin? (Department Qs)				
A-	Fixation	B-	Rehydration	C-	Staining
D-	Slicing	E-	Dehydration		

17-	During the preparation of a routine H&E slide, what step occurs after the tissue is sliced? (Department Qs)				
A-	Fixation	B-	Embedding in paraffin	C-	Staining
D-	Slicing	E-	Dehydration		

18-	Which of the following is an acid dye? (Department Qs)				
A-	Azure II	B-	Toluidine blue	C-	Methylene blue
D-	Eosin	E-	Thionine		

19-	Which of the following stains is used for routine histological examination? (Department Qs)		
A-	Wright's stain	B-	Hematoxylin and eosin stain
C-	Sudan stain	D-	Silver impregnation
E-	Masson's trichrome stain		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
D	D	D	B	E	D	B	C	E	C
11	12	13	14	15	16	17	18	19	20
A	B	C	A	C	D	C	D	B	-

SNU Bank

1-	Haematoxylin stains what structure: (SNU 1 Bank)						
A-	Cytoplasm	B-	Cell membrane	C-	Collagen fibers	D-	Nucleus

2-	Haematoxylin is: (SNU 1 Bank)		
A-	Acidic dye	B-	Basic dye
C-	Special stain	D-	Histochemical stain

3-	Eosin is: (SNU 1 Bank)		
A-	Acidic dye	B-	Basic dye
C-	Special stain	D-	Histochemical stain

4-	Electron microscope: (SNU 1 Bank)
A-	Has only one type
B-	Its power of magnification is higher than light microscope
C-	Type of lens is glass
D-	Its medium is air

Answers - SNU Bank									
1	2	3	4	5	6	7	8	9	10
D	B	A	B	-	-	-	-	-	-

M3WAN Bank

1-	**Magnification of a microscope is best defined as the: (M3WAN - Core)?**
A-	power to distinguish fine details
B-	increase in linear dimensions of an object without optical defects
C-	shortest distance between two details below which they appear as one
D-	product of numerical aperture and wavelength

2-	**The limit of resolution of a microscope is defined as: (M3WAN - Core)?**
A-	the total magnification produced by the objective and ocular lenses
B-	the increase in linear dimensions of the specimen
C-	the shortest distance between two details below which they appear as one or cannot be seen
D-	the wavelength of the illuminating source

3-	**The limit of resolution of the ordinary light microscope is approximately: (Important M3WAN - Core)?**						
A-	200 um	B-	2 um	C-	0.24 um	D-	0.24 nm

4-	**The limit of resolution of the electron microscope is approximately: (M3WAN - Core)?**						
A-	0.24 um	B-	2.4 nm	C-	200 nm	D-	0.24 nm

5-	**The optical components of the ordinary light microscope consist of three systems of lenses, namely: (M3WAN - Core)?**
A-	condenser, objective and ocular (eye-piece) lenses
B-	objective, prism and mirror
C-	condenser, diaphragm and filter
D-	objective, polarizer and ocular

6-	**Which type of microscope permits direct observation of living cells, based on the difference in refractive index between cellular structures? (Important M3WAN - Core)?**		
A-	Fluorescent microscope	B-	Scanning electron microscope
C-	Phase contrast microscope	D-	High voltage electron microscope

7-	**The principle of fluorescence microscopy depends on: (M3WAN - Core)?**
A-	differences in refractive index between structures
B-	emission of visible light by certain substances after ultraviolet irradiation
C-	scanning of a narrow electron beam across the specimen surface
D-	combination of heavy metals with tissue components

8-	Which fluorescent stain has an affinity to combine with DNA and RNA? (M3WAN - Hard)?						
A-	Osmic acid	B-	Janus greenB	C-	Uranyl acetate	D-	Acridine orange

9-	The illumination source of the transmission electron microscope (TEM) is: (M3WAN - Core)?
A-	an ordinary tungsten lamp
B-	ultraviolet light
C-	an electron beam emitted by a heated cathode (tungsten filament)
D-	a laser beam

10-	A microscope that uses a high-penetrating electron beam and can be used to examine relatively thick sections (5 um) and living cells is the: (M3WAN - Hard)?		
A-	transmission electron microscope	B-	high voltage electron microscope
C-	scanning electron microscope	D-	phase contrast microscope

11-	The scanning electron microscope (SEM) is mainly used to study: (Important M3WAN - Core)?		
A-	the fine internal ultrastructure of organelles	B-	the details of surfaces and contours of cells
C-	living cells in real time	D-	fine chromosomal banding patterns

12-	Examination of living cells after dissociation of tissue with trypsin and culture in a sterile medium is termed: (M3WAN - Core)?		
A-	freeze drying	B-	frozen sectioning
C-	tissue culture (in vitro study)	D-	vital staining

13-	The main function of fixation in tissue processing is to: (Important M3WAN - Core)?
A-	replace water gradually with alcohol
B-	render the tissue transparent
C-	infiltrate the tissue with paraffin
D-	prevent postmortem autolysis and putrefaction and harden the tissue

14-	During preparation of a paraffin section, the correct order of steps is: (Important M3WAN - Core)?
A-	fixation, dehydration, clearing, embedding, sectioning and staining
B-	clearing, fixation, embedding, staining and sectioning
C-	staining, washing, fixation, dehydration and embedding
D-	cutting, fixation, washing, dehydration and staining

15-	In the paraffin technique, xylol is used mainly as? (M3WAN - Core)				
A-	fixative	B-	dehydrating agent	C-	clearing agent
D-	decalcifying agent				

16-	A recognized drawback of the frozen section technique is: (M3WAN - Hard)?		
A-	excessive tissue shrinkage	B-	formation of large ice crystals inside the cell
C-	inability to preserve enzyme activity	D-	prolonged embedding time

17-	In the freeze-drying technique, the tissue is immediately frozen in liquid helium at approximately: (M3WAN - Hard)?						
A-	0°C	B-	-20°C	C-	-70°C	D-	-170°C

18-	In the preparation of tissue for electron microscopy, cold glutaraldehyde fixation is typically followed by: (Important M3WAN - Hard)?
A-	dehydration in ascending grades of alcohol directly
B-	fixation in 1% osmic acid for one hour
C-	embedding in epoxy resin
D-	staining with uranyl acetate

19-	Osmic acid, used in electron microscopy preparation, functions as: (Important M3WAN - Hard)?
A-	a dehydrating agent only
B-	an embedding medium
C-	both a fixative and a stain with affinity for lipid structures and the outer membrane layers
D-	a decalcifying agent

20-	Ultra-thin sections (60–80 nm) intended for electron microscopy are cut using an ultramicrotome fitted with? (M3WAN - Core)				
A-	glass knife	B-	steel blade	C-	diamond knife
D-	razor blade				

21-	Uranyl acetate and lead citrate are used in electron microscopy preparation to: (M3WAN - Core)?
A-	fix the tissue
B-	render tissue components more electron dense
C-	dehydrate the tissue
D-	decalcify bone tissue

22-	All of the following are correctly matched special tissue preparations EXCEPT: (Important M3WAN - Hard)?
A-	Filming – examination of tissue fluids such as blood
B-	Teasing – examination of muscle and nerve fibers
C-	Grinding – examination of bone

D-	Smearing – examination of loose connective tissue

23-	**The essential difference between vital and supravital staining is that: (Important M3WAN - Hard)?**
A-	vital staining is done on tissues in the living state inside the body, while supravital staining is done on tissues in the living state outside the body
B-	vital staining requires a histochemical reaction, while supravital staining does not
C-	supravital staining is done inside the living body, while vital staining is done outside it
D-	there is no real difference between the two techniques

24-	**Janus greenB is used as a supravital stain to demonstrate: (Important M3WAN - Core)?**		
A-	glycogen granules	B-	lipid droplets
C-	mitochondria in the fresh state	D-	nucleic acids

25-	**Hematoxylin is a basic dye that stains acidic cell structures, such as the nucleus; such stained structures are described as: (Important M3WAN - Core)?**						
A-	acidophilic	B-	basophilic	C-	chromophobic	D-	amphophilic

26-	**Eosin is an acidic dye that stains basic cytoplasmic structures pink to red; such stained structures are described as: (M3WAN - Core)?**		
A-	basophilic	B-	chromophobic
C-	acidophilic (eosinophilic)	D-	neutrophilic

27-	**The specific chemical group of a dye that allows it to be retained by tissue components, by reacting with an acidic or basic group in the tissue, is the: (M3WAN - Hard)?**						
A-	chromophore	B-	polar region	C-	fluorophore	D-	auxochrome

Answers - M3WAN Bank									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	**C**	**C**	**D**	**A**	**C**	**B**	**D**	**C**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	**C**	**D**	**A**	**C**	**B**	**D**	**B**	**C**	**C**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	**D**	**A**	**C**	**B**	**C**	**D**	**-**	**-**	**-**

Lecture 2: Plasma membrane

Pre Exams & Formatives

1-	The cell coat is formed of ? (MID 33, Formative 36, Lecture Qs)		
A-	Lipids and proteins	B-	Glycoproteins and glycolipids
C-	Calcium ions	D-	glycolipids only

2-	What is the polysaccharide coating that is sometimes found on the cell membrane? (MID 30, FINAL 30)				
A-	Plasma lemma	B-	Glycocalyx	C-	Protoplasm
D-	Cristae	E-	Ground substance		

3-	Microionic environment ? (FINAL 30)				
A-	Cell coat	B-	Caveolae	C-	Tight junction
D-	Cilia	E-	microvilli		

4-	The following is not related to the cell coat? (MID 31)				
A-	Antigenicity	B-	Filtration	C-	cellular recognition
D-	Mitosis	E-	enzymatic activity		

5-	How thick is the plasma membrane? (FINAL 31, FINAL 32)				
A-	7-10 angstroms	B-	7-10 nanometers	C-	7-10 micrometers
D-	7-10 millimeters	E-	7-10 um		

6-	Molecular Structure Of The Cell Membrane is.....? (MID 32, Formative 37)
A-	75% proteins, 20% lipids and 5% carbohydrates
B-	50% proteins, 30% lipids and 20% carbohydrates
C-	60% proteins, 35% lipids and 5% carbohydrates
D-	None of the above

7-	The major molecular component of cell membrane is? (2ND TRIAL 32)						
A-	Lipids	B-	Proteins	C-	Carbohydrates	D-	Glycolipids

8-	Particles leaves the cell by? (2ND TRIAL 32)						
A-	Pinocytosis	B-	Phagocytosis	C-	Cyclosis	D-	Exocytosis

9-	Which of the following is NOT a component of the cell plasma membrane? (2ND TRIAL 32)						
A-	cholesterol	B-	proteins	C-	microfilaments	D-	phospholipids

10-	**Which of the following types of molecules are the major structural components of the cell membrane? (2ND TRIAL 32)**		
A-	phospholipids and cellulose	B-	nucleic acids and proteins
C-	phospholipids and proteins	D-	proteins and cellulose
E-	glycoproteins and cholesterol		

11-	**In order for a protein to be an integral membrane protein it would have to be which of the following? (2ND TRIAL 32)**		
A-	hydrophilic	B-	hydrophobic
C-	spin the whole thickness of the cell membrane	D-	completely covered with phospholipids
E-	exposed on only one surface of the membrane		

12-	**Which of these often serve as ion channels across the cell membrane? (2ND TRIAL 32)**				
A-	phospholipids	B-	integral proteins	C-	peripheral proteins
D-	Integrins	E-	glycoproteins		

13-	**Solid particles enter the cells by? (MID 33, Formative 36, Formative 32)**						
A-	pinocytosis	B-	exocytosis	C-	phagocytosis	D-	mitosis

14-	**Plasma membrane? (MID 34)**
A-	Ceal coat at the external surface

15-	**About plasma membrane?(FINAL 34)**				
A-	passage of protein	B-	lipid synthesis	C-	glycogen synthesis

16-	**Glucose is transported by? (2ND TRIAL 35)**
A-	Facilitated diffusion and Secondary active transport

17-	**Active structure in cell membrane?(FINAL 36)**		
A-	Glycocalyx	B-	Glycolipid
C-	Glycoprotein	D-	None of the above

18-	**Transport of solid materials through the cell membrane is called ? (MID 37)**						
A-	phagocytosis	B-	exocytosis	C-	pinocytosis	D-	passive diffusion

19-	**The part of the cell membrane that gives tight fixation to cells is: (MID 37)**		
A-	Glycocalyx	B-	Glycolipid
C-	Glycoprotein	D-	Glycated hemoglobin

20-	What is the primary function of the Cell Membrane? (FINAL SNU 1)
A-	Protein synthesis and modification
B-	Regulation of substance transport between the intracellular and extracellular environments

21-	All common peripheral protein except?(FINAL 37)						
A-	Antigen	B-	Enzymes	C-	Transport protein	D-	Receptors

22-	The intracellular fluid has a higher concentration of which of the following in comparison with the extracellular fluid? (FINAL SNU-M 1)						
A-	Sodium	B-	Potassium	C-	Chloride	D-	Proteins

23-	Pinocytosis is best defined as: ..... (2ND TRIAL 37)
A-	Transport of fluid material
B-	Transport of solid material
C-	Expulsion of residual bodies
D-	Specific receptor-mediated transport

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
B	B	A	D	B	C	B	D	C	C
11	12	13	14	15	16	17	18	19	20
C	B	C	A	A	A	C	A	A	B
21	22	23	24	25	26	27	28	29	30
C	B	A	-	-	-	-	-	-	-

Lecture Qs

1-	The ionic pore in the plasma membrane is: (Lecture Qs, Department Qs)?		
A-	Peripheral protein	B-	Transmembrane protein
C-	Formed of hydrophilic part of phospholipids	D-	Glycolipid

2-	One of the following characteristics is not true for P'M. (Lecture Qs)		
A-	fluid moasiac	B-	trilaminar structure
C-	free permeable to all substances	D-	have specific receptors

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
B	C	-	-	-	-	-	-	-	-

Department Qs

1-	The mobility of integral proteins can be measured by physical state of the? (Department Qs)		
A-	Amino acids	B-	External phospholipids
C-	Membrane phospholipids	D-	Membrane appendage

2-	This is not the function of plasma membrane? (Department Qs)		
A-	Energy transduction	B-	Intercellular interactions
C-	Responding to external stimul	D-	Assisting in chromosome segregation

3-	In the carbohydrates, plasma membrane? (Department Qs)
A-	Always faces outwards extracellular space
B-	Directed to all sides in the membrane randomly
C-	always faces to the lumen of cells
D-	Always faces inward to the nonpolar portion of membrane

4-	In the plasma membrane, Glycolipids are usually situated in? (Department Qs)
A-	Cannot be predicted, it varies according to the cell types
B-	Inner leaflet of plasma membrane
C-	Outer leaflet of plasma membrane
D-	Evenly distributed in both outer and inner leaves of plasma membrane

5-	The major interaction responsible for stabilizing plasma membrane? (Department Qs)		
A-	Hydrophobic interactions	B-	Hydrophilic interactions
C-	Covalent bonds	D-	Ionic bonds

6-	In plasma membrane, lipid molecules are arranged in? (Department Qs)						
A-	Head parallel	B-	Alternate	C-	Scattered	D-	Series

7-	Ion carriers are located in? (Department Qs)		
A-	Plasma membranes	B-	Cell wall
C-	Nucleus	D-	Cellular space

8-	Plasma membrane is made up of? (Department Qs)
A-	A protein, a lipid and a cellulose layer
B-	Bimolecular lipid layer surrounded by protein layers
C-	A protein layer between two lipid layers
D-	A lipid layer between two protein layers

9-	The thickness of the PM is? (Department Qs)								
A-	10um	B-	100 nm	C-	10 nm	D-	75 um	E-	7mm

10-	Integral proteins? (Department Qs)		
A-	Occupy only a part of the P'M	B-	Extend across the P'M
C-	Are present on the surface of the P'M	D-	A and B
E-	A and C		

11-	One of the following characteristics is not true .for P'M? (Department Qs)		
A-	Fluid moasic	B-	Trilaiminar structure
C-	Free permeable to all substances	D-	Have specific receptors
E-	All of the above		

12-	The plasma membrane: the following is not related to the cell coat? (Department Qs)								
A-	Antiginicity	B-	Filtration	C-	Receptors	D-	Protection	E-	Mitosis

13-	A solid particle enters the cell by? (Department Qs)						
A-	Exocytosis	B-	Pinocytosis	C-	Phagocytosis	D-	Cyclosis

14-	Carriers envolved in active transport are? (Department Qs)						
A-	Lipids	B-	Proteins	C-	Carbohydrates	D-	Phagosomes

15-	Protein pores are envolved in? (Department Qs)		
A-	Pinocytosis	B-	Transport of lipid soluble material
C-	lonic transportation	D-	Phagocytosis

16-	Some integral proteins are envolved in? (Department Qs)				
A-	lonic transportation	B-	Active transportation	C-	Formation of the nexus
D-	All of the above	E-	None of the above		

17-	Integral proteins? (Department Qs)		
A-	Are present on the inner surface of P'M	B-	Cause the fluidity of P'M
C-	Are present on the outer surface of P'M	D-	All of the above

18-	The fluid mosaic model of membrane structure? (Department Qs)
A-	contains actin filaments
B-	has hydrophobic elements directed away from the center
C-	has hydrophilic elements directed away from the center

D-	contains intermediate filaments
E-	none of the above

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	D	A	C	A	A	A	B	C	D
11	12	13	14	15	16	17	18	19	20
C	E	C	B	C	D	D	C	-	-

SNU Bank

1-	Cell membrane: (SNU 1 Bank)		
A-	unilipid layer	B-	bilipid layer
C-	consists of proteins only	D-	trilipid layer

2-	Cell membrane: (SNU 1 Bank)		
A-	Does not contain cholesterol	B-	Has no carbohydrates
C-	Contains lipid and proteins and carbohydrates	D-	has no proteins

3-	Integral proteins are best described as proteins that: (SNU 1 Bank)		
A-	Are loosely attached to membrane surface	B-	Are embedded within the lipid bilayer
C-	Are found only on the outer surface	D-	Are made only of carbohydrates

4-	Which part of a phospholipid is hydrophobic? (SNU 1 Bank)		
A-	Phosphate head	B-	Fatty acid tails
C-	Glycerol backbone	D-	Carbohydrate chains

5-	Glycoproteins in the cell membrane function mainly in: (SNU 1 Bank)		
A-	Energy production	B-	Cell recognition
C-	Protein synthesis	D-	DNA replication

6-	Which process involves the cell taking in particles by engulfing them? (SNU 1 Bank)					
A-	Exocytosis	B-	Endocytosis	C-	Diffusion	D-	Filtration

Answers - SNU Bank									
1	2	3	4	5	6	7	8	9	10
B	C	B	B	B	B	-	-	-	-

M3WAN Bank

1-	The plasma membrane (plasmalemma) primarily functions to: (Important M3WAN - Core)?
A-	store genetic material
B-	regulate the passage of materials between the cell and its surroundings
C-	generate ATP for cellular activities
D-	synthesize secretory proteins

2-	The average thickness of the plasma membrane is: (Important M3WAN - Core)?		
A-	7-10 angstroms	B-	7-10 micrometers
C-	7-10 nanometers	D-	7-10 millimeters

3-	The plasma membrane cannot be distinguished by the light microscope mainly because: (M3WAN - Core)?
A-	it does not take up any stain
B-	its thickness is below the limit of resolution of the light microscope
C-	it is dissolved during fixation
D-	it is located deep within the cytoplasm

4-	By electron microscopy, the plasma membrane appears as: (M3WAN - Core)?
A-	a single homogeneous dense line
B-	three layers: outer and inner dark lines with a light middle layer (trilaminar)
C-	a granular meshwork without distinct layers
D-	two dark lines only, without a middle layer

5-	The observation that all cellular membranes share the same trilaminar structure by electron microscopy is known as: (Important M3WAN - Core)?		
A-	the fluid mosaic model	B-	the glycocalyx theory
C-	the unit membrane theory	D-	the symbiont theory

6-	The molecular composition of the plasma membrane is approximately: (Important M3WAN - Core)?
A-	75% proteins, 20% lipids, 5% carbohydrates
B-	50% proteins, 30% lipids, 20% carbohydrates
C-	35% proteins, 60% lipids, 5% carbohydrates
D-	60% proteins, 35% lipids, 5% carbohydrates

7-	The lipid bilayer of the plasma membrane is mainly formed of: (M3WAN - Core)?		
A-	phospholipid and cholesterol molecules	B-	glycoprotein and glycolipid molecules
C-	collagen and elastin molecules	D-	integral and peripheral protein molecules

8-	The hydrophilic (polar) part of the phospholipid molecule: (M3WAN - Core)?
A-	forms the tail and faces the center of the membrane
B-	forms the head and is directed towards the intracellular and extracellular surfaces
C-	is composed of non-charged lipid
D-	has no affinity for aqueous solutions

9-	The hydrophobic (non-polar) part of the phospholipid molecule: (M3WAN - Core)?
A-	forms the head and faces the membrane surfaces
B-	is composed of charged lipid
C-	forms the tail, and the tails face each other in the center of the membrane
D-	has a great affinity for aqueous solutions

10-	Cholesterol molecules in the plasma membrane: (M3WAN - Core)?
A-	are mainly present on the extracellular surface and increase membrane fluidity
B-	are mainly present on the cytoplasmic aspect and stabilize the packing of phospholipid molecules
C-	form the outer dark line seen by electron microscopy
D-	are absent from most cell membranes

11-	Protein molecules that are loosely bound to either surface of the plasma membrane are called: (M3WAN - Core)?		
A-	intrinsic or integral proteins	B-	transmembrane proteins
C-	extrinsic or peripheral proteins	D-	glycoproteins

12-	Large intrinsic protein molecules that extend through the entire thickness of the plasma membrane are known as: (Important M3WAN - Core)?		
A-	peripheral proteins	B-	glycolipids
C-	extrinsic proteins	D-	transmembrane proteins

13-	Integral membrane proteins that often serve as ion channels across the plasma membrane must: (Important M3WAN - Hard)?
A-	be loosely attached to only one surface of the membrane
B-	span the entire thickness of the cell membrane
C-	be composed only of carbohydrate
D-	be completely covered by phospholipid molecules

14-	Oligosaccharides attached to membrane lipids form: (M3WAN - Core)?		
A-	glycoproteins	B-	glycolipids
C-	peripheral proteins	D-	transmembrane channels

15-	The cell coat, or glycocalyx, projecting from the outer surface of the plasma membrane is formed of: (M3WAN - Core)?		
A-	phospholipids and cholesterol only	B-	extrinsic proteins only
C-	glycolipids and glycoproteins	D-	intermediate filaments

16-	Passive diffusion across the plasma membrane allows entry of substances such as: (M3WAN - Core)?		
A-	large sugar molecules requiring a carrier	B-	dissolved gases and essential ions
C-	solid particulate matter	D-	droplets of extracellular fluid

17-	Active transport of large molecules such as sugars, amino acids, and fatty acids across the plasma membrane requires: (M3WAN - Core)?
A-	receptor recognition only
B-	combination with a catalyst and energy supplied by mitochondrial enzymatic reactions
C-	formation of a phagosome
D-	no energy expenditure

18-	Transport of drugs, hormones, bacteria, and viruses that act specifically on their target cells occurs mainly through: (M3WAN - Core)?		
A-	passive diffusion	B-	the sodium-potassium pump
C-	selective transport using membrane receptors	D-	exocytosis

19-	Engulfment of solid particulate material by the plasma membrane, forming an intracytoplasmic phagosome, is called: (Important M3WAN - Core)?						
A-	pinocytosis	B-	phagocytosis	C-	exocytosis	D-	passive diffusion

20-	Engulfment of droplets of extracellular fluid by the plasma membrane, forming a pinocytic vesicle, is called: (M3WAN - Core)?		
A-	Phagocytosis	B-	active transport
C-	pinocytosis	D-	selective transport

21-	After lysosomal digestion of engulfed material, expulsion of the residual bodies outside the cell occurs by: (M3WAN - Core)?						
A-	pinocytosis	B-	passive diffusion	C-	phagocytosis	D-	exocytosis

22-	Continuous pumping of sodium ions out of the cell, together with uptake of potassium ions into the cell, is performed by the: (M3WAN - Core)?		
A-	sodium-potassium pump	B-	calcium pump
C-	proton pump	D-	glucose transporter

23-	The component of the plasma membrane that is mainly responsible for tight adhesion between adjacent cells is the: (Important M3WAN - Core)?		
A-	cholesterol molecule	B-	glycocalyx (cell coat)
C-	phospholipid bilayer	D-	peripheral protein

24-	The glycocalyx plays an important role in cell recognition, which is clinically relevant in: (Important M3WAN - Core)?
A-	oxygen diffusion across capillaries
B-	transplantation compatibility of organs such as kidney, heart, and skin
C-	generation of the mitotic spindle
D-	formation of secretory vesicles

25-	Besides adhesion and recognition, another important function of the glycocalyx is: (M3WAN - Core)?
A-	energy production
B-	development of cell immunity against infections
C-	formation of the mitotic spindle
D-	packaging of hydrolytic enzymes

26-	The plasma membrane may be modified to form specialized structures such as microvilli, cilia, and flagella, whose main shared purpose is to: (M3WAN - Core)?
A-	store genetic information
B-	serve specific functions in certain specialized cells
C-	generate ATP by oxidative phosphorylation
D-	synthesize lysosomal enzymes

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	C	B	B	C	D	A	B	C	B
11	12	13	14	15	16	17	18	19	20
C	D	B	B	C	B	B	C	B	C
21	22	23	24	25	26	27	28	29	30
D	A	B	B	B	B	-	-	-	-

Lecture 3: Membranous organelles (Endoplasmic reticulum, Mitochondria, Peroxisomes)

Pre Exams & Formatives

1-	**Which organelle is considered the powerhouse of the cell? (FINAL 35, MID 34, FINAL SNU-M 1, Formative 37, Department Qs)**		
A-	Rough endoplasmic reticulum	B-	Smooth endoplasmic reticulum
C-	Lysosome	D-	Golgi apparatus
E-	Mitochondria		

2-	**One of the functions of smooth endoplasmic reticulum is? (Formative 36, Formative 32, Department Qs)**		
A-	Regulation of intracellular calcium distribution	B-	Endocytosis
C-	Excretion	D-	Protein synthesis

3-	**Concerning rough endoplasmic reticulum, it is? (Formative 37, Department Qs)**
A-	The site of protein synthesis that will be secreted by the cell
B-	Its surface membrane is studded with ribosomes
C-	Composed of membranous tubules and sacs
D-	All of the above

4-	**What are the folds on the inner mitochodrial membrane called? (Formative 37, Department Qs)**						
A-	Cristae	B-	Protoplasm	C-	Glycocalyx	D-	Plasmalemma

5-	**Which organelle produces protein for export? (Formative 37, Department Qs)**		
A-	Rough endoplasmic reticulum	B-	Golgi bodies
C-	Lysosome	D-	Smooth endoplasmic reticulum

6-	**Which organelle is studded with ribosomes? (Formative 37, Department Qs)**		
A-	Lysosome	B-	Smooth endoplasmic reticulum
C-	Rough endoplasmic reticulum	D-	Golgi apparatus

7-	**What does the rough endoplasmic reticulum produce? (Formative 37, Department Qs)**						
A-	Fats	B-	Oxygen	C-	Proteins	D-	Carbon dioxide

8-	**The following are the characters of the mitochondria EXCEPT? (FINAL 31, Formative 36, Department Qs)**		
A-	they need special stain to be seen	B-	they have double membranes
C-	they can divide	D-	they contain DNA

E-	they contain some hydrolytic enzymes

9-	**Concerning characters of mitochondria? (MID 30, FINAL 30)**
A-	Surrounded by single unit membrane
B-	cannot move or divide
C-	Both: a&b
D-	contain electron dense matrix granules in the outer space
E-	Inner membrane forms tubular projections contain enzymes for electron transport

10-	**Peroxisomes are? (MID 30)**
A-	Spherical bodies not membrane bounded cell organelle
B-	Contain enzymes of glycolysis
C-	Contain lipase enzymes
D-	Hydrogen peroxide is formed inside peroxisomes
E-	Both: a&b

11-	**The mitochondrial DNA is? (MID 30, FINAL 30)**				
A-	Similar to nuclear DNA	B-	Inherited from the mother	C-	Helical structure
D-	Devoid of genes	E-	Can't divide		

12-	**All of the following are true of the endoplasmic reticulum EXCEPT? (MID 30)**
A-	It is involved only in the synthesis of the exportable protein
B-	It is continuous with the nuclear membrane
C-	It varies in the density depending upon the functional activity of the cell
D-	It is involved in protein synthesis
E-	One of the membranous organelles

13-	**Synthesis of the enzymes occurs in? (MID 30, FINAL 30)**						
A-	nucleus	B-	Golgi bodies	C-	R'E'R	D-	both
E-	b			F-	mitochondria		

14-	**Steroid secreting cell is particularly rich in?(FINAL 30)**		
A-	Peroxisome	B-	Rough endoplasmic reticulum
C-	Golgi apparatus	D-	smooth endoplasmic reticulum
E-	Lysosomes		

15-	**Peroxisomes are? (FINAL 30)**
A-	membrane bounded cell organelle
B-	contain enzymes of glycolysis
C-	contain lipase enzymes
D-	hydrogen peroxide is formed inside peroxisomes
E-	both:a&d

16-	**SER ? (FINAL 30)**				
A-	kreb's cycle	B-	microtubles	C-	pinocytosis
D-	lipid synthesis	E-	electron transfer		

17-	**elementary particles ? (FINAL 30)**				
A-	kreb's cycle	B-	microtubles	C-	pinocytosis
D-	lipid synthesis	E-	electron transfer		

18-	**Mitochondrial matrix ? (FINAL 30)**				
A-	kreb's cycle	B-	microtubles	C-	pinocytosis
D-	lipid synthesis	E-	electron transfer		

19-	**The elementary particles of the inner mitochondrial membrane are the site? (MID 31)**		
A-	where the Kreb's cycle takes place	B-	of ATP formation
C-	of glycolysis	D-	of mitochondrial division
E-	of oxidation of fatty acids		

20-	**Calcium is present in? (MID 31)**				
A-	S'E'R	B-	R'E'R	C-	mitochondria
D-	lysosomes	E-	ribosomes		

21-	**Smooth endoplasmic reticulum? (MID 31)**
A-	has ribosomes attached to its membranes
B-	is highly found in steroid secrete cells
C-	cereain localized basophilia indicate its presence
D-	is present in cells where drug detoxification is taking place
E-	Both B&D

22-	**All the following statement concerning peroxisomes are true EXCEPT? (MID 31)**
A-	spherical membrane bounded cell organelle
B-	contain enzymes of fatty acid oxidation

C-	contain catalase enzymes
D-	hydrogen peroxide does not form inside peroxisomes
E-	contain amino acid oxidase

23-	**Cytoplasmic basophilia is characteristic of cells that? (FINAL 31, 2ND TRIAL 32)**		
A-	are actively producing steroid hormones	B-	have abundant DNA in the cytoplasm
C-	are in theS stage	D-	are synthesizing large amounts of protein
E-	exhibit a large Golgi complex		

24-	**Each of the following statements describing mitochondrial cristae is true EXCEPT: ? (MID 32)**		
A-	Are folds in the inner mitochondrial membrane	B-	They contain enzymes for electron transport
C-	Variable in number from one cell to other	D-	They contain enzymes of Krebs' cycle
E-	both:a&c		

25-	**Smooth endoplasmic reticulum ? (FINAL 32)**
A-	often occurs in the form of branching anastomosing tubules
B-	house of energy
C-	is present in cells where drug detoxification is taking place
D-	is rarely found in skeletal muscle cells
E-	both:a&c

26-	**ß oxidation takes place in: ? (FINAL 32)**				
A-	R'E'R	B-	Golgi body	C-	Cell nucleus
D-	Lysosome	E-	Peroxisomes		

27-	**Which structure within the cell produces ATP (adenosine triphosphate)? (2ND TRIAL 32)**		
A-	The mitochondria	B-	the nucleus
C-	Peripheral Proteins	D-	the endoplasmic reticulum

28-	**The inner mitochondrial membrane? (MID 33)**
A-	contains porins protein
B-	is thick permeable
C-	is surrounding the matrix
D-	is thrown into folds with constant number from cell to other

29-	**Protein secreting cells are rich in?(FINAL 33)**						
A-	Free ribosomes	B-	sER	C-	RER	D-	Mitochondria

30-	Rough endoplasmic reticulum?(FINAL 33)
A-	Occurs in the form of un-connected tubules with attached ribosomes
B-	Is connected with outer nuclear membrane
C-	Forms only protein for outside the cell
D-	Both:a&b

31-	Which of structure localized basophil? (MID 34, Formative 36, Formative 37)
A-	RER

32-	Which of the following is true about? (MID 34)
A-	Mitochondria

33-	Which of these has catalase enzyme?(FINAL 34)						
A-	RER	B-	microbody	C-	SER	D-	mitochondria

34-	Which of the following helps in detoxification of drugs? (FINAL 35, Formative 37)						
A-	RER	B-	Peroxisomes	C-	Lysosomes	D-	Golgi

35-	Smooth endoplasmic reticulum? (2ND TRIAL 35)
A-	Has ribosomes attached to its membranes
B-	Is rarely found in skeletal muscle cells
C-	Areas of localized basophilia indicate its presence
D-	Is present in cells where drug detoxification is taking place

36-	Related to mitochondria functions?(FINAL 36)		
A-	Storage of Ca	B-	Initiating apoptosis

37-	Smooth endoplasmic reticulum ? (MID 37)		
A-	Appears as dense granules	B-	ribosomes attached to its surface
C-	the site of drug detoxification	D-	it is not found in skeletal muscles

38-	Functions of mitochondria include: ? (MID 37) وظيفة الميتوكوندريا انها تعمل Regulation لتركيز ال Ca اما المسئول عن تخزينه SER		
A-	Storage of Ca+2	B-	Catabolism of toxic substances
C-	Secretion of granzymes	D-	Initiation of apoptosis

39-	Which of the following physiological processes is a major function of the smooth endoplasmic reticulum? (FINAL SNU-M 1, Formative 37)		
A-	Protein synthesis and extracellular packaging	B-	Lipid metabolism and drug detoxification
C-	Intracellular digestion of metabolic debris	D-	Generation of adenosine triphosphate (ATP)

40-	Organelle called symbion? (Formative 36, Formative 32)						
A-	Golgi apparatus	B-	RER	C-	mitochondria	D-	SER

41-	The attached ribosomes to RER synthesize? (Formative 37)
A-	Cytoplasmic protein necessary for the cell own existence
B-	Carbohydrate
C-	Lipid
D-	Protein that will be secreted by the cell

42-	What is structure 1? (Formative 32)				
A-	Nucleus	B-	Mitochondria	C-	RER
D-	SER	E-	Golgi body		

43-	Which of the following is synthesized by the smooth endoplasmic reticulum? (2ND TRIAL 37)						
A-	Proteins	B-	Glycogen	C-	Ribosomes	D-	ATP

44-	Oxidation of long-chain fatty acids with the production of H2O2 is the function of which of the following organelles? (2ND TRIAL 37)				
A-	Mitochondria	B-	Lysosomes	C-	Peroxisomes
D-	Smooth endoplasmic reticulum				

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
E	A	D	A	A	C	C	E	E	D
11	12	13	14	15	16	17	18	19	20
B	A	C	D	E	D	E	A	B	C
21	22	23	24	25	26	27	28	29	30
E	D	D	D	E	E	A	C	C	B
31	32	33	34	35	36	37	38	39	40
A	A	B	B	D	B	C	D	B	C
41	42	43	44	45	46	47	48	49	50
D	C	B	C	-	-	-	-	-	-

Lecture Qs

1-	SER is not related to: (Lecture Qs)		
A-	Lipid metabolism	B-	Impulse transporation
C-	Protein synthesis	D-	Minizal metabolism

2-	Respiratory chain enzymes in mitochondria are present in: (Lecture Qs)		
A-	In the matrix	B-	On the outer membrane
C-	On the Ca particles	D-	On the inner membrane

3-	The localized cytoplasmic basophilia indicates the presence of: (Lecture Qs)						
A-	mitochondria	B-	SER	C-	lysosomes	D-	RER

4-	Which of the following activities are characteristics of peroxisomes? (Lecture Qs)		
A-	Catalase activity	B-	Possess a glyoxylate cycle
C-	Oxidation of very long-chain fatty acids	D-	Produce dihydrogen

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
C	D	D	A&C	-	-	-	-	-	-

Department Qs

1-	With respect to their surrounding membrane system, which is the odd one out? (Department Qs)		
A-	Nucleus	B-	Endoplasmic reticulum
C-	Mitochondria	D-	Chloroplasts

2-	The organoid that can duplicate themselves is? (Department Qs)		
A-	Rough endoplasmic reticulum	B-	Smooth endoplasmic reticulum
C-	Lysosome	D-	None of the above
E-	Mitochondria		

3-	The number of mitochondria present in a cell depends on its? (Department Qs)				
A-	Size	B-	Shape	C-	Energy requirements
D-	Capability of division	E-	None of the above		

4-	The localized cytoplasmic basophilia indicates the presence of? (Department Qs)		
A-	Mitochondria	B-	Golgi apparatus
C-	Rough endoplasmic reticulum	D-	Smooth endoplasmic reticulum
E-	Lysosomes		

5-	The mitochondria? (Department Qs)
A-	Replicate independently from the cell
B-	Are the main sites for oxidative phosphorylation

C-	Contain their own DNA & genetic materials
D-	All of the above
E-	None of the above

6-	**Which organelle is involved in lipid metabolism? (Department Qs)**		
A-	Rough endoplasmic reticulum	B-	Smooth endoplasmic reticulum
C-	Lysosome	D-	Golgi apparatus
E-	Mitochondria		

7-	**Which organelle contains detoxifying enzymes? (Department Qs)**				
A-	Ribosomes	B-	Peroxisome	C-	Microfilaments
D-	Centrioles	E-	Nucleoli		

8-	**The following are the functions of the mitochondria EXCEPT? (Department Qs)**		
A-	Heat production	B-	Sites of Krebs's cycle
C-	Site of ß oxidation of fatty acids	D-	Synthesis of protein and lipids
E-	Homeostasis of calcium		

9-	**Organelles most notable for producing hydrogen peroxide are? (Department Qs)**				
A-	Lysosomes	B-	Mitochondria	C-	Peroxisomes
D-	Peroxisomes	E-	Golgi bodies		

10-	**Which of the following contains the mitochondrial genome? (Department Qs)**		
A-	Matrix granules	B-	Outer mitochondrial membrane
C-	Ribosome	D-	Mitochondrial matrix
E-	Inner mitochondrial membrane		

11-	**Which of the following contains the enzymes responsible for the respiratory chain? (Department Qs)**		
A-	Matrix granules	B-	Outer mitochondrial membrane
C-	Ribosome	D-	Mitochondrial matrix
E-	Inner mitochondrial membrane		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
D	E	C	C	D	B	B	D	D	D
11	12	13	14	15	16	17	18	19	20
E	-	-	-	-	-	-	-	-	-

SNU Bank

1-	Detoxification occurs within which of the following structures? (SNU 1 Bank)		
A-	Mitochondria	B-	Rough endoplasmic reticulum
C-	Golgi apparatus	D-	Smooth endoplasmic reticulum

2-	One of the functions of smooth endoplasmic reticulum is: (SNU 1 Bank)						
A-	Excretion	B-	Endocytosis	C-	Protein synthesis	D-	Lipid metabolism

3-	Which of the following is INCORRECT concerning SER? (SNU 1 Bank)
A-	involved in detoxification of drugs
B-	abundant in steroid-secreting cells
C-	forms the sarcoplasmic reticulum of skeletal muscles
D-	abundant in protein-secreting cells

4-	The endoplasmic reticulum (ER) consists of which two main components? (SNU 1 Bank)		
A-	Rough ER and Mitochondria	B-	Smooth ER and Golgi Apparatus
C-	Rough ER and Smooth ER	D-	Ribosomes and Lysosomes

5-	What are the folds of the inner mitochondrial membrane called? (SNU 1 Bank)						
A-	Matrix	B-	Outer membrane	C-	Inner membrane	D-	Cristae

6-	Organelle called symbion: (SNU 1 Bank)						
A-	Golgi apparatus	B-	RER	C-	SER	D-	mitochondria

7-	Respiratory chain enzymes in mitochondria are present: (SNU 1 Bank)		
A-	In the matrix	B-	On the outer membrane
C-	On the Ca particles	D-	On the inner membrane

8-	Concerning characters of mitochondria: (SNU 1 Bank)
A-	Surrounded by a single-unit membrane
B-	cannot move or divide
C-	contain electron-dense matrix granules in the outer space
D-	Inner membrane forms tubular projections that contain enzymes for electron transport

9-	Synthesis of the enzymes occurs in: (SNU 1 Bank)						
A-	nucleus	B-	Golgi bodies	C-	R'E'R'	D-	mitochondria

Answers - SNU Bank									
1	2	3	4	5	6	7	8	9	10
D	D	D	C	D	D	D	D	C	-

M3WAN Bank

1-	By electron microscopy, the rough endoplasmic reticulum (RER) appears as: (M3WAN - Core)?
A-	branching, anastomosing tubules without ribosomes
B-	tubules, vesicles, or cisternae studded on their surface with ribosomes
C-	spherical bodies with a dense central core
D-	parallel curved stacks with a convex and a concave surface

2-	The rough endoplasmic reticulum is functionally specialized for: (Important M3WAN - Core)?
A-	lipid and steroid synthesis
B-	formation, packing, and transfer of secretory proteins and lysosomal enzymes to the Golgi
C-	detoxification of drugs and toxins
D-	storage of calcium ions only

3-	Unlike the rough endoplasmic reticulum, the smooth endoplasmic reticulum (SER): (M3WAN - Core)?
A-	is connected with the outer nuclear membrane and studded with ribosomes
B-	consists of branching, anastomosing tubules not studded with ribosomes
C-	appears as a localized area of basophilia by light microscopy
D-	is the principal site of secretory protein synthesis

4-	Cells rich in smooth endoplasmic reticulum characteristically show, by light microscopy: (M3WAN - Hard)?		
A-	localized basophilia	B-	a negative Golgi image
C-	highly acidophilic cytoplasm	D-	a deeply basophilic nucleus

5-	The smooth endoplasmic reticulum is particularly abundant in cells specialized for lipid and steroid synthesis, such as: (Important M3WAN - Core)?		
A-	hepatocytes	B-	adrenal cortical cells
C-	osteoblasts	D-	plasma cells

6-	The smooth endoplasmic reticulum plays a major role in detoxification of drugs and toxins, a function that is particularly prominent in: (Important M3WAN - Core)?		
A-	liver cells	B-	lacrimal gland cells

C-	skeletal muscle cells	D-	adrenal cortical cells

7-	**Besides steroid synthesis and detoxification, the smooth endoplasmic reticulum is also involved in: (M3WAN - Hard)?**
A-	packaging of lysosomal hydrolytic enzymes
B-	glycogen synthesis and impulse transmission within skeletal and cardiac muscle cells
C-	synthesis of ribosomal RNA
D-	formation of the mitotic spindle

8-	**All of the following statements regarding the endoplasmic reticulum are true EXCEPT: (Important M3WAN - Hard)?**
A-	it is one of the membranous cytoplasmic organelles
B-	the rough type is continuous with the outer nuclear membrane
C-	it is involved only in the synthesis of exportable (secretory) protein
D-	its density varies according to the functional activity of the cell

9-	**Mitochondria are surrounded by two membranes; regarding these membranes, the outer membrane: (Important M3WAN - Core)?**
A-	is thin, less permeable, and contains the respiratory chain enzymes
B-	is thick, permeable, and contains transmembrane channel proteins called porins
C-	is thrown into folds called cristae
D-	contains the elementary particles responsible for ATP generation

10-	**The small particles projecting from the membrane of the mitochondrial cristae, each having a globular head and a connecting stalk, represent: (Important M3WAN - Core)?**		
A-	ribosomes	B-	porins
C-	elementary particles (ATP synthetase)	D-	matrix granules

11-	**The mitochondrial matrix characteristically contains all of the following EXCEPT: (Important M3WAN - Hard)?**
A-	ring-like DNA inherited from the mother
B-	small ribosomes
C-	enzymes of the Krebs cycle and beta oxidation of fatty acids
D-	attached ribosomes on its outer surface

12-	**Mitochondria are sometimes referred to as "symbions" because they: (Important M3WAN - Core)?**
A-	resemble bacteria in structure and multiply by binary fission
B-	are formed inside the Golgi apparatus

C-	are surrounded by a single unit membrane only
D-	are devoid of any genetic material

13-	The main functions of mitochondria include energy production and: (Important M3WAN - Core)?		
A-	packaging of secretory proteins	B-	regulation of cytosolic calcium concentration
C-	decomposition of hydrogen peroxide	D-	synthesis of hydrolytic enzymes

14-	Each of the following statements concerning the mitochondria is true EXCEPT: (Important M3WAN - Hard)?
A-	they are surrounded by a double membrane
B-	they can divide by binary fission
C-	they contain DNA in their matrix
D-	they contain hydrolytic enzymes active at acidic pH

15-	By electron microscopy, peroxisomes are morphologically similar to lysosomes and appear finely granulated with a dense core known as the: (Important M3WAN - Core)?		
A-	matrix granule	B-	elementary particle
C-	nucleoid	D-	residual body

16-	Peroxisomes contain a large number of enzymes (about 50), most notably: (Important M3WAN - Core)?		
A-	acid phosphatase and proteases	B-	catalase, urate oxidase, and fatty acid oxidase
C-	respiratory chain enzymes	D-	alkaline phosphatase

17-	Regarding peroxisomes, all of the following statements are true EXCEPT: (Important M3WAN - Hard)?
A-	they are spherical, membrane-bounded organelles believed to arise from the RER
B-	they contain enzymes for beta oxidation of fatty acids
C-	they contain the catalase enzyme
D-	hydrogen peroxide is never formed inside them

18-	The catalase enzyme found within peroxisomes functions to: (M3WAN - Core)?
A-	generate ATP by oxidative phosphorylation
B-	decompose hydrogen peroxide by converting it to water
C-	synthesize secretory proteins
D-	replicate mitochondrial DNA

19-	Compared with mitochondria, peroxisomes are distinguished by: (M3WAN - Hard)?		
A-	possessing a double membrane with cristae	B-	containing ring-like maternally inherited DNA
C-	having a dense nucleoid core and no cristae	D-	being the principal site of the Krebs cycle

20-	Peroxisomes are described as being particularly numerous in which cell type? (M3WAN - Core)?						
A-	Osteoclasts	B-	Liver cells	C-	Chondrocytes	D-	Neutrophils

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	B	B	C	B	A	B	C	B	C
11	12	13	14	15	16	17	18	19	20
D	A	B	D	C	B	D	B	C	B

«يَجلسُ الرَّجلُ عِندمَا يُريدُ النَّومَ للهِ سَاعَة، يُحَاسِبُ نَفسَهُ فِيهَا عَلَى مَا خسِرَهُ وربحَهُ فِي يَومِه،
ثُـمَّ يُجدِّدُ لَهُ تَوبةً نَصُوحَةً بينهُ وبينَ اللهِ، فينامُ عَلَى تِلكَ التَّوبةِ.

ويَعزمُ عَلَى أنْ لَا يُعاودَ الذَّنبَ إذا استَيقظَ، ويَفعلُ هَذا كُلَّ ليلةٍ، فإن مَاتَ مِن لَيلتِهِ ماتَ عَلَى توبةٍ،
وإنِ استَيقظَ، استَيقظَ مُستقبلًا لِلعملِ مَسرُورًا مِن تَأخيرِ أجلِهِ حتَّى يَستقبلَ ربَّه، ويَستدركَ مَا
فَاتَهُ، وليسَ لِلعبدِ أنفعُ مِن هذهِ التَّوبةِ، ولا سيَّما إذا أعقبَ ذَلِك بذِكرِ اللهِ، واستعمَالِ السُّننِ التِي
وردَت عَن رسُولِ اللهِ ﷺ عِندَ النَّومِ، حتَّى يَغلبهُ النَّوم.

فمَن أرادَ اللهُ بِهِ خيرًا وفَّقهُ لِذلِك».

-ابنُ القيِّمِ -رَحِمَهُ اللهُ-.

[كِتابُ الرُّوحِ | | ٧٩]

Lecture 4: Membranous organelles (continued) (Golgi apparatus, Secretory vesicles, Lysosomes)

Pre Exams & Formatives

1-	The following are features of Golgi apparatus EXCEPT? (Formative 37, Department Qs)
A-	Chemical modification of some secretory products
B-	Concentration of proteins
C-	Synthesis of proteins
D-	packaging of hydrolytic enzymes into lysosomes

2-	Which organelle sorts and packages proteins within a cell? (FINAL 31, Formative 37, Lecture Qs, Department Qs)		
A-	rough endoplasmic reticulum	B-	smooth endoplasmic reticulum
C-	lysosome	D-	Golgi apparatus
E-	mitochondria		

3-	Golgi apparatus? (MID 30, FINAL 30, MID 33, Formative 36)		
A-	Stores digested foods	B-	receives proteins from theSER
C-	can't seen by silver stain	D-	Modifies chemicals to make them functional
E-	Digests bacteria and foreign bodies		

4-	Negative Golgi image is seen by H&E in?(FINAL 30)				
A-	chondrocyte	B-	Plasma cells	C-	mesenchymal cells
D-	Fat cells	E-	Both:b&c		

5-	Which of the following is a feature of Golgi apparatus: ? (MID 32)
A-	synthesis of proteins
B-	concentration of fatty acids
C-	package of glycogen
D-	svnthesis of some secretory products
E-	segregation of hydrolytic enzymes into lysosomes

6-	Intracellular hydrolytic enzymes are stored in the? (MID 32, 2ND TRIAL 35, MID 37, Formative 36, Formative 37, Formative 32)		
A-	Golgi apparatus	B-	smooth endoplasmic reticulum
C-	lysosomes	D-	residual bodies
E-	mitochondria		

7-	Which of the following organelles contain intracellular hydrolytic enzymes? (2ND TRIAL 32)		
A-	Ribosomes	B-	Mitochondria
C-	Golgi bodies	D-	None of the above

8-	Which of the following structures transfers proteins and lipids to the Golgi from the endoplasmic reticulum? (Formative 36, Formative 32)		
A-	Cis-compartment	B-	Secretory vesicles
C-	Medial compartment	D-	Transfer vesicles

9-	The Golgi apparatus is (Formative 37)?		
A-	The site of protein concentration	B-	all of the above
C-	modification of protein	D-	site for packaging enzymes for lysosomes

10-	At LM level Golgi apparatus can be demonstrated using? (Formative 32)		
A-	Sudan black	B-	Silver impregnation
C-	Negative golgi area with HX andE	D-	Both:BandC

11-	What is structure 2? (Formative 32)				
A-	Nucleus	B-	Mitochondria	C-	RER
D-	SER	E-	Golgi body		

12-	The acrosome is formed from? (Formative 34, Formative 33)						
A-	Nucleus	B-	Golgi bodies	C-	Centrosome	D-	Mitochondria

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
C	D	D	B	E	C	D	D	B	D
11	12	13	14	15	16	17	18	19	20
E	B	-	-	-	-	-	-	-	-

Lecture Qs

1-	Which organelle contains digestive enzymes? (Lecture Qs, Department Qs)		
A-	Rough endoplasmic reticulum	B-	Smooth endoplasmic reticulum
C-	Lysosome	D-	Golgi apparatus

2-	Which of the following organelles originated from the Golgi apparatus? (Lecture Qs)		
A-	Ribosomes	B-	Microtubules
C-	Secretory vesicles	D-	Glycogen granules

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
C	C	-	-	-	-	-	-	-	-

Department Qs

1-	**The Golgi apparatus is? (Department Qs)**
A-	The site of protein concentration
B-	Has transfer vesicles associated with its outer convex surface (cis) face
C-	Has secretory vesicles associated with its inner surface (trans) face
D-	None of the above
E-	All of the above

2-	**At LM level Golgi apparatus can be demonstrated using? (Department Qs)**				
A-	Sudan black	B-	Best's carmine	C-	Silver impregnation
D-	Osmic acid	E-	None of the above		

3-	**The Golgi apparatus is? (Department Qs)**
A-	The site of packing of secretory products of the cell
B-	Capable of providing energy
C-	The site of protein synthesis
D-	Continuous with smooth endoplasmic reticulum
E-	Visible only by E/M

4-	**Hydrolytic enzymes are found in? (Department Qs)**				
A-	Golgi apparatus	B-	RER	C-	SER
D-	Lysosomes	E-	Ribsosomes		

5-	**The site of lysosomes can be seen using a specific histochemical reaction called? (Department Qs)**				
A-	Alkaline phosphatase	B-	PAS	C-	Succinic dehydrogenase
D-	Acid phosphatas	E-	Peroxidase		

6-	**Which of the following statements best describes the functional characteristics of lysosomes? (Department Qs)**
A-	They only function within the phagocytic pathway of the cell
B-	They contain oxidative enzymes
C-	They function at alkaline pH

D-	Primary type can be detected by electron microscope only
E-	They arise from Golgi body and RER

7-	Which of the following structures transfers proteins and lipids to the Golgi from the endoplasmic reticulum? (Department Qs)				
A-	Transport vesicles	B-	Cis-compartment	C-	Medial compartment
D-	Secretory vesicles	E-	None of the above		

8-	Azurophilic non specific granules represent? (Department Qs)				
A-	Lysosomes	B-	Ribosomes	C-	Mitochondria
D-	Glycogen	E-	Fat droplets		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
E	C	A	D	D	E	A	A	-	-

SNU Bank

1-	Golgi Body: (SNU 1 Bank)		
A-	Modifies proteins	B-	Generates ATP
C-	Digests and recycles materials	D-	Synthesize Protein

2-	Which of the following structures transfers proteins to the Golgi from the endoplasmic reticulum? (SNU 1 Bank)		
A-	Transfer vesicles	B-	Cis-compartment
C-	Medial compartment	D-	Secretory vesicles

3-	Which organelle sorts and packages proteins within a cell? (SNU 1 Bank)		
A-	Rough endoplasmic reticulum	B-	Smooth endoplasmic reticulum
C-	Lysosome	D-	Golgi apparatus

4-	Hydrolytic enzymes are found in: (SNU 1 Bank)						
A-	Golgi apparatus	B-	RER	C-	SER	D-	Lysosomes

Answers - SNU Bank									
1	2	3	4	5	6	7	8	9	10
A	A	D	D	-	-	-	-	-	-

M3WAN Bank

1-	**The Golgi apparatus is typically located at the apical surface in protein-secreting cells such as epididymal cells, or in a perinuclear position in cells such as: (M3WAN - Core)?**						
A-	osteoclasts	B-	nerve cells	C-	fibroblasts	D-	erythrocytes

2-	**In H&E stained sections, deeply basophilic cells such as plasma cells and osteoblasts show an unstained area corresponding to the Golgi apparatus; this appearance is termed: (Important M3WAN - Core)?**		
A-	a residual body	B-	a phagosome
C-	a negative Golgi image	D-	a nucleoid

3-	**Demonstration of the Golgi apparatus in histological sections classically requires: (Important M3WAN - Core)?**		
A-	osmic acid	B-	silver stain
C-	acid phosphatase histochemistry	D-	Janus greenB

4-	**By electron microscopy, the Golgi apparatus is formed of membrane-bound cisternae arranged as curved parallel stacks; its convex surface, associated with transfer vesicles budded from the RER, is called the: (Important M3WAN - Core)?**		
A-	maturation (trans) surface	B-	formation (cis) surface
C-	residual surface	D-	nucleoid surface

5-	**The concave surface of the Golgi apparatus, from which secretory vesicles bud, is called the: (M3WAN - Core)?**		
A-	formation (cis) surface	B-	transfer surface
C-	maturation (trans) surface	D-	nucleoid surface

6-	**One of the principal functions of the Golgi apparatus is: (Important M3WAN - Core)?**
A-	synthesis of ring-like mitochondrial DNA
B-	condensation and packing of proteins transferred from the RER into secretory vesicles
C-	decomposition of hydrogen peroxide
D-	generation of ATP by the respiratory chain

7-	**The Golgi apparatus contributes to the formation of glycoproteins mainly by: (M3WAN - Core)?**
A-	synthesizing new protein chains
B-	adding carbohydrate groups to proteins transferred from the RER
C-	attaching ribosomes to its cisternal surface

D-	degrading old cytoplasmic organelles

8-	Besides protein packaging and glycosylation, the Golgi apparatus is also the site of: (M3WAN - Core)?		
A-	sulfation of some secretory products	B-	Krebs cycle enzyme activity
C-	DNA replication	D-	beta oxidation of fatty acids

9-	All of the following are recognized functions of the Golgi apparatus EXCEPT: (Important M3WAN - Hard)?
A-	packaging of the hydrolytic enzymes of primary lysosomes
B-	acting as a main center for membrane traffic and recycling
C-	synthesis of new secretory proteins from amino acids
D-	condensation and packing of proteins received from the RER

10-	Secretory vesicles, which transport integral membrane proteins to the plasma membrane and release stored soluble materials by exocytosis, are formed in the endoplasmic reticulum and are then: (M3WAN - Core)?
A-	processed and packed in the Golgi apparatus
B-	processed directly in the nucleus
C-	degraded immediately by peroxisomal enzymes
D-	fused permanently with mitochondria

11-	Secretory vesicles that specifically contain digestive enzymes are known as: (M3WAN - Core)?		
A-	residual bodies	B-	primary lysosomes
C-	zymogen granules	D-	phagosomes

12-	Lysosomes, which are rich in hydrolytic enzymes, are invisible by light microscopy but can be demonstrated histochemically by detecting: (Important M3WAN - Core)?		
A-	cytochrome oxidase	B-	acid phosphatase
C-	alkaline phosphatase	D-	catalase

13-	Regarding the electron density of lysosomes, primary lysosomes are typically: (Important M3WAN - Core)?
A-	heterogeneous, while secondary lysosomes are homogeneous
B-	homogeneous, while secondary lysosomes are heterogeneous
C-	both homogeneous
D-	both heterogeneous

14-	**Lysosomal hydrolytic enzymes are synthesized in the rough endoplasmic reticulum and are then delivered to the Golgi apparatus, from which lysosomes are: (Important M3WAN - Core)?**
A-	budded into the cytoplasm with the enzymes inside
B-	extruded directly outside the cell
C-	transformed directly into residual bodies
D-	converted into peroxisomes

15-	**Lysosomal hydrolytic enzymes, which are active in an acidic medium, include proteases, lipases, nucleases, and: (M3WAN - Core)?**
A-	amylases, which hydrolyze polysaccharides into glucose
B-	catalase, which decomposes hydrogen peroxide
C-	porins, which form transport channels
D-	elementary particles, which generate ATP

16-	**When a primary lysosome fuses with an old cytoplasmic organelle such as an old mitochondrion, the resulting structure is called a secondary lysosome, also known as: (Important M3WAN - Hard)?**		
A-	a phagosome	B-	an autophagic vacuole (cytosome)
C-	a zymogen granule	D-	a transfer vesicle

17-	**The lysosomal process by which exogenous material taken into the cell by phagocytosis or pinocytosis is digested after fusion with a lysosome is called: (M3WAN - Core)?**						
A-	autophagy	B-	autolysis	C-	heterophagy	D-	exocytosis

18-	**The normal physiological process by which lysosomes degrade the cell's own worn-out organelles, essential for cellular homeostasis and turnover, is called: (M3WAN - Core)?**						
A-	heterophagy	B-	autophagy	C-	autolysis	D-	fertilization

19-	**Killing of an entire cell due to breakdown of the lysosomal membrane and release of its enzymes into the cytoplasm is termed: (M3WAN - Hard)?**						
A-	heterophagy	B-	autophagy	C-	autolysis	D-	phagocytosis

20-	**During fertilization, the acrosome of the sperm head functions as a giant lysosome that ruptures to release enzymes, allowing sperm entry by: (Important M3WAN - Core)?**		
A-	digesting the zona pellucida/egg membrane	B-	generating ATP for sperm motility
C-	forming the mitotic spindle of the zygote	D-	synthesizing new secretory proteins

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	C	B	B	C	B	B	A	C	A
11	12	13	14	15	16	17	18	19	20
C	B	B	A	A	B	C	B	C	A

✿

قلّةُ المعرفةِ بالمعبودِ، وضعفُ اليقينِ بالمُجازي :قيامُ اللّيلِ صعبٌ على أكثرِ الخلقِ؛ لوجوهٍ؛ أعظمُها والجزاءِ، ومتى قويتِ المعرفةُ واليقينُ يسهُلُ قيامُ اللّيلِ؛ إمّا لطلبِ الثوابِ، وإمّا لسؤالِ العفوِ عن الذّنوبِ، وإمّا للخلوةِ بالحقّ ولذّةِ مناجاتِه؛ ولكلِّ قومٍ شِرْبٌ.

إذا غربتِ الشّمسُ فرحتُ بخلوتي بربّي، وإذا طلعتْ حزنتُ لدخولِ النّاسِ عليّ :قالَ الفُضَيلُ.

أهلُ اللّيلِ في ليلهم ألذُّ من أهلِ اللّهوِ في لهوهم وقالَ عليُّ بنُ بكّارٍ: منذُ :وقالَ أبو سُليمانَ أربعينَ سنةً ما أحزنني إلا طلوعُ الفجرِ.

ابنُ الجوزي | قيامُ اللّيل 📖

Lecture 5: Non-membranous organelles (Ribosomes, Cytoskeleton, Centrioles) & Cell inclusions (Glycogen, Fat, Pigments)

Pre Exams & Formatives

1-	Which of the following is NON a membranous organelle? (Formative 37, Department Qs)						
A-	Lysosomes	B-	Microtubules	C-	Mitochondria	D-	Peroxisomes

2-	Where is the site of protein synthesis? (Formative 37, Department Qs)						
A-	Centrioles	B-	Microfilaments	C-	Peroxisome	D-	Ribosomes

3-	Which structure contributes to the cells cytoskeleton? (2ND TRIAL 32, Formative 37, Department Qs)						
A-	Peroxisome	B-	Centrioles	C-	Microfilaments	D-	Ribosomes

4-	Iron containing pigment is? (FINAL 31, Lecture Qs, Department Qs)								
A-	bilirubin	B-	lipochrome	C-	hemosiderin	D-	melanin	E-	carotene

5-	The mitotic spindle is composed mostly of? (MID 30, FINAL 30)				
A-	flattened	B-	SER	C-	microtubules
D-	intermediate filaments	E-	fibrillin		

6-	Which of the following is a non-membranous organelle?(FINAL 30)				
A-	Lysosomes	B-	ribosomes	C-	Peroxisomes
D-	Mitochondria	E-	Endoplasmic reticulum		

7-	Centriole ? (FINAL 30)				
A-	kreb's cycle	B-	microtubles	C-	pinocytosis
D-	lipid synthesis	E-	electron transfer		

8-	Ribosomes? (MID 31)
A-	are attached to the surface of the inner nuclear membrane to form the nuclear protein
B-	are organized into polysomes in cells synthesizing intracellular protein
C-	are associated with Golgi cisternae
D-	are stained with acidic stain
E-	are basic structures

9-	All of the following is true regarding lipofucin pigments EXCEPT? (MID 31)		
A-	present in cardiac muscles	B-	present in nerve cells
C-	not surrounded by a membrane	D-	increase with age

E-	they are cell inclusions

10-	**Free polysomes are mainly concerned with?(FINAL 31)**		
A-	Formation of steroids	B-	Formation of cytoplasmic proteins
C-	Conduction of material in the cytoplasm	D-	Synthesis of secretory proteins
E-	both:b&d		

11-	**Centrioles are associated with each of the following EXCEPT?(FINAL 31)**		
A-	ciliogenesis (cilia formation)	B-	nine triplets of microtubules
C-	dyinine arms that possesses ATP ase activity	D-	the ability to duplicate themselves
E-	development of mitotic spindle		

12-	**Microtubules does NOT share in?(FINAL 31)**		
A-	Cell support	B-	Intracellular transport
C-	Support of microvilli	D-	Movement and support of axoneme of cilia
E-	Formation of mitotic spindle		

13-	**Ribosomes are characterized by all the following EXCEPT: ? (MID 32)**
A-	Formed of RNA and protein
B-	Organized into polysomes in cells synthesizing intracellular protein
C-	Associated with outer nuclear envelope
D-	Consist of a large and a small subunits
E-	Have double membrane

14-	**Hemosiderin particle ? (MID 32)**
A-	considered one of the monocyte function
B-	related to carbohydrate metabolism
C-	seen with L'M. as dark brown color
D-	usually observed in long lived cells as skeletal muscle cell
E-	both:a&d

15-	**Centrioles are related to each of the following EXCEPT: ? (FINAL 32)**
A-	Basal body of cilia and flagella
B-	Nine doublets of microtubules and one central single
C-	The ability to duplicate themselves
D-	Development of mitotic spindle
E-	Nine triplet of microtubules

16-	Glycogen granules ? (FINAL 32)		
A-	Only in steroid secreting cells	B-	observed only with electron microscope
C-	in muscle as dark small granules	D-	seen near smooth endoplasmic reticulum
E-	in liver as large rosette like granules		

17-	Cytoskeleton under the plamalemma in erythrocytes to: (2ND TRIAL 32)?		
A-	transport of vesicles	B-	Maintain biconcave discoid shape
C-	Carry oxygen	D-	blood grouping

18-	Free ribosomes synthesize? (2ND TRIAL 32, Formative 37, Formative 32)
A-	Cytoplasmic protein necessary for the cell own existence
B-	Intra cellular Lipid
C-	Protein that will be secreted by the cel
D-	Carbohydrate that will be secreted by the cell

19-	Glycogen is? (MID 33)		
A-	endogenous pigment	B-	stained by Sudan black
C-	not surrounded with membrane	D-	secondary to intercellular digestion

20-	Which structure contributes to the cell's cytoskeleton? (MID 33)		
A-	Microsomes	B-	Intermediate filament
C-	Phospholipid	D-	Ribosomes

21-	Microtubules are formed from protein called? (FINAL 33, Formative 36)						
A-	Tubulin	B-	Actin	C-	Globulin	D-	Myosin

22-	Hemosidrin particles are?(FINAL 33)
A-	Related to carbohydrate metabolism
B-	Seen with L'M. as brown granular material
C-	Usually observed in long lived cells as nerve cells
D-	Both:b&c

23-	Which of the following is not a cell inclusion? (MID 34)
A-	Pigments

24-	About microtubule?(FINAL 34)
A-	consists of 10 protofilaments
B-	formed of protein globulin

C-	undergoes polymerization and depolymerization
D-	consists of 12 protofilaments

25-	**Non membranous organelle? (2ND TRIAL 35)**		
A-	Ribosomes	B-	Endoplasmic reticulum
C-	Peroxisome	D-	Mitochondria

26-	**Centrioles are characterized by: ? (MID 37)**		
A-	Cell inclusions	B-	Duplicate duringS phase
C-	Formed of microtubules	D-	Small membranous organelles

27-	**EM of glycogen inside cells appears as ? (MID 37)**						
A-	Rosettes	B-	Tubules	C-	Cisternae	D-	Vesicles

28-	**Ribosomes are: (Formative 37)?**		
A-	Surrounded by a double membrane	B-	Surrounded by a single membrane
C-	makes cell basophilic	D-	Attached to Golgi body

29-	**Glycogen is considered a: (2ND TRIAL37)?**		
A-	Cell inclusion	B-	Non-membranous organelle
C-	Membranous organelle	D-	Cytoskeletal element

Answers - Pre Exams & Formatives									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	**D**	**C**	**C**	**C**	**B**	**B**	**B**	**C**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	**C**	**E**	**C**	**B**	**D**	**B**	**A**	**C**	**B**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	**B**	**A**	**C**	**A**	**C**	**A**	**C**	**A**	**-**

Lecture Qs

1-	**All are endogenous pigment EXCEPT: (Lecture Qs, Department Qs)?**						
A-	Hemoglobin	B-	Carbon	C-	Lipofuscin	D-	Melanin

2-	**Which of the following is NOT a membranous organelle? (Lecture Qs)**						
A-	Lysosomes	B-	Peroxisomes	C-	Mitochondria	D-	Ribosomes

3-	The mitotic spindle is composed mostly of: (Lecture Qs)		
A-	Golgi flattened sacs	B-	Smooth-surfaced endoplasmic reticulum
C-	Microtubules	D-	Intermediate filaments

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
B	D	C	-	-	-	-	-	-	-

Department Qs

1-	Which of the following is an organelle? (Department Qs)				
A-	Pigment	B-	Glycogen	C-	Lipid
D-	Hemosiderin	E-	Mitochondria		

2-	Ribosomes? (Department Qs)
A-	Are attached to the surface of the inner nuclear membrane
B-	Are organized into polysomic in cells synthesizing intracellular proteins
C-	Are always associated with cell growth and regeneration of organelles
D-	Consist of a large and a small subunits
E-	:B&D

3-	Ribosomes are? (Department Qs)
A-	Attached to the surface of the outer nuclear membrane
B-	Organized into polysomes in cells synthesizing intracellular carbohydrates
C-	Always associated with a strand of rRNA
D-	Surrounded by a delicate thin membrane
E-	None of the above

4-	What structure is at the base of cilia? (Department Qs)				
A-	Ribosomes	B-	Peroxisome	C-	Microfilaments
D-	Centrioles	E-	Nucleoli		

5-	Free ribosomes synthesize? (Department Qs)
A-	Cytoplasmic protein necessary for the cell own existence
B-	Protein that will be secreted by the cell
C-	Lipid
D-	Carbohydrate

E-	None of the above

6-	**The attached ribosomes to RER synthesize? (Department Qs)**
A-	Cytoplasmic protein necessary for the cell own existence
B-	Protein that will be secreted by the cell
C-	Lipid
D-	Carbohydrate
E-	Glycogen

7-	**Ribosomes are? (Department Qs)**		
A-	Basophilic organoids	B-	Surrounded by a single membrane
C-	Surrounded by a double membrane	D-	Attached to immature face of Golgi body
E-	None of the above		

8-	**Centrioles? (Department Qs)**
A-	Self-replicate
B-	Nine doublets of microtubules and one central single
C-	Surrounded by a double membraned
D-	Nine doublets of microtubules and two central single
E-	Do not self-replicate

9-	**Centrioles are associated with each of the following EXCEPT? (Department Qs)**
A-	Ciliogenesis (cilia formation)
B-	Nine doublets of microtubules and one central single
C-	The ability to duplicate themselves
D-	Development of mitotic spindle
E-	Dynine arms that possesses ATP ase activity

10-	**Microtubules are absent in? (Department Qs)**				
A-	Mitochondria	B-	Flagella	C-	Centrioles
D-	Spindle fibres	E-	:A&c		

11-	**At E/M level the centrioles shows? (Department Qs)**
A-	Nine triplets of microtubules and two central singles
B-	Nine doublets of microtubules and one central single
C-	Nine doublets of microtubules and two central singles
D-	Nine triplets of microtubules and no central singles

12-	The cytoskeleton includes? (Department Qs)				
A-	Microtubules	B-	The microtrabecular lattice	C-	Intermediate filaments
D-	Actin filaments	E-	All of the above		

13-	Microtubules have important role in? (Department Qs)		
A-	Carbohydrate metabolism	B-	The microtrabecular lattice
C-	Protein synthesis	D-	Oxidative phosphorylation
E-	Formation of cell spindle during mitosis		

14-	Microtubules are formed from? (Department Qs)								
A-	Tubulin	B-	Actin	C-	Globulin	D-	Myosin	E-	Albumin

15-	Microtubules function is? (Department Qs)
A-	Synthesis of protein
B-	Formation of cell spindle during mitosis
C-	Transportion of substances within the cytoplasm
D-	A & b
E-	B & c

16-	Function of intermediate filament, involved in providing? (Department Qs)		
A-	Formation of cell spindle during mitosis	B-	Structural support to cell
C-	Mobility	D-	Reproduction
E-	Apoptosis		

17-	Centrioles are made of? (Department Qs)				
A-	Microtubules	B-	Microfilaments	C-	Intermediate filaments
D-	All of the above	E-	None of the above		

18-	Functions of microtubules? (Department Qs)
A-	Cell Shape
B-	To Move things around in the Cell
C-	Move Specialized Surface Extensions (Cilia/Flagella)
D-	All of the above
E-	None of the above

19-	The function of Cytoskeleton is? (Department Qs)
A-	Providing cell shape
B-	Providing cell motility

C-	Fixing position of certain cytoskeletal elements
D-	Stabilizes the cell itself
E-	All of the above

20-	**They form irregular networks in the cytoplasm, which allows them to perform very specialized functions for the cell? (Department Qs)**				
A-	Microtubules	B-	Microfilaments	C-	Intermediate filaments
D-	None of the above	E-	All of the above		

21-	**Form large pipes/ hollow rods/cables? (Department Qs)**				
A-	Microtubules	B-	Microfilaments	C-	Intermediate filaments
D-	None of the above	E-	All of the above		

22-	**The primary function of intermediate filaments is to? (Department Qs)**		
A-	Generate movement	B-	Provide mechanical stability
C-	Carry out nucleation of microtubules	D-	Stabilize microtubules against disassembly
E-	Transport organelles within the cell		

23-	**Which of the following is NOT considered an inclusion? (Department Qs)**				
A-	Pigment	B-	Glycogen	C-	Lipid
D-	Lipochrome	E-	Mitochondria		

24-	**Glycogen can be demonstrated using? (Department Qs)**				
A-	H&E	B-	Sudan III	C-	PAS
D-	Best's carmine	E-	Osmic acid		

25-	**The most important and essential pigment in the body is? (Department Qs)**				
A-	Melanin	B-	Bilirubin	C-	Hemoglobin
D-	Carotene	E-	Carbon particles		

26-	**Which set of items below includes only those that are either cytoplasmic inclusions or that are found within some kind of cytoplasmic inclusions? (Department Qs)**
A-	Lipofuscin, glycogen, karyosome, lipid droplets
B-	Endoplasmic reticulum, melanin, lipid droplets, Golgi bodies
C-	Melanin, lipid droplets, endoplasmic reticulum, glycogen
D-	Secretory granules, lipofuscin, melanosome, autosome
E-	Hemosiderin, glycogen, lipofuscin

27-	The following are regarded as cellular inclusions EXCEPT? (Department Qs)				
A-	Carbon particles	B-	Pigments	C-	Glycogen
D-	Lipofuscin	E-	Endosomes		

28-	Lipofuscin? (Department Qs)
A-	Is the wear- and -tear pigments
B-	Is stored as residual body in lysosomes
C-	Is perinuclear brownish yellow granules that indicate function of age
D-	Is not injurious to cells but is a marker of past free radical injury
E-	All of the above

29-	Melanin? (Department Qs)				
A-	Is an endogenous pigment	B-	Is formed by melanocytes	C-	Is an exogenous pigment
D-	Both:a&b	E-	Both:c&b		

30-	Hemosiderin is? (Department Qs)
A-	Is a hemoglobin-derived pigment radical injury
B-	Accumulates in tissues when there is excess of iron
C-	Is a golden yellow granules normally found in small amount in liver, spleen and bone marrow
D-	Secondary to the breakdown of RBC's
E-	All of the above

31-	Color of skin is mainly due to presence of? (Department Qs)				
A-	Melanin	B-	Lipofuscin	C-	Hemoglobin
D-	Carotenes	E-	None of the above		

32-	Cell inclusions are? (Department Qs)
A-	Non-living materials present in the cytoplasm
B-	Another name of cell organelle
C-	Cytoskeletal framework of cell
D-	Combined name for cell wall and plasma membrane
E-	None of the above

33-	Which of the following is responsible for production of lipofuscin? (Department Qs)				
A-	Golgi complex	B-	Lysosomes	C-	Mitochondria
D-	Peroxisomes	E-	None of the above		

34-	Which two macromolecules offer energy storage to the cell? (Department Qs)		
A-	Lipids and nucleic acids	B-	Proteins and carbohydrates
C-	Nucleic acids and proteins	D-	Lipids and carbohydrates
E-	Proteins and nucleic acid		

35-	Cytoskeleton under the plamalemma in erythrocytes to? (Department Qs)		
A-	Help amoeboid movement	B-	Maintain biconcave discoid shape
C-	Carry oxygen	D-	All of the above
E-	None of the above		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
E	E	A	D	A	B	A	A	B	A
11	12	13	14	15	16	17	18	19	20
D	E	E	A	E	B	A	D	E	E
21	22	23	24	25	26	27	28	29	30
A	B	E	C	C	E	E	E	D	E
31	32	33	34	35	36	37	38	39	40
A	A	B	D	B	-	-	-	-	-

SNU Bank

1-	Intermediate filaments are: (SNU 1 Bank)		
A-	Membranous organelles	B-	Non membranous organelles
C-	Cell inclusion	D-	None of the above

2-	Glycogen is: (SNU 1 Bank)		
A-	Membranous organelles	B-	Non membranous organelles
C-	Cell inclusion	D-	None of the above

3-	Lipid droplets are: (SNU 1 Bank)		
A-	Membranous organelles	B-	Non membranous organelles
C-	Cell inclusion	D-	None of the above

4-	Haemoglobin is: (SNU 1 Bank)		
A-	Membranous organelles	B-	Non membranous organelles
C-	Cell inclusion	D-	None of the above

5-	**Haemosiderin is: (SNU 1 Bank)**		
A-	Membranous organelles	B-	Non membranous organelles
C-	Cell inclusion	D-	None of the above

6-	**Ribosomes are: (SNU 1 Bank)**		
A-	Membranous organelles	B-	Non membranous organelles
C-	Cell inclusion	D-	None of the above

7-	**Cytoskeleton represents: (SNU 1 Bank)**		
A-	Membranous organelles	B-	Non membranous organelles
C-	Cell inclusion	D-	None of the above

8-	**Signet ring shape cell: (SNU 1 Bank)**						
A-	fat cell	B-	goblet cell	C-	muscle cell	D-	nerve cell

9-	**Microtubules share in formation in which of the following: (SNU 1 Bank)**						
A-	microvilli	B-	mitotic spindle	C-	microbodies	D-	cell coat

10-	**Intermediate filaments differ from microtubules and microfilaments because they: (SNU 1 Bank)**
A-	Are hollow cylinders
B-	Are composed of tubulin
C-	Provide tensile strength and resist mechanical stress
D-	Help in chromosome movement

11-	**The major function of the cytoskeleton is to: (SNU 1 Bank)**
A-	Synthesize proteins
B-	Provide structural support, movement, and intracellular transport
C-	Store genetic material
D-	Produce energy

12-	**Ribosomes are mainly involved in: (SNU 1 Bank)**		
A-	Lipid synthesis	B-	Protein synthesis
C-	Carbohydrate breakdown	D-	ATP production

Answers - SNU Bank									
1	2	3	4	5	6	7	8	9	10
B	C	C	C	C	B	B	A	B	C
11	12	13	14	15	16	17	18	19	20
B	B	-	-	-	-	-	-	-	-

M3WAN Bank

1-	**Ribosomes are ribonucleoprotein particles formed of two subunits; both the RNA and protein components are assembled in the: (M3WAN - Core)?**						
A-	mitochondria	B-	Golgi apparatus	C-	nucleolus	D-	centriole

2-	**Free ribosomes, organized into polysomes with the aid of mRNA, are mainly concerned with synthesis of proteins used for: (M3WAN - Core)?**		
A-	cell growth and differentiation	B-	export outside the cell
C-	lysosomal enzyme packaging	D-	glycoprotein formation

3-	**Ribosomes attached to the outer surface of the rough endoplasmic reticulum are concerned with synthesis of: (M3WAN - Core)?**		
A-	proteins for internal cell use only	B-	ribosomal RNA
C-	mitochondrial DNA	D-	secretory proteins exported outside the cell

4-	**All of the following statements correctly characterize ribosomes EXCEPT: (Important M3WAN - Hard)?**		
A-	formed of RNA and protein	B-	surrounded by a double membrane
C-	consist of a large and a small subunit	D-	may be organized into polysomes

5-	**Microtubules are rigid tubular structures, 25 nm in diameter, formed of the protein tubulin arranged into: (Important M3WAN - Core)?**		
A-	13 protofilaments	B-	9 triplet microtubules
C-	actin monomers only	D-	keratin chains

6-	**Assembly of microtubules occurs mainly in the: (M3WAN - Core)?**
A-	nucleolus
B-	Golgi apparatus
C-	rough endoplasmic reticulum
D-	microtubule organizing center near the centriole

7-	**The length of a microtubule can change rapidly through the process of: (M3WAN - Core)?**
A-	glycosylation and sulfation
B-	oxidative phosphorylation
C-	polymerization and depolymerization of tubulin subunits
D-	binary fission

8-	All of the following are recognized functions of microtubules EXCEPT: (Important M3WAN - Hard)?
A-	displacement of macromolecules and organelles in the cytoplasm
B-	support of microvilli
C-	mechanical support and determination of cell shape
D-	support of the ciliary axoneme and formation of the mitotic spindle

9-	Thin microfilaments, present under the plasma membrane and highly organized/contractile in muscle cells, are formed of the protein: (M3WAN - Core)?						
A-	tubulin	B-	keratin	C-	myosin	D-	actin

10-	Keratin intermediate filaments, which form tonofilaments, are characteristic of: (M3WAN - Core)?		
A-	stratified epithelia	B-	smooth muscle cells
C-	fibroblasts	D-	nerve cells

11-	Desmin is the intermediate filament protein characteristically present in: (Important M3WAN - Core)?		
A-	fibroblasts	B-	glial cells
C-	smooth muscle cells	D-	nerve cells

12-	Vimentin, an intermediate filament protein, is characteristically present in: (M3WAN - Hard)?		
A-	glial cells	B-	fibroblasts
C-	stratified epithelial cells	D-	skeletal muscle cells

13-	Thick (myosin) filaments, as a component of the cytoskeleton, are present in: (M3WAN - Core)?		
A-	fibroblasts and glial cells	B-	stratified epithelia only
C-	smooth muscle only	D-	skeletal and cardiac muscle cells

14-	General functions of cytoskeletal filaments include mechanical support, distribution of tensile force, and a role in: (M3WAN - Core)?		
A-	muscle contraction and cell division	B-	ribosomal RNA transcription
C-	lysosomal enzyme synthesis	D-	oxidative phosphorylation

15-	By light microscopy, using a special stain such as iron hematoxylin, centrioles appear as: (M3WAN - Core)?
A-	a single dark ring around the nucleus
B-	fine granular clusters throughout the cytoplasm
C-	two dots close to the nucleus
D-	a negative image within basophilic cytoplasm

16-	By electron microscopy, each centriole appears as a hollow cylinder, the wall of which is formed of: (Important M3WAN - Core)?		
A-	13 protofilaments of tubulin	B-	9 bundles of triplet microtubules
C-	a single ring of actin filaments	D-	a 9+2 arrangement of doublet microtubules

17-	Centrioles are associated with all of the following EXCEPT: (Important M3WAN - Hard)?		
A-	ciliogenesis (formation of cilia)	B-	the ability to duplicate themselves
C-	development of the mitotic spindle	D-	dynein arms possessing ATPase activity

18-	Centrioles duplicate during the G2 phase and serve as the basal body of: (Important M3WAN - Core)?		
A-	cilia and flagella	B-	microvilli only
C-	the mitotic spindle only	D-	the nuclear pore complex

19-	Cell inclusions are best described as: (M3WAN - Core)?
A-	permanent, living, membrane-bound structures with metabolic activity
B-	non-living substances stored in the cytoplasm, not bound by a membrane
C-	membranous organelles concerned with digestion
D-	structures formed exclusively of ribonucleoprotein

20-	By electron microscopy, glycogen appears in the liver as individual dense particles (beta-particles), while in muscle it aggregates into larger rosette-like clusters called: (Important M3WAN - Core)?						
A-	residual bodies	B-	matrix granules	C-	alpha-particles	D-	zymogen granules

21-	Glycogen granules, which can be localized histochemically by the PAS reaction, are typically located near the: (Important M3WAN - Core)?		
A-	Golgi apparatus	B-	nucleolus
C-	mitochondria	D-	smooth endoplasmic reticulum

22-	In fresh tissue sections, lipid droplets can be specifically demonstrated as black granules using: (M3WAN - Core)?						
A-	Sudan black stain	B-	the PAS reaction	C-	silver stain	D-	iron hematoxylin

23-	Hemosiderin, a brown granular pigment resulting from red blood cell breakdown, characteristically accumulates in the cytoplasm of phagocytes in the: (Important M3WAN - Core)?		
A-	skin and cornea	B-	spleen, liver, and bone marrow
C-	lens and vitreous	D-	cartilage and tendon

24-	On the ultrastructural level, melanin, synthesized by melanocytes, appears as: (M3WAN - Core)?
A-	a homogenous dense granule separated from the cytoplasm by a membrane unit
B-	finely granulated material with a dense nucleoid core
C-	a rosette-like cluster of particles
D-	an electron-lucent vacuole

25-	Lipofuscin, the "wear-and-tear" pigment that increases with age and is easily seen at the poles of cardiac nuclei, is stored inside: (Important M3WAN - Core)?		
A-	mitochondria	B-	peroxisomes
C-	lysosomes	D-	the nuclear envelope

26-	Carbon and dust particles taken up by lung macrophages are examples of: (M3WAN - Core)?		
A-	endogenous pigments	B-	exogenous pigments
C-	cell organelles	D-	cell matrix components

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
C	A	D	B	A	D	C	B	D	A
11	12	13	14	15	16	17	18	19	20
C	B	D	A	C	B	D	A	B	C
21	22	23	24	25	26	27	28	29	30
D	A	B	A	C	B	-	-	-	-

قال الإمام ابن القيم رحمه الله :

إذا أراد أن يُهلك أعداءه ، ويمحقهم ؛ قيّض لهم الأسباب التي يستوجبون بها هلاكهم (ومحقهم ، ومِن أعظمها : مبالغتهم في أذى أوليائه !) .

زاد المعاد (٣/١٩٩)

Lecture 6: Nucleus

Pre Exams & Formatives

1-	**Non-membrane bound body of the nucleus which disappears in the late prophase and reappears in telophase? (FINAL 31, Department Qs)**				
A-	nucleolus	B-	chromosome	C-	nucleoplasm
D-	nuclear pore	E-	nuclear envelope		

2-	**Chromatin is composed of? (Formative 32, Department Qs)**		
A-	DNA	B-	DNA and proteins
C-	DNA, RNA and proteins	D-	None

3-	**Which of the following has beads on a string structure? (Formative 32, Department Qs)**						
A-	Chromosomes	B-	Chromatin	C-	Nucleosomes	D-	Heterochromatin

4-	**Each unit of a nucleic acid consisting of a sugar, attached phosphate group, and base is? (MID 30, Department Qs)**								
A-	Nucleolus	B-	Nucleotide	C-	Nucleosome	D-	Histone	E-	non histone

5-	**Each unit of a nucleic acid consisting of a sugar, a phosphate group and a base is?(FINAL 30)**								
A-	Nucleolus	B-	Nucleotide	C-	Nucleosome	D-	Histone	E-	non-histone

6-	**Euchromatin is? (MID 31)**		
A-	inactive chromatin	B-	related to protein synthesis
C-	related to one X'chromosome	D-	associated with the nucleolus
E-	associated with the inner nuclear membrane		

7-	**Which region of chromatin is transcriptionally silent?(FINAL 31)**		
A-	nucleoid	B-	centromere
C-	euchromatin	D-	heterochromatin
E-	nucleolus associated chromatin		

8-	**Each of the following statements concerning the nucleolar organizer region are true EXCEPT?(FINAL 31)**
A-	it is located in the nucleolus
B-	it contains DNA
C-	it contains the genes necessary for rRNA synthesis
D-	both: a & b
E-	It responsible for initiation of mitosis

9-	Which one is wrong regarding the nuclear matrix? (MID 32)
A-	has fibrillar appearance
B-	acts as mechanical support to the nuclear pores
C-	support for chromatin
D-	is colloidal solution
E-	acts as medium for protein synthesis

10-	What is the type of chromatin that is transcriptionally active during interphase: ? (MID 32)				
A-	Euchromatin	B-	Nuclear envelope	C-	Heterochromatin
D-	Nucleolus	E-	Nuclear pores		

11-	Which is true regarding nucleolus in mitotic division? (FINAL 32, Formative 32)				
A-	disappear in prophase	B-	disappear in telophase	C-	reappear in metaphase
D-	reappear in telophase	E-	both: a & d		

12-	Which is true regarding the nucleus?(FINAL 32)				
A-	It is acidophilic structure	B-	It is basophilic structure	C-	It is acidic in nature
D-	It is basic in nature	E-	both: b & c		

13-	Where are ribosomes constructed? (FINAL 32, Formative 37)		
A-	Cytoskeleton	B-	Peroxisome
C-	Microtubule organizing center	D-	Centrioles
E-	Nucleolus		

14-	Euchromatin is characterized by all the following EXCEPT (2ND TRIAL 32)?		
A-	Present in liver cells	B-	Lightly stained by LM
C-	Dark basophilic clumps	D-	Active

15-	The nuclear pore complex: (2ND TRIAL 32)?		
A-	is cylindrical in shape	B-	is a transient structure
C-	partially covered by fibrillar material	D-	Both: a & c

16-	All these are nitrogen bases enter in structure of DNA EXCEPT? (MID 33)						
A-	Thiamine	B-	Cytosine	C-	Uracil	D-	Guanine

17-	Ribosomal RNA is formed in? (FINAL 33, Formative 37)						
A-	Heterochromatin	B-	Euchromatin	C-	Nucleolus	D-	Golgi apparatus

18-	Chromatin component of? (MID 34)
A-	Nucleus

19-	Nucleus is? (MID 34)
A-	enclosed in nuclear envelope

20-	About heterochromatin?(FINAL 34)		
A-	less coiled	B-	appears acidophilic
C-	predominant in inactive cells		

21-	Nucleus is basophilic when stained by H&E? (2ND TRIAL 35)				
A-	Its content is acidic	B-	Its content is basic	C-	RER

22-	The nucleus of active cell has? (MID 37, Formative 36)		
A-	euchromatin only	B-	heterochromatin only
C-	more euchromatin	D-	more Heterochromatin

23-	Which of the following is always basophilic with H&E stain? (Formative 36, Formative 32)						
A-	Cell membrane	B-	Nucleus	C-	Collagen fibers	D-	Cytoplasm

24-	All of the following are true about the nucleolus EXCEPT: (2ND TRIAL 37)		
A-	Composed of pars fibrosa and pars granulosa	B-	Site of transcription of ribosomal RNA
C-	Present in active protein synthesizing cells	D-	Limited by a membrane

25-	DNA is organized around histones into: (2ND TRIAL 37)						
A-	Nucleoli	B-	Nucleotides	C-	Centromeres	D-	Nucleosomes

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
A	C	C	B	B	B	D	E	E	A
11	12	13	14	15	16	17	18	19	20
E	E	E	C	D	C	C	A	A	C
21	22	23	24	25	26	27	28	29	30
A	C	B	D	D	-	-	-	-	-

Lecture Qs

1-	The perinuclear space is contiguous with the inside of the: (Lecture Qs)		
A-	endoplasmic reticulum	B-	Golgi apparatus
C-	nucleus	D-	lysosome

2-	Which part of the nuclear membrane has ribosomes? (Lecture Qs)		
A-	Inner nuclear membrane	B-	Outer nuclear membrane
C-	Perinuclear space	D-	Nucleoplasm

3-	DNA consists of four different nitrogenous bases, which are: (Lecture Qs)		
A-	Guanine, adenine, thymine and ribonucleic	B-	Guanine, adenine, thymine and cytosine
C-	Guanine, adenine, uracil and cytosine	D-	Guanine, deoxynucleic, thymine and cytosine

4-	Where are ribosomes RNA constructed? (Lecture Qs)						
A-	Peroxisome	B-	Microfilaments	C-	Centrioles	D-	Nucleoli

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	B	B	D	-	-	-	-	-	-

Department Qs

1-	Where are ribosomes constructed? (Department Qs)				
A-	Cytoskeleton	B-	Peroxisome	C-	Microfilaments
D-	Centrioles	E-	Nucleoli		

2-	Ribosomal RNA is formed in? (Department Qs)				
A-	The euchromatin	B-	The nucleolus	C-	The SER
D-	The hetrochromatin	E-	Golgi bodies		

3-	Name the control center of the eukaryotic cell? (Department Qs)						
A-	Nucleus	B-	Ribosome	C-	Cytoplasm	D-	Golgi complex

4-	Which of the following is not a component of the nucleus? (Department Qs)						
A-	Chromosome	B-	Nucleolus	C-	Cytoplasm	D-	Nuclear envelope

5-	Name the structure which is used to transfer macromolecules between the cytoplasm and nucleus? (Department Qs)						
A-	Microtubules	B-	Nuclear pores	C-	Cilia	D-	Centrioles

6-	Nucleoporins are? (Department Qs)		
A-	Nuclear pores	B-	Ribosomes on nuclear membranes
C-	rRNAs in the nucleolus	D-	None of the mentioned

7-	The nucleus is? (Department Qs)
A-	Always found one per cell
B-	An essential organelle present in some cells
C-	Enclosed within a nuclear membrane
D-	Enclosed within inner, intermediate, and outer unit membrane
E-	Completely covered externally by the fibrous lamina

8-	Where Barr bodies are typically found? (Department Qs)		
A-	Male gametes	B-	Male somatic cells
C-	Female somatic cells	D-	Male and female somatic cells
E-	All of the above		

9-	What structure is visible at the light microscopic level and is generally believed to represent chromatin that is transcriptionally inactive during interphase? (Department Qs)				
A-	Euchromatin	B-	Nuclear envelope	C-	Heterochromatin
D-	Nucleolus	E-	Nuclear pores		

10-	Which of the following is not a characteristic of the nucleus? (Department Qs)		
A-	Bound by a nuclear envelope	B-	Contains chromatin
C-	May be multilobulated	D-	Must be spherical or ovoid
E-	All of the above		

11-	Which of the following best describes euchromatin? (Department Qs)
A-	Abundant in active cells, basophilic clumps
B-	Abundant in active cells, lightly stained, uniformly dispersed
C-	Abundant in inactive cells, lightly stained, uniformly dispersed
D-	Basophilic clumps, predominant in inactive cells
E-	Acidophilic clumps, predominant in active cells

12-	**The nucleus of inactive cell has? (Department Qs)**		
A-	Hetrochromatin only	B-	Euchromatin only
C-	Equal amounts hetro & euchromatin	D-	Euchromatin is predominant
E-	Hetrochromatin is predominant		

13-	**Hetrochromatin is predominant in? (Department Qs)**
A-	Present in the nuclei of metabolically active cells
B-	Present in the nuclei of metabolically inactive cells
C-	The cell cytoplasm
D-	Type of cell organoids
E-	Type of cell inclusions

14-	**Chromatin is a component of the? (Department Qs)**				
A-	Nucleolus	B-	Nucleus	C-	Cytoplasm
D-	Intercellular space	E-	None of the above		

15-	**The nuclear pore? (Department Qs)**
A-	Is hexagonal in shape
B-	Is a transient structure
C-	Allows for communication between the nucleus and the cytoplasm
D-	Is bridged by a unit membrane
E-	All of the above

16-	**DNA contains the following nitrogenous bases EXCEPT? (Department Qs)**								
A-	Thymine	B-	Cytosine	C-	Uracil	D-	Adenine	E-	Guanine

17-	**Bar body? (Department Qs)**
A-	Is an inactive (coiled) X'chromosomes
B-	Is a sex chromatin in somatic female cells
C-	During the interphase, appears as a drumstick stained with basic dyes Hx and Feulgon +ve
D-	All of the above
E-	None of the above

18-	**Bar body is? (Department Qs)**		
A-	An inactive (coiled) X'chromosomes in females	B-	X'chromosomes in males
C-	Y'chromosomes in males	D-	XX-chromosomes in females
E-	XY-chromosomes in males		

19-	Cells containing single nucleus are called as? (Department Qs)				
A-	Mononucleated	B-	Binucleated	C-	Single nucleated
D-	Multinucleated	E-	None nucleated		

20-	A section of chromosomes that codes for a trait can be called a? (Department Qs)						
A-	nucleotide	B-	base-pair	C-	gene	D-	nucleus

21-	Which of the following is NOT a characteristic of DNA? (Department Qs)		
A-	Composed of nucleotides	B-	Complementary
C-	Contains ribose	D-	Double-stranded

22-	The function of the nucleosomes is to? (Department Qs)
A-	form pores for bilateral nuclear-to-cytoplasmic transport
B-	hold together adjacent chromatids
C-	form the nuclear matrix
D-	package genetic material in a condensed form
E-	transcribe the DNA

23-	The nucleolus is formed of? (Department Qs)		
A-	hetrochromatin and euchromatin	B-	granular and fibrillar parts
C-	nucleolus associated and peripheral chromatin	D-	condensed chromatin
E-	extended chromatin		

24-	The nuclear envelope is studded with ribosomes and continuous with the rough ER (Department Qs)		
A-	True	B-	False

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
E	B	A	C	B	A	C	C	C	D
11	12	13	14	15	16	17	18	19	20
B	E	B	B	C	C	D	A	A	C
21	22	23	24	25	26	27	28	29	30
C	D	B	A	-	-	-	-	-	-

SNU Bank

1-	A structure that is continuous with RER is the: (SNU 1 Bank)		
A-	nuclear pore complex	B-	nucleolus
C-	heterochromatin	D-	outer nuclear membrane

2-	Where are ribosomes constructed? (SNU 1 Bank)						
A-	In the Golgi	B-	In the peroxisome	C-	In the centrioles	D-	In the nucleoli

3-	Which statement correctly describes a key feature of the nuclear envelope? (SNU 1 Bank)
A-	It is composed of a double lipid bilayer
B-	It contains nuclear pores that regulate the flow of traffic
C-	It is continuous with the endoplasmic reticulum
D-	All of the above

4-	Which of the following best describes the nucleolus? (SNU 1 Bank)
A-	It is involved in the synthesis of ribosomal RNA
B-	It stores DNA during cell division
C-	It is a site of lipid synthesis
D-	It regulates gene expression

5-	What is the fundamental repeating unit of chromatin? (SNU 1 Bank)						
A-	Chromosome	B-	Nucleosome	C-	Histone	D-	DNA strand

6-	Which proteins help in the packing of DNA into the nucleosomes? (SNU 1 Bank)						
A-	Histones	B-	Enzymes	C-	RNA polymerases	D-	Peptides

Answers - SNU Bank									
1	2	3	4	5	6	7	8	9	10
D	D	D	A	B	A	-	-	-	-

M3WAN Bank

1-	The nucleus is best described as the: (M3WAN - Core)?
A-	site of ATP production
B-	primary site of lipid synthesis
C-	control center of the cell containing genes that regulate structure and function
D-	main site of hydrolytic digestion

2-	Two nuclei (binucleated cells) may normally be observed in some: (M3WAN - Core)?						
A-	neurons	B-	osteocytes	C-	chondrocytes	D-	liver cells

3-	Multiple nuclei (multinucleation) are a normal feature of: (M3WAN - Core)?		
A-	skeletal muscle cells	B-	hepatocytes
C-	fibroblasts	D-	osteocytes

4-	The nucleus is usually located in the: (M3WAN - Core)?		
A-	narrowest area of the cytoplasm	B-	widest (central) area of the cytoplasm
C-	periphery of the cell only	D-	apical pole exclusively

5-	The basophilic staining property of the nucleus with hematoxylin is due to its: (Important M3WAN - Core)?		
A-	glycogen content	B-	neutral lipid content
C-	rich content of nucleic acids (DNA and RNA)	D-	hydrolytic enzyme content

6-	A vesicular or open-face nucleus, indicating an active cell, is characterized by a high content of: (Important M3WAN - Core)?		
A-	euchromatin	B-	heterochromatin
C-	lipofuscin	D-	nucleolar-associated chromatin only

7-	By electron microscopy, the nucleus is composed of four components: nuclear envelope, chromatin, nucleolus, and: (M3WAN - Core)?		
A-	nuclear pore complex only	B-	centriole
C-	nucleoplasm exclusively	D-	nuclear matrix

8-	The nuclear envelope is formed of two unit membranes, each about 8 nm thick, separated by a space of approximately: (M3WAN - Core)?						
A-	8 nm	B-	25 nm	C-	60 nm	D-	100 nm

9-	The outer membrane of the nuclear envelope is continuous with the rough endoplasmic reticulum and is covered with: (M3WAN - Core)?		
A-	peripheral chromatin	B-	histones
C-	ribosomes	D-	pars granulosa

10-	The inner membrane of the nuclear envelope is lined, on its nucleoplasmic surface, by: (M3WAN - Core)?		
A-	peripheral chromatin	B-	ribosomes
C-	pars fibrosa	D-	nuclear matrix only

11-	Nuclear pores in the nuclear envelope mainly regulate: (Important M3WAN - Core)?
A-	ATP transport into mitochondria
B-	lysosomal enzyme secretion
C-	glycogen storage
D-	entry of protein into the nucleus and exit of ribonucleoprotein

12-	The nuclear envelope characteristically: (M3WAN - Core)?		
A-	persists unchanged throughout mitosis	B-	disappears during cell division
C-	is duplicated before the S phase	D-	fuses permanently with the Golgi apparatus

13-	Chromatin materials are best defined as the basophilic particles and threads from which the following are formed during cell division: (M3WAN - Core)?						
A-	ribosomes	B-	nucleoli	C-	chromosomes	D-	centrioles

14-	Euchromatin, the more active and less coiled form of chromatin, is characteristically abundant in: (Important M3WAN - Core)?		
A-	lymphocytes	B-	osteoclasts
C-	adipocytes	D-	liver and nerve cells

15-	Heterochromatin, appearing as dark, coarse basophilic clumps, represents chromatin that is: (Important M3WAN - Core)?		
A-	inactive and more coiled	B-	actively transcribed and less coiled
C-	exclusively confined to the nucleolus	D-	devoid of DNA

16-	The nucleolar-associated heterochromatin (nucleolar organizer chromosomes) corresponds to chromosome numbers: (Important M3WAN - Hard)?						
A-	1, 2, 3, 4, 5	B-	13, 14, 15, 21, 22	C-	XandY only	D-	6, 7, 8, 9, 10

17-	Sex chromatin, or the Barr body, represents one of the: (M3WAN - Core)?		
A-	Y chromosomes present in the male	B-	autosomes present in both sexes
C-	X chromosomes present in the female	D-	nucleolar organizer chromosomes

18-	DNA is organized around histone proteins to form structures called nucleosomes, which are further wound into a helix to form: (M3WAN - Core)?		
A-	the nucleolus	B-	the nuclear matrix
C-	ribosomes	D-	chromatin

19-	The nucleolus, by electron microscopy, is composed of a central pars fibrosa and a surrounding: (Important M3WAN - Core)?
A-	pars granulosa, representing accumulated RNA-protein
B-	peripheral chromatin layer
C-	nuclear matrix
D-	nuclear pore complex

20-	The main function of the nucleolus is: (Important M3WAN - Core)?		
A-	DNA replication	B-	transcription of ribosomal RNA
C-	packaging of secretory proteins	D-	generation of ATP

21-	The nucleolus is a non-membrane-bound body that characteristically disappears in late prophase and reappears in: (Important M3WAN - Core)?						
A-	metaphase	B-	anaphase	C-	telophase	D-	early prophase

22-	Regarding the nuclear matrix, which of the following statements is INCORRECT? (Important M3WAN - Hard)?
A-	it has a fibrillar appearance
B-	it acts as mechanical support to the nuclear pores and chromatin
C-	it provides a medium for chromosomal duplication and gene transcription
D-	it acts as the main site of protein synthesis

23-	In the DNA double helix, obligatory base pairing occurs between adenine and thymidine, and between guanine and: (M3WAN - Core)?						
A-	uracil	B-	cytosine	C-	ribose	D-	deoxyribose

24-	Compared with DNA, RNA is distinguished by containing the sugar ribose instead of deoxyribose, and the base: (M3WAN - Core)?		
A-	uracil instead of thymidine	B-	thymidine instead of uracil
C-	cytosine instead of guanine	D-	adenine instead of thymidine

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
C	D	A	B	C	A	D	B	C	A
11	12	13	14	15	16	17	18	19	20
D	B	C	D	A	B	C	D	A	B
21	22	23	24	25	26	27	28	29	30
C	D	B	A	-	-	-	-	-	-

Lecture 7: Structure of chromosomes & karyotyping

Pre Exams & Formatives

1-	In acrocentric chromosomes the centromere is? (MID 33, MID 34, Department Qs)						
A-	terminal	B-	subterminal	C-	absent	D-	median

2-	A picture of a person's chromosomes is called? (Formative 36, Formative 32, Department Qs)
A-	syndrome
B-	karyotype
C-	fingerprint
D-	chromatin

3-	Karyotype preparation is used to study? (MID 31)				
A-	nuclear structure	B-	chromosomal defects	C-	cytoplasmic contents
D-	cell division	E-	tissue structure		

4-	In submetacentric chromosomes the centromere is: ? (MID 32)				
A-	Terminal	B-	Absent	C-	Subterminal
D-	Near to the center	E-	None of the above		

5-	From the abnormality in the number of chromosomes? (FINAL 32)				
A-	Deletion	B-	Duplication	C-	Inversion
D-	Aneuploidy	E-	Non of the above		

6-	The chromosome is formed of two chromatids which are connected together at one point known as (2ND TRIAL 32)?						
A-	Centriole	B-	Spheriole	C-	Centromere	D-	Kinetochore

7-	All chromosomes other than sex chromosome are called? (FINAL 33, Formative 36)						
A-	Autosomes	B-	Acrocentric	C-	Subterminal	D-	Submedian

8-	Number of autosomes in human somatic cells?(FINAL 34)						
A-	22	B-	23 pairs	C-	2	D-	22 pairs

9-	Chromosome telocentric morphology? (FINAL SNU 1)
A-	the centromere is at the Terminal ends of chromatin
B-	centromere Near the terminal

10-	Submetacentric is ...?(FINAL 37)		
A-	Near to the center	B-	Midway between center and end

11-	Polyploidy means: (Formative 36)?
A-	loss of an arm of a chromosome
B-	addition of an extra piece of a chromosome
C-	division of a chromosome in a transverse manner
D-	abnormal number of chromosomes

12-	Which of the following conditions is characterized by a cell containing more than 2 haploid numbers of chromosomes, such as in cancer cells? (2ND TRIAL 37)						
A-	Aneuploidy	B-	Polyploidy	C-	Inversion	D-	Deletion

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
B	B	B	E	D	C	A	D	A	B
11	12	13	14	15	16	17	18	19	20
D	B	-	-	-	-	-	-	-	-

Lecture Qs

1-	Submetacentric chromosomes have: (Lecture Qs)		
A-	centromere located near the center	B-	centromere located median in position
C-	centromere located near one end	D-	none of the above

2-	Loss of an arm of a chromosome abnormality called: (Lecture Qs)						
A-	deletion	B-	Isochromosome	C-	Duplication	D-	inversion

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
D	A	-	-	-	-	-	-	-	-

Department Qs

1-	The DNA threads which appear inside the nucleus at the time of cell division? (Department Qs)						
A-	Spindle fibers	B-	Centrioles	C-	Asters	D-	Chromosomes

2-	The number of autosomes in human somatic cell is? (Department Qs)		
A-	22 chromosomes	B-	22 pairs of identical chromosomes
C-	23 pairs of identical chromosomes	D-	23 chromosomes

3-	In telocentric chromosomes the centromere is? (Department Qs)						
A-	Terminal	B-	Absent	C-	Subterminal	D-	Median

4-	Chromosomal study helps in clinical diagnosis in? (Department Qs)		
A-	multiple congenital malformation	B-	abnormal sexual development
C-	infertility	D-	all of the above

5-	A karyotype is a spread of chromosomes arranged according to their? (Department Qs)						
A-	Size	B-	Shape	C-	Banding pattern	D-	All of the above

6-	Each chromosome consists of two identical? (Department Qs)						
A-	genes	B-	nuclei	C-	chromatids	D-	base

7-	All chromosomes other than 'sex' chromosome are called? (Department Qs)		
A-	Autosomes	B-	Acentric chromosomes
C-	Metacentric Chromosomes	D-	Telocentric chromosomes

8-	A person who receives an extra chromosome, could have? (Department Qs)		
A-	Heightened intelligence	B-	Down syndrome
C-	Red eyes	D-	all of the above

9-	Two chromosomes in a pair are classified as? (Department Qs)		
A-	heterologous chromosomes	B-	homologous chromosomes
C-	homozygous chromosomes	D-	heterozygous chromosomes

10-	Chromosomes are made up of all these bases except? (Department Qs)						
A-	cytosine	B-	thymine	C-	chromatin	D-	adenine

11-	is a chromosome rearrangement in which a segment of a chromosome is reversed end to end? (Department Qs)						
A-	Duplication	B-	Inversion	C-	Deletion	D-	Translocation

12-	Diploid chromosome number is found in normal body cell whereas haploid is found in? (Department Qs)				
A-	Gametes	B-	brain cells	C-	hair cells
D-	sensory cells	E-	nerve cell		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
D	B	A	D	D	C	A	B	B	C
11	12	13	14	15	16	17	18	19	20
B	A	-	-	-	-	-	-	-	-

SNU Bank

1-	**Isochromosome means: (SNU 1 Bank)**
A-	Loss of an arm of a chromosome
B-	Addition of an extra piece to a chromosome
C-	Chromosome that divides transversely
D-	Breaking and re-attachment of a chromosome in a reverse position

2-	**Morphology of the chromosomes is studied during: (SNU 1 Bank)**						
A-	Anaphase	B-	Prophase	C-	Metaphase	D-	Interphase

3-	**A picture of a person's chromosomes is called: (SNU 1 Bank)**				
A-	karyotype	B-	syndrome	C-	chromatin
D-	fingerprint				

4-	**Polyploidy means: (SNU 1 Bank)**
A-	addition of an extra piece of a chromosome
B-	loss of an arm of a chromosome
C-	division of a chromosome in a transverse manner
D-	abnormal number of chromosomes

5-	**What is a karyotype? (SNU 1 Bank)**
A-	A genetic mutation
B-	The visual representation of chromosomes arranged by size and shape
C-	A type of DNA sequencing
D-	An enzyme that replicates DNA

6-	**How many pairs of chromosomes are typically found in a normal human karyotype? (SNU 1 Bank)**						
A-	20 pairs	B-	23 pairs	C-	46 pairs	D-	50 pairs

Answers - SNU Bank									
1	2	3	4	5	6	7	8	9	10
C	C	A	D	B	B	-	-	-	-

M3WAN Bank

1-	Each somatic cell nucleus normally contains 46 chromosomes, arranged as: (M3WAN - Core)?
A-	23 pairs of autosomes only
B-	22 pairs of autosomes and one pair of sex chromosomes
C-	46 pairs of sex chromosomes
D-	23 unpaired autosomes and 23 unpaired sex chromosomes

2-	In female somatic cells, the pair of sex chromosomes is: (M3WAN - Core)?		
A-	identical, and called XX	B-	non-identical, called XY
C-	absent	D-	represented by a singleX chromosome

3-	In male somatic cells, the sex chromosomes consist of: (M3WAN - Core)?
A-	two identicalX chromosomes
B-	one longX chromosome and one much smallerY chromosome
C-	twoY chromosomes
D-	two chromosomes of equal size

4-	Each mature gamete (ovum or sperm) normally contains: (M3WAN - Core)?						
A-	46 chromosomes	B-	44 chromosomes	C-	23 chromosomes	D-	22 chromosomes

5-	Regarding the sex chromosome content of gametes, sperms are of two types (X'bearing and Y'bearing), while all mature ova carry: (M3WAN - Core)?		
A-	aY chromosome only	B-	anX chromosome only
C-	bothX andY chromosomes	D-	no sex chromosome

6-	Karyotyping is best defined as the study of the: (Important M3WAN - Core)?
A-	rate of DNA replication in a cell
B-	sequence of nitrogenous bases in DNA
C-	distribution of euchromatin and heterochromatin
D-	number, type, and morphology of the chromosomes of an individual

7-	In the preparation of a karyotype, dividing cells are treated with colchicine mainly to: (Important M3WAN - Core)?		
A-	arrest the cells in the metaphase stage	B-	stain the chromosomes with a basic dye
C-	induce chromosomal deletion	D-	fix the tissue for electron microscopy

8-	The most convenient cells used for chromosomal (karyotype) studies are: (M3WAN - Core)?		
A-	red blood cells	B-	hepatocytes
C-	white blood cells	D-	skeletal muscle cells

9-	Each chromosome is formed of two chromatids connected together at one point known as the: (M3WAN - Core)?						
A-	kinetochore	B-	spherule	C-	nucleoid	D-	centromere

10-	The centromere appears as a light-stained region except at its center, where a dense small granule called the ___ is present: (Important M3WAN - Core)?		
A-	spherule	B-	nucleoid
C-	kinetochore	D-	nucleolar organizer

11-	The kinetochore is a disc-shaped structure present in the centromere that functions as the: (Important M3WAN - Core)?
A-	site of DNA replication
B-	site of ribosomal RNA transcription
C-	microtubule organizing center during cell division
D-	attachment point for the nuclear envelope

12-	A chromosome in which the centromere is located centrally is classified as: (M3WAN - Core)?						
A-	acrocentric	B-	telocentric	C-	submetacentric	D-	metacentric

13-	A chromosome in which the centromere lies between the center and the end of the chromatids is classified as: (Important M3WAN - Core)?						
A-	metacentric	B-	submetacentric	C-	acrocentric	D-	telocentric

14-	A chromosome in which the centromere is located near the terminal ends of the chromatids is classified as: (Important M3WAN - Core)?						
A-	metacentric	B-	submetacentric	C-	acrocentric	D-	telocentric

15-	A chromosome in which the centromere is located at the very terminal end of the chromatids is classified as: (Important M3WAN - Core)?						
A-	telocentric	B-	acrocentric	C-	submetacentric	D-	metacentric

16-	Chromosomes other than the sex chromosomes are collectively termed: (Important M3WAN - Core)?						
A-	gonosomes	B-	autosomes	C-	isochromosomes	D-	kinetochores

17-	A numerical chromosomal abnormality in which the cell contains more than two haploid sets of chromosomes, as seen in some cancer cells, is called: (M3WAN - Core)?		
A-	aneuploidy	B-	isochromosome formation
C-	polyploidy	D-	inversion

18-	Trisomy 21, resulting in Down syndrome, is an example of a numerical chromosomal abnormality called: (Important M3WAN - Core)?						
A-	aneuploidy	B-	polyploidy	C-	deletion	D-	duplication

19-	Loss of an arm of a chromosome is termed deletion, while addition of an extra piece to a chromosome is termed: (M3WAN - Hard)?		
A-	inversion	B-	isochromosome formation
C-	aneuploidy	D-	duplication

20-	A structural chromosomal abnormality that occurs when a chromosome breaks and re-attaches in a reverse position is called: (M3WAN - Core)?						
A-	deletion	B-	inversion	C-	duplication	D-	isochromosome

21-	An abnormality that occurs when a chromosome divides in a transverse manner instead of longitudinally, producing a chromosome with identical arms, is called: (Important M3WAN - Core)?						
A-	deletion	B-	inversion	C-	isochromosome	D-	polyploidy

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	A	B	C	B	D	A	C	D	A
11	12	13	14	15	16	17	18	19	20
C	D	B	C	A	B	C	A	D	B
21	22	23	24	25	26	27	28	29	30
C	-	-	-	-	-	-	-	-	-

لا تمسِ حتى تنظر فيما مضى من يومك، فإن رأيت زلة فامحها بتوبة، أو خرقًا فارقعه باستغفار.

وإذا أصبحت فتأمل ما مضى في ليلك، وإياك والتسويف فإنه أكبر جنود إبليس.

- ابن الجوزي البغدادي

Lecture 8: Cell cycle & cell death

Pre Exams & Formatives

1-	**Which of the following is a characteristic of G1 phase? (Formative 32, Department Qs)**		
A-	DNA is replicated	B-	Cell growth and preparation for mitosis
C-	Nuclear membrane reforms	D-	Cell growth and normal function
E-	All of the above		

2-	**Which of the following is a characteristic of metaphase? (Formative 32, Department Qs)**
A-	Sister chromatids are pulled to opposite sides of the cell
B-	Nuclear membrane breaks down
C-	Chromosomes line up in the middle of the cell
D-	The cytoplasm splits
E-	Chromosomes condense

3-	**Which of the following would you find DNA just replicated? (Formative 32, Department Qs)**								
A-	Metaphase	B-	Interphase	C-	Prophase	D-	G1	E-	G2

4-	**Which of the following phases would you find the cell creating ribosomes for protein synthesis? (MID 30, FINAL 30, Department Qs)**								
A-	G1	B-	G2	C-	prophase	D-	Sstage	E-	interphase

5-	**Which of the following processes starts the stages of mitosis? (MID 30, Formative 32)**		
A-	Condensation of chromatin	B-	Development of spindle fibers
C-	DNA replication	D-	Dissolving of the nuclear membrane
E-	cytokinesis		

6-	**Which of the following stages is prior to mitosis?(FINAL 30)**								
A-	G2	B-	G1	C-	G0	D-	S'stage	E-	cytokinesis

7-	**Unreplicated DNA check point occurs in? (MID 31)**								
A-	G1'phase	B-	G2'phase	C-	M'phase	D-	S'phase	E-	prophase

8-	**Cell differentiation means? (MID 31)**		
A-	increase in the number of cells	B-	increase in the size of cells
C-	acquiring new features	D-	acquiring new functions
E-	both:c&d		

9-	All of the following are related to apoptosis EXCEPT? (MID 31)
A-	occurs in physiological and pathological conditions
B-	the cell swells and ruptures
C-	occurs due to activation of certain genes
D-	known as suicidal death
E-	nuclear condensation

10-	Phase of the cell, where it undergoes growth and DNA replication is called?(FINAL 31)				
A-	mitosis phase	B-	G1'phase	C-	Interphase
D-	M'phase	E-	G2'phase		

11-	How does a somatic cell that has just completed theS phase of its cell cycle compare with a gamete of the same species?(FINAL 31)
A-	It has twice the number of chromosomes and twice the amount of DNA
B-	It has the same number of chromosomes but twice the amount of DNA
C-	It has twice the number of chromosomes and four times the amount of DNA
D-	It has four times the number of chromosomes and twice the amount of DNA
E-	It has twice the number of chromosomes and the same amount of DNA

12-	Apoptosis is associated with?(FINAL 31)				
A-	cell membrane damage	B-	inflamation	C-	cytoplasmic swelling
D-	DNA fragmentation	E-	Mitochondrial swelling		

13-	Chromosomes are less visible during?(FINAL 31)								
A-	Anaphase	B-	interphase	C-	Metaphase	D-	Telophase	E-	Prophase

14-	Which is correct regarding cell cycle? (MID 32)		
A-	The amount of DNA in G2 is doubleS phase	B-	The amount of DNA inS phase is double G1
C-	The amount of DNA inS phase is half in G1	D-	The amount of DNA in G2 is half in G1
E-	The amount of DNA is similar in G1 and G2		

15-	Apoptosis is characterized by all the following EXCEPT: ? (FINAL 32)				
A-	Inflammation	B-	Condensation of chromatin	C-	Decrease in cell size
D-	Membrane blebbing	E-	Intact organelles		

16-	which is true regardings cell necrosis (2ND TRIAL 32)?		
A-	swelling with destruction of organelles	B-	shrinkage with destruction of organ!les
C-	swelling with normal organelles	D-	shrinkage with normal organlles

17-	cell differentiation characterized by: (2ND TRIAL 32)?		
A-	increase in number	B-	acquire a specialized function
C-	decrease in size	D-	increase in size

18-	Example of cells which enter G 0 stage for its life span? (MID 33)						
A-	Epithelial cell	B-	Stem cell	C-	Cardiac muscle	D-	Cartilage cell

19-	Life cycle of cell is? (MID 34)
A-	consist of interphase and cell division

20-	In which phase chromosome is single chromatid? (MID 34)
A-	G1 phase

21-	Which type of cell death is due to exogenous factors such as injury , toxins or hypoxia?(FINAL 35)		
A-	Apoptosis	B-	Necrosis

22-	Necrosis is caused by: ? (MID 37)
A-	Normal cell functions
B-	Injury or trauma causing cell swelling and rupture
C-	Cell proliferation
D-	Cell differentiation

23-	Apoptosis is: (Formative 36)?		
A-	cell division	B-	mechanical death
C-	Programmed cell death	D-	all of the above

24-	The phase at which nuclear envelop disappear is:- (Formative 36)?						
A-	prophase	B-	Telophase	C-	Anaphase	D-	Metaphase

25-	Apoptosis is characterized by which of the following? (2ND TRIAL 37)		
A-	Condensation and fragmentation of DNA	B-	Cell swelling
C-	Destruction of organelles	D-	Inflammation

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
D	C	E	A	A	A	B	E	B	C
11	12	13	14	15	16	17	18	19	20
C	D	B	B	A	A	B	C	A	A

21	22	23	24	25	26	27	28	29	30
B	B	C	A	A	-	-	-	-	-

Lecture Qs

1-	DNA is replicated during: (Lecture Qs, Department Qs)?						
A-	Prophase	B-	G1 of interphase	C-	Sof interphase	D-	G2 of interphase

2-	The cell cycle is the period between: (Lecture Qs, Department Qs)?
A-	prophase and telophase
B-	end of mitosis to the end of the successive mitosis
C-	Interphase and metaphase
D-	G1 and G2

3-	Cell differentiation means: (Lecture Qs)		
A-	increase in the number of cells	B-	increase in the size of the cells
C-	acquiring of new features and functions	D-	all of the above

Answers - Lecture Qs

1	2	3	4	5	6	7	8	9	10
C	B	C	-	-	-	-	-	-	-

Department Qs

1-	Nucleoli disappears during which mitotic event? (Department Qs)				
A-	prometaphase	B-	anaphase	C-	prophase
D-	telophase	E-	anaphase		

2-	Which of the following processes starts the stages of mitosis? (Department Qs)		
A-	Shorten of chromatin	B-	Development of mitotic spindle
C-	DNA replication	D-	Dissolving of the nuclear membrane
E-	cytokinesis		

3-	The central alignment of chromosomes by the spindle microtubules occurs in which of the following stages of mitosis? (Department Qs)								
A-	telophase	B-	anaphase	C-	metaphase	D-	prophase	E-	cytokinesis

4-	The development of the nuclear envelope occurs in which of the following stages of mitosis? (Department Qs)								
A-	telophase	B-	anaphase	C-	metaphase	D-	prophase	E-	cytokinesis

5-	The contraction of microtubules to separate sister chromatids occurs in which of the following stages of mitosis? (Department Qs)								
A-	telophase	B-	anaphase	C-	metaphase	D-	prophase	E-	cytokinesis

6-	Apoptosis usually involve? (Department Qs)		
A-	single cells	B-	shrinkage of cells
C-	condensed nuclear chromatin	D-	peripheral aggregation of chromatin
E-	all of the above		

7-	During anaphase of mitosis? (Department Qs)
A-	the chromosomes uncoil becoming indistinct
B-	the chromosomes become visible and condense
C-	the chromatids separate, moving away from each other
D-	the cell contains intermediate filaments. e'none of the above

8-	The 2 chromatides of a chromosome separate and migrate to the opposite pole of the mitotic spindle during? (Department Qs)				
A-	prophase stage	B-	metaphase stage	C-	anaphase stage
D-	telophase stage	E-	both:b&c		

9-	The mitotic spindle is composed mostly of? (Department Qs)		
A-	Golgi flattened sacs	B-	smooth endoplasmic reticulum
C-	microtubules	D-	intermediate filaments
E-	none of the above		

10-	The S'phase of the cell cycle is the phase where? (Department Qs)		
A-	DNA is duplicated	B-	DNA filaments are combined with proteins
C-	protein synthesis occurs	D-	organoids are formed
E-	the cell perform its function		

11-	In the G2 phase of the cell cycle? (Department Qs)				
A-	DNA is synthesized	B-	proteins are formed	C-	RNA are formed
D-	Both: b and c	E-	Both: a and c		

12-	The amount of DNA in the cell at G2 phase is? (Department Qs)		
A-	similar to the cell at G1 phase	B-	similar to the cell at Go phase
C-	double the cell atS phase	D-	double the cell at G1 phase
E-	half the cell atG phase		

13-	Cell death by apoptosis? Performs a useful physiological role in the body (Department Qs)		
A-	True	B-	False

14-	Cell death by apoptosis? Involves a decrease in cell size (Department Qs)		
A-	True	B-	False

15-	Cell death by apoptosis? Involves break down of the cell into fragments (Department Qs)		
A-	True	B-	False

16-	Cell death by apoptosis? May help the body get rid of cells with damaged DNA (Department Qs)		
A-	True	B-	False

17-	Cell death by apoptosis? Results in inflammation (Department Qs)		
A-	True	B-	False

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	A	C	A	B	E	C	C	C	A
11	12	13	14	15	16	17	18	19	20
B	D	A	A	A	A	B	-	-	-

SNU Bank

1-	Which of the following is a characteristic of metaphase? (SNU 1 Bank)
A-	Sister chromatids are pulled to opposite sides of the cell
B-	Nuclear membrane breaks down
C-	Chromosomes line up in the middle of the cell
D-	Chromosomes condense

2-	The chromosome is formed of a single chromatid in: (SNU 1 Bank)						
A-	G1 phase	B-	G2 phase	C-	S phase	D-	Interphase

3-	The mitotic spindle is composed mostly of: (SNU 1 Bank)		
A-	Microfilaments	B-	Microtubules

C-	Intermediate filaments	D-	Thick filaments

4-	**The cell cycle is defined as the period between: (SNU 1 Bank)**
A-	prophase and telophase
B-	end of mitosis to the end of the successive mitosis
C-	Interphase and metaphase
D-	G1 and G2

5-	**Which phase of the cell cycle involves DNA replication? (SNU 1 Bank)**						
A-	G1 phase	B-	S phase	C-	G2 phase	D-	M phase

6-	**Necrosis is caused by: (SNU 1 Bank)**
A-	Normal cell functions
B-	Injury or trauma causing cell swelling and rupture
C-	Cell proliferation
D-	Cell differentiation

Answers - SNU Bank									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	**A**	**B**	**B**	**B**	**B**	**-**	**-**	**-**	**-**

M3WAN Bank

1-	**The cell cycle is defined as the interval between: (M3WAN - Core)?**		
A-	two successiveS phases	B-	start and end of interphase only
C-	the end of one mitosis and the end of the next	D-	prophase and telophase

2-	**Interphase is: (M3WAN - Core)?**
A-	the interval between divisions, with specialized function
B-	the resting phase, with no activity
C-	the four phases of mitosis
D-	synonymous with mitosis

3-	**In the G1 phase, chromosomes are: (M3WAN - Core)?**		
A-	double chromatids	B-	absent
C-	condensed at the equator	D-	single chromatids

4-	**The G0 (resting) stage is typical of: (M3WAN - Core)?**		
A-	epithelial cells	B-	nerve and cardiac muscle cells

C-	fibroblasts	D-	osteoblasts

5-	**In the S phase: (M3WAN - Core)?**		
A-	cytoplasmic mass increases only	B-	chromosomes remain single
C-	the nuclear envelope disappears	D-	DNA/histones replicate; chromatids double

6-	**Once a cell enters the S phase, it: (M3WAN - Hard)?**		
A-	must complete the cycle	B-	may exit to G0 freely
C-	can revert to G1	D-	arrests permanently

7-	**The amount of DNA in the S phase, compared to G1, is: (Important M3WAN - Core)?**						
A-	half	B-	equal	C-	double	D-	quadruple

8-	**In the G2 phase: (M3WAN - Core)?**		
A-	DNA replicates	B-	centrioles duplicate; cytoplasm increases
C-	chromosomes become single	D-	the nuclear envelope disappears

9-	**Mitosis is best described as: (M3WAN - Core)?**		
A-	occurring in gametes only	B-	occurring only in G0 cells
C-	synonymous with interphase	D-	somatic cell division, in four phases

10-	**In prophase: (M3WAN - Core)?**		
A-	chromosomes align at the equator	B-	chromatids separate
C-	centrioles migrate; the envelope disappears	D-	cytokinesis is completed

11-	**In metaphase: (M3WAN - Core)?**		
A-	chromosomes align at the equator	B-	centrioles begin migration
C-	the envelope reassembles	D-	chromatids separate

12-	**In anaphase: (M3WAN - Core)?**		
A-	chromosomes align at the equator	B-	the envelope disappears
C-	the cleavage furrow forms	D-	the centromere splits; chromatids separate

13-	**In telophase: (M3WAN - Core)?**		
A-	chromosomes align at the equator	B-	cleavage completes; the envelope reassembles
C-	centrioles migrate	D-	chromatids separate

14-	**During mitosis, the nucleolus: (Important M3WAN - Core)?**
A-	disappears in metaphase, reappears in anaphase

B-	persists throughout
C-	disappears in late prophase, reappears in telophase
D-	disappears in telophase, reappears in prophase

15-	"Growth," as an aspect of cell development, refers to: (M3WAN - Core)?		
A-	new function acquisition	B-	harmony of growth and differentiation
C-	programmed cell death	D-	increase in cell mass by mitosis

16-	"Differentiation," as an aspect of cell development, refers to: (M3WAN - Core)?		
A-	increase in cell number	B-	new features/functions arising
C-	harmony of growth and death	D-	decrease in cell mass

17-	"Integration," as an aspect of cell development, refers to: (M3WAN - Core)?		
A-	harmony of growth and differentiation	B-	programmed cell death
C-	DNA replication	D-	an exclusive property of stem cells

18-	Necrosis is caused by: (Important M3WAN - Core)?		
A-	gene activation	B-	normal physiology
C-	differentiation	D-	exogenous injury, toxin, or hypoxia

19-	Apoptosis is: (Important M3WAN - Core)?		
A-	exogenous injury with cell swelling	B-	bursting with scattered fragments
C-	programmed, gene-activated cell death	D-	clumped chromatin, destroyed organelles

20-	All are features of apoptosis EXCEPT: (Important M3WAN - Hard)?		
A-	DNA condensation/fragmentation	B-	swelling, with organelle destruction
C-	cell shrinkage	D-	preservation of normal organelles

21-	Apoptotic bodies are: (Important M3WAN - Core)?		
A-	left extracellular permanently	B-	strongly inflammatory
C-	cleared by neutrophils only	D-	phagocytized by neighboring cells

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
C	A	D	B	D	A	C	B	D	C
11	12	13	14	15	16	17	18	19	20
A	D	B	C	D	B	A	D	C	B
21	22	23	24	25	26	27	28	29	30
D	-	-	-	-	-	-	-	-	-

Lecture 9: Stem cells and tissue renewal

Pre Exams & Formatives

1-	**A common source of stem cells is?(FINAL 30)**				
A-	A blastocyst	B-	A zygote	C-	An embryo
D-	a mesoderm	E-	a chorion		

2-	**Stem cell ? (FINAL 30)**				
A-	Antigen presenting cell	B-	Immunity	C-	Growth
D-	Differentiation	E-	remodeling		

3-	**Genetic reprogramming occurs in? (MID 31)**		
A-	adult stem cells	B-	multipotent stem cells
C-	induced pluripotential stem cells	D-	embryonic stem cells
E-	totipotent stem cells		

4-	**Which of the following terms describes the potential of the inner mass cells from the blastocyst? (FINAL 31, 2ND TRIAL 35)**								
A-	Totipotent	B-	Pluripotent	C-	Multipotent	D-	unipotent	E-	bipotent

5-	**What is a stem cell?(FINAL 31)**
A-	A cell that can make copies of itself and make more specialized types of cell
B-	A cell that helps to fight against infections
C-	A cell that is specialized
D-	A cell that can produce all the cell types of the body
E-	A cell which is genetically reprogrammed

6-	**Pluripotent stem cells can differentiate into? (MID 32, Formative 32)**				
A-	ectoderm	B-	endoderm	C-	mesoderm
D-	none of the above	E-	all of the above		

7-	**Stem Cells are important in? (FINAL 32)**		
A-	Replacing diseased cells in sick patients	B-	research on new drugs
C-	research on disease mechanism	D-	Only: A and B
E-	All of the above		

8-	**Easy and least invasive source of stem cells (2ND TRIAL 32)?**		
A-	Cord blood	B-	Adipose tissue
C-	MSCs in bone marrow	D-	nervous tissue

9-	**Sources of stem cell include? (2ND TRIAL 32, Formative 36, Formative 32)**		
A-	Umbilical cord blood	B-	embryonic cells
C-	bone marrow	D-	Fall of the above

10-	**Stem cells is? (MID 34)**
A-	Cells can proliferate and differentiate

11-	**The inner cell mass in the blastocyst can differentiate into? (MID 34)**		
A-	Only one type of cells in brain	B-	All types of body tissues

12-	**What are the Induced pluripotent stem cells (iPSCs)? (FINAL 35)**
A-	stem cells that are created in the laboratory through genetic reprogramming
B-	stem cells found in human amniotic fluid
C-	stem cells are used in hematopoietic stem cell transportation
D-	stem cells derived from bone marrow stem cells are created in the laboratory through

13-	**The ability of stem cells to divide and produce both stem cells and differentiated cells is called? (MID 37)**		
A-	Differentiation	B-	Self-renewal
C-	Apoptosis	D-	Trans differentiation

14-	**Totipotent differ from pluripotent in …? (FINAL 37)**
A-	Totipotent give embryo and placenta

15-	**Embryonic stem cells are primarily derived from which of the following structures during embryonic development? (FINAL SNU-M 1)**						
A-	Trophoblast	B-	Inner cell mass	C-	Zona pellucida	D-	Corona radiata

16-	**Embryonic stem cell: (Formative 32)?**
A-	Unable to divide
B-	It is an ethical issue since it kills the embryo, which is a potential life
C-	Present in umbilical cord blood
D-	All of the above
E-	None of the above

17-	**A stem cell is best described as: (2ND TRIAL 37)**		
A-	Undifferentiated	B-	Specialized
C-	Genetically reprogrammed only	D-	Unable to self-renew

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
A	D	C	B	A	E	E	A	D	A
11	12	13	14	15	16	17	18	19	20
B	A	B	A	B	B	A	-	-	-

Lecture Qs

1-	**What are the roles of stem cells in our bodies? (Lecture Qs, Department Qs)**
A-	We are not sure what roles stem cells play in the body
B-	They produce new specialized cells to replace cells that die or are used up
C-	They fight against infections
D-	They perform specialized roles in the body (e'g' produce insulin, transmit signals in the nervous system, ...)

2-	**What is a stem cell? (Lecture Qs)**
A-	A cell that can make copies of itself AND make more specialized types of cells
B-	A cell that helps to fight against infections
C-	A cell that is specialized
D-	A cell that can produce all the cell types of the body

3-	**Embryonic stem cells can differentiate into which types of cells? (Lecture Qs)**
A-	Only brain stem cells and specialized brain cells
B-	All types of specialized cells in the body
C-	Only cells that can produce insulin
D-	Only cells that can produce artificial skin

4-	**A blastocyst is... (Lecture Qs)**		
A-	A very early-stage embryo	B-	A type of stem cell
C-	Part of the blood system	D-	A type of brain cell

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
B	A	B	A	-	-	-	-	-	-

Department Qs

1-	In a developing embryo, stem cells can differentiate into? (Department Qs)				
A-	ectoderm	B-	endoderm	C-	mesoderm
D-	none of the above	E-	all of the above		

2-	In terms of types of stem cells, define "totipotent"? (Department Qs)
A-	cell type able to give rise to all cell types including extraembryonic cell lineages
B-	cell type able to give rise to all cells found in the embryo proper
C-	cell type able to give rise to more than one lineage of cells - lineage specific tissue stem cells
D-	all of the above
E-	none of the above

3-	Where can Stem Cells come from? (Department Qs)				
A-	Umbilical cord blood	B-	embryonic cells	C-	bone marrow
D-	all of the above	E-	none of the above		

4-	What are Stem Cells used for? (Department Qs)		
A-	Replacing diseased cells in sick patients	B-	Growing into a new organ that a patient needs
C-	Making a clone of cells	D-	Only:AandB
E-	All of the above		

5-	What do you think a stem cell is? (Department Qs)
A-	It is a cell out of which the stem of the tree is made up of
B-	It is a cell that can divide and give rise to specialized cells
C-	It is a part of the tissue that forms the outer layer of the skin in human beings
D-	all of the above
E-	none of the above

6-	One of the most important disadvantages of deriving embryonic stem cell from the embryo is? (Department Qs)
A-	the subject cannot get pregnant again
B-	It is an ethical issue since it kills the embryo, which is a potential life
C-	It increases inflammation in the subject
D-	all of the above
E-	none of the above

7-	Which diseases should be treated by cell-based therapies in the future? (Department Qs)		
A-	Parkinson's and Alzheimer's Diseases	B-	common cold
C-	HIV	D-	all of the above
E-	none of the above		

8-	Adult stem cells are described as... (Department Qs)				
A-	Somatic	B-	Syngenic	C-	Specialized
D-	all of the above	E-	none of the above		

9-	What is the least invasive source of stem cells from the human body? (Department Qs)				
A-	Cord blood	B-	Adipose tissue	C-	Bone Marrow
D-	nervous tissue	E-	none of the above		

10-	Name two defining feature of embryonic stem cells? (Department Qs)				
A-	specialized cells	B-	Self-renewal	C-	degeneration
D-	Proliferation	E-	Meiosis		

11-	What are iPS cells, and what do they mean to stem cell research? (Department Qs)
A-	Induced purpose stem cells, mean embryo cells were made into adult-like cells
B-	Induced purpose stem cells, mean adult cells were reversed to baby cells
C-	Induced pluripotent stem cells, mean adult cells were made into embryo-like cells
D-	Induced pluripotent stem cells, mean cells were totipotent and are now pluripotent
E-	none of the above

12-	Stem cells can be induced physiologically or in the laboratory to become a specific cell for special functions. (Department Qs)		
A-	True	B-	False

13-	Stem cells can undergo cell division even after long periods of inactivity (Department Qs)		
A-	True	B-	False

14-	Basically, two types of stem cells - embryonic and adult - from human beings have been worked with by scientists. (Department Qs)		
A-	True	B-	False

15-	Stem cell research can help treat a number of diseases successfully in the near future. (Department Qs)		
A-	True	B-	False

16-	The primary role of adult stem cell is to maintain and repair the tissue in which they are found. (Department Qs)		
A-	True	B-	False

17-	Is it possible to store stem cells for future use by the donor. (Department Qs)		
A-	True	B-	False

18-	ESC can differentiate into all cell types (more than 200) in the body whereas differentiation in ASC is limited to the cell types of their tissue of origin. (Department Qs)		
A-	True	B-	False

19-	Adult stem cells are found only in bone marrow. (Department Qs)		
A-	True	B-	False

20-	Scientists can reprogram skin cells to act like stem cells. (Department Qs)		
A-	True	B-	False

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
E	A	D	E	B	B	A	A	A	D
11	12	13	14	15	16	17	18	19	20
C	A	A	A	A	A	A	A	B	A

SNU Bank

1-	Stem cells are best characterized as: (SNU 1 Bank)		
A-	Specialized cells with fixed functions	B-	Cells that do not divide
C-	Unspecialized cells capable of differentiation	D-	Cells found only in embryonic tissue

2-	Hematopoietic stem cells, responsible for blood cell formation, are found in: (SNU 1 Bank)						
A-	Liver	B-	Bone marrow	C-	Epidermis	D-	Muscle tissue

3-	The ability of stem cells to divide and produce both stem cells and differentiated cells is called: (SNU 1 Bank)		
A-	Differentiation	B-	Self-renewal
C-	Apoptosis	D-	Trans differentiation

4-	Cells of the inner cell mass of blastocyst can differentiate into: (SNU 1 Bank)		
A-	epithelial cells	B-	Endocrine cells
C-	Connective tissue cells	D-	All of the above

5-	All of the following are stem cell types according to the potentiality EXCEPT: (SNU 1 Bank)						
A-	Totipotent	B-	Pluripotent	C-	Multipotent	D-	None potent

6-	Cells of the inner cell mass of the blastocyst can differentiate into which types of cells? (SNU 1 Bank)		
A-	Brain cells	B-	insulin-producing cells
C-	Skin cells	D-	All types of specialized cells in the body

7-	Sources of stem cells include: (SNU 1 Bank)		
A-	Umbilical cord blood	B-	embryonic cells
C-	bone marrow	D-	all of the above

8-	What are the induced pluripotential stem cells (iPSCs)? (SNU 1 Bank)
A-	Stem cells derived from bone marrow
B-	Stem cells created in the laboratory through genetic reprogramming
C-	Stem cells found in human amniotic fluid
D-	Stem cells used in hematopoietic stem cell transplantation

Answers - Stem cells and tissue renewal									
1	2	3	4	5	6	7	8	9	10
C	B	B	D	D	D	D	B	-	-

M3WAN Bank

1-	A stem cell is defined as: (M3WAN - Core)?
A-	a fully differentiated cell
B-	a cell incapable of division
C-	a cell exclusive to bone marrow
D-	a cell dividing and differentiating into other cell types

2-	"Undifferentiated," as a stem cell property, means: (M3WAN - Core)?		
A-	unspecialized	B-	self-renewal capacity
C-	forms new tissue types	D-	divides indefinitely

3-	**"Self-renewal," as a stem cell property, means: (Important M3WAN - Core)?**		
A-	being unspecialized	B-	giving rise to specialized cells
C-	dividing and renewing for long periods	D-	forming a different tissue type

4-	**"Differentiation," as a stem cell property, means: (M3WAN - Core)?**		
A-	being unspecialized	B-	giving rise to specialized cell types
C-	dividing for long periods	D-	forming a different tissue type

5-	**"Plasticity," as a stem cell property, means: (M3WAN - Core)?**		
A-	being unspecialized	B-	self-renewal capacity
C-	differentiation into specialized cells	D-	one cell type forming another tissue type

6-	**A totipotent cell e.g. the fertilized egg, can form: (Important M3WAN - Core)?**
A-	the three germ layers only
B-	a complete individual, embryo and placenta
C-	a single tissue type
D-	blood cells only

7-	**A pluripotent cell e.g. the inner cell mass, can form: (Important M3WAN - Core)?**
A-	the three germ layers, without placenta
B-	a complete individual with placenta
C-	blood lineages only
D-	one organ only

8-	**A multipotent cell e. g. the hematopoietic stem cell, can form: (Important M3WAN - Core)?**
A-	a complete individual
B-	the three germ layers
C-	cell types of its specific tissue
D-	its own type only

9-	**An oligopotent cell e. g. a lymphoid/myeloid progenitor, can form: (M3WAN - Core)?**
A-	a complete individual
B-	the three germ layers
C-	its own type only
D-	a few related cell types

10-	A bipotent cell e.g. the hepatic progenitor cell, can form: (Important M3WAN - Core)?
A-	red and white blood cells
B-	two specific cell types
C-	all digestive cell types
D-	ectoderm and mesoderm

11-	A unipotent cell e.g. an epidermal stem cell, is characterized by: (M3WAN - Core)?
A-	producing only its own type, self-renewing
B-	forming multiple unrelated types
C-	forming a complete embryo
D-	lacking self-renewal

12-	Embryonic stem cells are derived from the: (Important M3WAN - Core)?						
A-	trophoblast	B-	amniotic fluid	C-	inner cell mass	D-	umbilical cord

13-	Compared with embryonic stem cells, adult stem cells are: (M3WAN - Core)?		
A-	less specialized, totipotent	B-	found only in the early embryo
C-	incapable of self-renewal	D-	more specialized, present throughout life

14-	Adult stem cell sites include bone marrow, umbilical cord, amniotic fluid, and: (M3WAN - Core)?		
A-	the zona pellucida	B-	the corona radiata
C-	adipose tissue	D-	the trophoblast

15-	Adipose tissue-derived stem cells (ADSCs) are a source of: (M3WAN - Core)?		
A-	multipotent stem cells	B-	totipotent stem cells
C-	unipotent stem cells	D-	pluripotent stem cells

16-	Induced pluripotent stem cells (iPSCs) are best defined as: (Important M3WAN - Core)?		
A-	cells from the inner cell mass	B-	cells from cord blood
C-	adult cells from bone marrow	D-	lab-reprogrammed somatic cells

17-	iPSCs were first discovered in: (M3WAN - Hard)?						
A-	1997	B-	2017	C-	2007	D-	1957

18-	Stem cell surface markers include CD45 and: (M3WAN - Core)?						
A-	CD3	B-	c'Kit (CD117)	C-	CD20	D-	CD68

19-	The only stem cell therapy currently approved by the FDA is: (Important M3WAN - Core)?		
A-	HSC transplantation	B-	embryonic stem cell therapy
C-	iPSC therapy	D-	ADSC injection

20-	HSC transplantation from an identical twin is termed: (M3WAN - Hard)?						
A-	autologous	B-	allogenic	C-	xenogeneic	D-	syngeneic

21-	HSCs for transplantation are derived from bone marrow, peripheral blood, or: (M3WAN - Core)?		
A-	adipose tissue	B-	umbilical cord blood
C-	amniotic membranes	D-	skin biopsies

22-	The umbilical cord is a rich source of: (Important M3WAN - Core)?		
A-	HSCs only	B-	totipotent cells
C-	mesenchymal stem cells	D-	iPSCs

23-	SHED stands for stem cells from: (M3WAN - Core)?		
A-	exfoliated deciduous teeth	B-	embryonic disc cells
C-	adipose depot cells	D-	hematopoietic donor cells

24-	The skin epidermis is renewed approximately every: (M3WAN - Core)?						
A-	one week	B-	three weeks	C-	ten days	D-	one month

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	A	C	B	D	B	A	C	D	B
11	12	13	14	15	16	17	18	19	20
A	C	D	C	A	D	C	B	A	D
21	22	23	24	25	26	27	28	29	30
B	C	A	B	-	-	-	-	-	-

كان عامر بن قيس إذا أصبح قال:

اللهم قد أصبح الصبح، وطفق الناس يغدون ويروحون، يبتغون من فضلك، وإن لكل منهم حاجة، وحاجة عامر أن تغفر له.

Lecture 10: Histological structure of the neurons

Pre Exams & Formatives

1-	All of the following present in dendrites except? (2ND TRIAL 35)						
A-	Golgi apparatus	B-	Mitochondria	C-	RER	D-	Neurofibrils

2-	Cells secretion myelin in peripheral nerve? (FINAL 37, Formative 36, Formative 37)
A-	Schwann cells

3-	Nissl granules in neurons are primarily composed of which of the following organelles? (FINAL SNU-M 1)		
A-	Smooth endoplasmic reticulum	B-	Rough endoplasmic reticulum
C-	Golgi apparatus	D-	Mitochondria

4-	Nodes of Ranvier are: (2ND TRIAL 37)		
A-	The spaces between adjacent Schwann cells	B-	Covered by thick myelin sheath
C-	Areas containing no ion channels	D-	Found exclusively in unmyelinated fibers

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
A	A	B	A	-	-	-	-	-	-

Lecture Qs

5-	Processes of neuron are: (Lecture Qs)		
A-	one dendrite and many axons	B-	may be dendrites only
C-	Axons only	D-	many dendrites and one Axon

6-	Axon hillock are all of the following except: (Lecture Qs)		
A-	is the point of axon origin from cell body	B-	contains mitochondria, microtubules
C-	contains free ribosomes	D-	contains large amount of neurofilaments

7-	Myelination of axons occur by: (Lecture Qs)						
A-	other perikaryon	B-	connective tissue	C-	Neurofilaments	D-	Schwann cell

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
D	C	D	-	-	-	-	-	-	-

M3WAN Bank

1-	**A neuron is best described as: (M3WAN - Core)?**		
A-	incapable of division, with no nucleus	B-	highly differentiated, with high conductivity
C-	a non-membranous organelle	D-	a connective tissue cell type

2-	**The neuron structurally consists of: (M3WAN - Core)?**		
A-	nucleus and cytoplasm only	B-	axon and dendrites only
C-	perikaryon and axon only	D-	perikaryon, axon, and dendrite(s)

3-	**By light microscopy, the neuron cell body shows: (M3WAN - Core)?**
A-	pale cytoplasm, basophilic Nissl bodies, vesicular nucleus
B-	deep basophilia throughout, pyknotic nucleus
C-	empty, vacuolated cytoplasm
D-	uniform acidophilia

4-	**Silver staining of the neuron demonstrates: (M3WAN - Core)?**						
A-	Nissl bodies	B-	lipofuscin	C-	neurofibrils	D-	the myelin sheath

5-	**Nissl bodies correspond ultrastructurally to: (Important M3WAN - Core)?**
A-	Golgi apparatus and secretory vesicles
B-	mitochondria and lysosomes
C-	microtubules and centrioles
D-	rough endoplasmic reticulum and polyribosomes

6-	**Lipofuscin pigment within neurons: (M3WAN - Core)?**		
A-	decreases with age	B-	becomes more numerous with age
C-	is confined to the axon hillock	D-	is absent from neurons

7-	**Dendrites are structurally characterized as: (M3WAN - Core)?**		
A-	one or many, thick at the base, tapering	B-	always single, straight, regular diameter
C-	devoid of cytoplasmic organelles	D-	the exclusive site of myelination

8-	**All are found in dendrites EXCEPT: (Important M3WAN - Hard)?**		
A-	mitochondria	B-	RER and ribosomes
C-	Golgi apparatus	D-	neurofibrils

9-	The axon is structurally characterized as: (M3WAN - Core)?		
A-	multiple per cell, tapering, branching	B-	irregular in diameter, rough surface
C-	devoid of cytoplasmic content	D-	single per cell, straight, regular diameter

10-	The axon hillock, the origin of the axon, is: (Important M3WAN - Core)?		
A-	rich in Nissl bodies	B-	free of Nissl bodies
C-	the exclusive site of myelination	D-	indistinguishable from the axon

11-	The axoplasm characteristically contains: (M3WAN - Core)?
A-	mitochondria, vesicles, neurofilaments, microtubules
B-	Nissl bodies and a well-developed Golgi
C-	glycogen and lipid droplets only
D-	nuclear pores and peripheral chromatin

12-	In peripheral myelination, the Schwann cell membrane wraps the axon up to: (Important M3WAN - Core)?						
A-	5 times	B-	15 times	C-	25 times	D-	50 times

13-	The myelin sheath is composed of: (M3WAN - Core)?		
A-	pure carbohydrate material	B-	collagen and elastin
C-	lipoprotein of the Schwann cell membrane	D-	hydrolytic enzymes

14-	Nodes of Ranvier represent: (M3WAN - Core)?		
A-	continuous, unbroken myelin	B-	gaps between adjacent Schwann cells
C-	specialized regions of the cell body	D-	sites of Nissl body formation

15-	The internodal segment is defined as: (M3WAN - Core)?		
A-	the distance between two nodes of Ranvier	B-	the total axon length
C-	the region from cell body to hillock	D-	the space between adjacent dendrites

16-	Regarding myelin thickness and nerve conduction: (M3WAN - Core)?		
A-	there is no relationship	B-	thinner myelin conducts faster
C-	conduction depends on axon length alone	D-	thicker myelin, faster conduction

17-	In H&E stained sections, the myelin sheath appears as: (M3WAN - Core)?
A-	deeply basophilic
B-	strongly PAS positive
C-	an empty layer, demonstrable with osmic stains

D-	indistinguishable from the axoplasm

18-	By electron microscopy, myelin shows: (M3WAN - Core)?		
A-	a single homogeneous dense layer	B-	concentric lamellae, alternating light/dark
C-	randomly scattered vesicles	D-	a trilaminar unit membrane only

19-	Unmyelinated (autonomic) fibers are: (M3WAN - Core)?
A-	within Schwann cells, one membrane layer, up to 12 axons
B-	always myelinated
C-	devoid of any Schwann cell
D-	found exclusively in the CNS

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	D	A	C	D	B	A	C	D	B
11	12	13	14	15	16	17	18	19	20
A	D	C	B	A	D	C	B	A	-

عشرةُ أشياء ضائعةٌ لا يُنتفع بها :

علمٌ لا يُعمَل به، وعملٌ لا إخلاصَ فيه ولا اقتداء، ومالٌ لا يُنفَقُ منه فلا يَستمتِعُ به جامعُه في الدُّنيا ولا يُقدّمُه أمامَه إلى الآخرة، وقلبٌ فارغٌ من محبة الله والشوق إليه والأنس به، وبدنٌ معطَّلٌ من طاعته وخدمته، ومحبةٌ لا تتقيَّدُ برِضى المحبوب وامتثال أوامره، ووقتٌ معطَلٌ عن استدراك فارطٍ أو اغتنام برٍّ وقُربةٍ، وفكرٌ يجولُ فيما لا ينفعُ، وخدمةُ من لا تُقرِّبُك خدمتُه إلى الله ولا تعودُ عليك بصلاح دُنياك، وخوفُك ورجاؤُك لمن ناصيته بيد الله وهو أسيرٌ في قبضته ولا يَملِك لنفسه ضرًّا ولا نفعًا ولا موتًا ولا حياةً ولا نشورًا .

وأعظمُ هذه الإضاعات إضاعتان هُما أصلُ كلِّ إضاعة: إضاعةُ القلب وإضاعةُ الوقت فإضاعة القلب من إيثار الدُّنيا على الآخرة، وإضاعةُ الوقت من طول الأمل.

فاجتمع الفسادُ كلُّه في اتباع الهوى وطول الأمل، والصلاحُ كلُّه في اتِّباع الهدى والاستعداد للِّقاء . والله المستعانُ .

ابن القيم | الفوائد صـ(١١١-١١٢)

Lecture 11: Epithelium: Characteristic features, basement membrane & intercellular junctions

Pre Exams & Formatives

1-	The basement membrane is best identified by? (FINAL 30, Lecture Qs)				
A-	pas	B-	orcein	C-	Toluidine blue
D-	Hx&E	E-	:AandB		

2-	The basement membrane is best identified by? (MID 30)				
A-	PAS	B-	Electron microscope	C-	Toluidine blue
D-	Hx andE	E-	both:aandb		

3-	Regarding epithelium all of the following are true EXCEPT? (MID 30)
A-	It lines body surfaces
B-	It is avascular
C-	It has basal lamina
D-	It contains relatively large amounts of intercellular substance
E-	both a ,c

4-	Hemi desmosome differs from spot desmosome in that hemi desmosome? (MID 30, FINAL 30)
A-	connects cell to cell
B-	connects extracellular matrix to extracellular matrix
C-	connects cell to extracellular matrix
D-	has tonofilament, made up of thick filaments
E-	Increases adhesiveness

5-	The cell junction which seals the intercellular spaces is the? (MID 30, FINAL 30)						
A-	Adherent junction	B-	Gap junction	C-	Tight junction	D-	Both B and C

6-	All of the following regarding tight junctions are true EXCEPT?(FINAL 30)
A-	are essential for ionic transport
B-	one type of junctional complexes
C-	prevent inter cellular fluid leakage
D-	Increases adhesiveness
E-	The cytoplasm of adjacent cells are rich in tonofilaments

7-	The statement about the general features of epithelial tissue is true EXCEPT?(FINAL 30)
A-	epithelia rest on basal lamina
B-	epithelia are avascular
C-	epithelia are derived only from ectoderm
D-	the epithelial cells are attached together by a special junctions
E-	bothb,c

8-	Cellular sealing? (FINAL 30)				
A-	Cell coat	B-	Caveolae	C-	Tight junction
D-	Cilia	E-	microvilli		

9-	The components that are not found between the epithelial cells is? (MID 31)				
A-	glycosaminoglycans	B-	nerve fibers	C-	blood capillaries
D-	desmosomes	E-	both:a&c		

10-	Which of the following prevents the passage of unwanted materials into the body via adjacent epithelial cells? (MID 31)				
A-	zonula adherens	B-	zonula occludens	C-	macula adherens
D-	macula occludens	E-	gap junction		

11-	What is a low resistance junction?(FINAL 31)				
A-	tight junction	B-	gap junction	C-	junctional epithelium
D-	junctional complex	E-	macula adherens		

12-	Concerning epithelium: ? (FINAL 32)		
A-	It is derived only from mesoderm	B-	It cannot regenerate
C-	It is vascular	D-	It is of different shape
E-	Both:a&d		

13-	Metaplasia: (2ND TRIAL 32)?
A-	A process of cell division
B-	Cell death
C-	Cell proliferation
D-	Change of type of cell to anther due to mechanical or chemical irritation

14-	Regarding epithelium, It? (FINAL 33)		
A-	Line body surfaces	B-	Is vascular
C-	Simple type has the character of polarity	D-	Both:a&c

15-	Gap junction consists of? (FINAL 34)				
A-	peripheral protein	B-	transmembrane protein	C-	tonofilaments

16-	Regarding Gap junction:? (MID 37)		
A-	Formed of phospholipid bilayer	B-	Formed of integral protein
C-	prevent intercellular diffusion	D-	intermediate junction

17-	Regarding Tight Junctions, which of the following is true? (FINAL SNU 1)
A-	They are primarily composed of phospholipids
B-	They facilitate the free movement of ions between adjacent cells
C-	They act as a barrier to prevent the paracellular exchange of substances
D-	Their intercellular matrix is composed of glycoprotein

18-	Collagen type four present in? (FINAL 37, Formative 36)				
A-	Basal lamina	B-	Bone	C-	Elastic cartilage

19-	Epithelial tissue is characterized by which of the following? (2ND TRIAL 37)		
A-	Highly vascularized	B-	Contains large intercellular spaces
C-	Traversed by numerous nerve endings	D-	Derived only from the embryonic ectoderm

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
A	E	D	C	C	A	C	C	E	B
11	12	13	14	15	16	17	18	19	20
B	D	D	D	B	B	C	A	C	-

Lecture Qs

1-	What connects the lamina densa to the subepithelial connective tissue? (Lecture Qs)						
A-	Laminalucida	B-	lamina reticular	C-	Anchoring fibrils	D-	cell membrane

2-	The statement about the general features of epithelial tissue is true EXCEPT. (Lecture Qs)
A-	epithelia rest on basal lamina
B-	epithelia is avascular
C-	epithelia are derived only from ectoderm
D-	the epithelial cells are attached together by a special junction

3-	**Epithelium: (Lecture Qs)**
A-	They only line body surfaces
B-	they are vascular
C-	they have basal lamina
D-	they contain relatively large amounts of intercellular substance

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
C	C	C	-	-	-	-	-	-	-

M3WAN Bank

1-	**Epithelial cells are characterized by: (M3WAN - Core)?**
A-	being widely spaced, with abundant intercellular substance
B-	rich vascularity
C-	derivation exclusively from mesoderm
D-	being closely packed, with minimal intercellular space

2-	**All are true about epithelium EXCEPT: (Important M3WAN - Hard)?**		
A-	it rests on a basal lamina	B-	it is derived exclusively from ectoderm
C-	cells are joined by special junctions	D-	it lines body surfaces/cavities

3-	**Regarding epithelium, all are true EXCEPT: (Important M3WAN - Hard)?**
A-	it contains large amounts of intercellular substance
B-	it lines body surfaces
C-	it is avascular
D-	it has a basal lamina

4-	**Regarding nerve supply, epithelial cells are: (M3WAN - Core)?**		
A-	completely devoid of nerves	B-	innervated only if stratified
C-	traversed by numerous nerve endings	D-	innervated only at the basement membrane

5-	**Epithelial polarity is best defined as: (M3WAN - Core)?**		
A-	the ability to regenerate	B-	the presence of junctions
C-	derivation from more than one germ layer	D-	cell functions directed toward one end

6-	A marker of polarity at the epithelial free surface is: (M3WAN - Core)?		
A-	a supranuclear nucleus	B-	basolateral secretory granules
C-	apical specializations	D-	g. microvilli
E-	a basally located Golgi		

7-	In a polarized epithelial cell, the Golgi apparatus is located: (M3WAN - Core)?		
A-	at the base of the cell	B-	in a supranuclear position
C-	with the nucleus at the apex	D-	outside the cell entirely

8-	Regarding biochemical polarity of membrane pumps: (M3WAN - Core)?
A-	H+ ATPase is apical, Na+/K+ ATPase basolateral
B-	Na+/K+ ATPase is apical, H+ ATPase basolateral
C-	both are distributed equally
D-	neither shows domain-specific distribution

9-	Metaplasia is best defined as: (M3WAN - Core)?		
A-	programmed cell death of epithelium	B-	an increase in cell layers
C-	loss of the basement membrane	D-	transformation of one epithelium into another

10-	By light microscopy, the basement membrane is: (Important M3WAN - Core)?		
A-	acidophilic with H&E	B-	PAS-positive
C-	best seen with Sudan black	D-	unstained (negative)

11-	By electron microscopy, the basement membrane shows: (M3WAN - Core)?		
A-	outer/inner dense layers, no light zone	B-	a single homogeneous layer
C-	lamina lucida and lamina densa	D-	pars fibrosa and pars granulosa

12-	The lamina densa is mainly formed of: (M3WAN - Core)?		
A-	collagen, proteoglycans, and laminin	B-	collagen fibers only
C-	keratin tonofilaments	D-	actin microfilaments

13-	The reticular lamina is best described as: (M3WAN - Core)?		
A-	part of the epithelial cell membrane	B-	formed exclusively by epithelium
C-	absent from all basement membranes	D-	reticular fibers from CT, near the basal lamina

14-	The collagen type present in the basal lamina is: (Important M3WAN - Core)?						
A-	Type IV	B-	Type I	C-	Type II	D-	Type III

15-	The tight junction (zonula occludens) is: (Important M3WAN - Core)?
A-	the least apical junction
B-	the most apical junction, sealing the space
C-	a junction with a 2 nm space, allowing ion movement
D-	responsible mainly for mechanical attachment

16-	All are true about tight junctions EXCEPT: (Important M3WAN - Hard)?		
A-	being one type of junctional complex	B-	preventing fluid leakage
C-	being essential for ionic transport	D-	being the most apical junction

17-	The zonula adherens is characterized by: (M3WAN - Core)?		
A-	being the strongest junction	B-	a 2 nm space with connexons
C-	being found only in cardiac muscle	D-	adhesive glycoprotein and a terminal web

18-	The macula adherens (desmosome) is: (Important M3WAN - Core)?
A-	the most apical junction
B-	allowing free ionic movement
C-	the strongest junction, at mechanical stress sites
D-	found only between endothelial cells

19-	The desmosome's attachment plaque is: (M3WAN - Core)?
A-	a dense discoid material, into which filaments insert
B-	a fluid-filled intercellular space
C-	a layer of proteoglycans and laminin
D-	a network of actin microfilaments

20-	A hemidesmosome differs from a desmosome in that it: (Important M3WAN - Hard)?		
A-	connects cell to cell	B-	connects the cell to the ECM
C-	is formed of thick filaments	D-	is the most apical junction

21-	The gap junction (nexus) is structurally: (Important M3WAN - Core)?		
A-	a wide space with adhesive glycoprotein	B-	an attachment plaque with looping filaments
C-	the most apical junctional component	D-	a 2 nm space, with connexons (ion channels)

22-	Functionally, the gap junction represents: (M3WAN - Core)?
A-	a barrier preventing any exchange
B-	low resistance, rapid impulse transmission (e.g. cardiac muscle)
C-	a junction for mechanical strength only

D-	a structure exclusive to stratified epithelium

23-	**Depending on its site, epithelium performs: (M3WAN - Core)?**
A-	protection exclusively
B-	secretion exclusively
C-	protection, absorption, secretion, filtration, diffusion
D-	absorption exclusively

24-	**Regarding renewal of epithelial cells: (M3WAN - Core)?**
A-	continual degeneration, replaced at varying rates
B-	only the basal layer regenerates
C-	epithelial cells are permanent
D-	replacement occurs only after injury

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	B	A	C	D	C	B	A	D	B
11	12	13	14	15	16	17	18	19	20
C	A	D	A	B	C	D	C	A	B
21	22	23	24	25	26	27	28	29	30
D	B	C	A	-	-	-	-	-	-

قال ابن الجوزي – رحمه الله: -

السعيدُ من ذل لله، وسأل العافية؛ فإنه لا يُوهَبُ العافية على الإطلاق؛ إذ لا بدَّ من بلاء، فلا ينزال العاقل يسأل العافية؛ لتغلب على جُمهُورِ أحواله، فيُقرِّبُ الصبر على يسير البلاء..

وفي الجملة؛ ينبغي للإنسان أن يعلم أنه لا سبيل الى محبوباته خالصةً، ففي كُلِّ جُرعةٍ غُصَصٌ، وفي كُلِّ لُقمةٍ شجًا..

وعلى الحقيقة؛ ما الصبرُ الا على الاقدار، وقلَّ أن تجري الأقدارُ إلا على خلافِ مُرادِ النَّفس؛ فالعاقل من داري نفسه في الصَّبرِ بوعدِ الأجر، وتسهيلِ الأمر؛ ليذهبَ زمانُ البلاء سالمًا من شكوى، ثم يستغيثُ بالله تعالى سائلًا العافية.

صيد الخاطر (٢٩٦)

Lecture 12: Epithelium: Apical cell specializations and covering & lining epithelium

Pre Exams & Formatives

1-	**Transitional epithelium is found at? (MID 30)**						
A-	The gall bladder	B-	The urinary tract	C-	The Skin	D-	The esophagus

2-	**Pseudo-stratified epithelium is one type of? (MID 30, Formative 36)**				
A-	simple epithelium	B-	glandular epithelium	C-	neuro-epithelium
D-	stratified epithelium	E-	germinal epithelium		

3-	**Simple columnar epithelium is found at? (FINAL 30)**						
A-	the gall bladder	B-	the urinary tract	C-	the Skin	D-	the esophagus

4-	**Taste buds epithelium is one type of? (FINAL 30)**				
A-	simple epithelium	B-	glandular epithelium	C-	neuro-epithelium
D-	stratified epithelium	E-	germinal epithelium		

5-	**The cornea is covered with keratinized stratified squamous epithelium? (FINAL 30)**		
A-	True	B-	False

6-	**Motility? (FINAL 30)**				
A-	Cell coat	B-	Caveolae	C-	Tight junction
D-	Cilia	E-	microvilli		

7-	**Endocytosis? (FINAL 30)**				
A-	Cell coat	B-	Caveolae	C-	Tight junction
D-	Cilia	E-	microvilli		

8-	**What is the type of epithelial tissue that lines the body cavities? (MID 31)**		
A-	Stratified columnar epithelium	B-	Simple squamous epithelium
C-	Stratified squamous epithelium	D-	Simple cubical epithelium
E-	Simple coloumnar epithelium		

9-	**Which epithelium is characteristic of most of the conducting passages of the respiratory system? (MID 31)**
A-	Stratified squamous non-keratinized epithelium
B-	Pseudo stratified columnar ciliated epithelium
C-	Transitional epithelium
D-	Stratified columnar epithelium

E-	Both B and C

10-	**Which of the following epithelial tissue types is best adapted for the rapid transport of materials across its membranes? (FINAL 31)**		
A-	Transitional	B-	Stratified squamous
C-	Simple epithelium	D-	Pseudostratified ciliated columnar
E-	Stratified cuboidal		

11-	**Which of the following types of epithelial tissue provides the greatest amount of protection? (FINAL 31)**		
A-	Stratified columnar	B-	Simple squamous
C-	Stratified squamous	D-	Pseudostratified ciliated columnar
E-	Transitional		

12-	**What forms the brush border? (FINAL 31)**				
A-	Microvilli	B-	Desmosomes	C-	Cilia
D-	Keratinization	E-	Both:a&b		

13-	**The epithelial lining of thyroid follicle is? (FINAL 32)**				
A-	Stratified squamous	B-	Stratified columnar	C-	Transitional
D-	Simple cuboidal	E-	Simple squamous		

14-	**Regarding the shaft of the cilia all of the following are true EXCEPT? (FINAL 32)**
A-	It is covered with plasma membrane
B-	Contains nine peripheral doublet of microtubules
C-	It contains central single microtubule
D-	Motile
E-	Both:b&c

15-	**The microvilli: (2ND TRIAL 32)?**		
A-	Are Present in the absorptive cells	B-	Contains microtubules
C-	Contains microfilaments	D-	Both: a&c

16-	**Cuticle? (FINAL 33)**
A-	Is lipid rich substance
B-	Increase absorptive surface
C-	Deposited in the inner lamina of the plasma membrane
D-	Present in transitional epithelium

17-	Axoneme of the cilia? (FINAL 33)		
A-	Rich in actin filaments	B-	Contains microtubules
C-	Increase absorptive surface	D-	Present in the basal part of the cell

18-	One of the following is true about microvilli? (FINAL 34)		
A-	consists of microtubule	B-	resulted from endocytosis
C-	increase absorptive area	D-	motile

19-	One of the following is a simple columnar epithelium? (FINAL 34)		
A-	skin	B-	cardia of the stomach
C-	urethra	D-	urinary bladder

20-	Epithelium in parietal layer of bowman's capsule? (FINAL 37)		
A-	Simple squamous	B-	Simple cuboidal
C-	Simple Columnar	D-	Stratified squamous

21-	Epithelium of the skin is (Formative 36)?		
A-	Non keratinized stratified	B-	Keratinized stratified squamous
C-	simple squamous	D-	Stratified columnar

22-	What structure is at the base of cilia? (Formative 37)						
A-	Centrioles	B-	Microfilaments	C-	Peroxisome	D-	Ribosomes

23-	Which of the following is true about stratified squamous epithelium? (2ND TRIAL 37)		
A-	Superficial cells are thin squamous	B-	Contains only one cell layer
C-	Apical cells are columnar	D-	Present in the lining of the stomach

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
B	A	A	C	B	D	B	B	B	C
11	12	13	14	15	16	17	18	19	20
C	A	D	C	D	D	B	C	B	A
21	22	23	24	25	26	27	28	29	30
B	A	A	-	-	-	-	-	-	-

Lecture Qs

1-	**Microvilli of epithelial cells: (Lecture Qs)**
A-	are restricted in occurrence to the lining of digestive system
B-	are not visible with light microscope
C-	develop into cilia
D-	increase the absorptive capacity of the cells
E-	are motile appendages

2-	**Microvilli have structural support formed of: (Lecture Qs)**						
A-	Actin filaments	B-	Microtubules	C-	Thick filaments	D-	keratin filament

3-	**Simple squamous epithelium lines all of the following structures EXCEPT: (Lecture Qs)**		
A-	the alveoli of the lung	B-	parietal layer of Bowman's capsule
C-	distal convoluted tubule	D-	lymphatic vessels

4-	**The epithelium which is well adapted for exchange of gases and metabolites is: (Lecture Qs)**		
A-	The simple squamous type	B-	The simple columnar type
C-	The stratified epithelium	D-	The simple cuboidal type

5-	**All of the following are parts of cilia EXCEPT: (Lecture Qs)**						
A-	Axoneme	B-	Basal body	C-	Basal lamina	D-	Rootlet

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
D	A	C	A	C	-	-	-	-	-

M3WAN Bank

1-	**By light microscopy, cilia appear as: (M3WAN - Core)?**		
A-	immotile stereocilia	B-	motile hair-like processes
C-	rigid protein spikes	D-	membrane invaginations

2-	**The ciliary axoneme is formed of: (Important M3WAN - Core)?**		
A-	9 triplets, no central pair	B-	actin filaments, 9+2 pattern
C-	9 single peripherals, no center	D-	9 doublets + 2 central singlets

3-	The basal body of the cilium is: (M3WAN - Core)?		
A-	9 triplet microtubules, from a centriole	B-	9 doublets + 2 central singlets
C-	a protein plaque with filaments	D-	absent in motile cilia

4-	The rootlet of the cilium functions to: (M3WAN - Core)?		
A-	synthesize ATP	B-	cover the shaft
C-	anchor the basal body to the cytoplasm	D-	form microvilli

5-	The main function of cilia is to: (M3WAN - Core)?		
A-	increase absorptive area	B-	beat, pushing fluids/particles
C-	seal the intercellular space	D-	form the mitotic spindle

6-	By light microscopy, microvilli give the surface? (Important M3WAN - Core)		
A-	smooth appearance	B-	scalloped/wavy appearance
C-	isolated, sparse projections	D-	brush-like appearance

7-	By electron microscopy, microvilli consist of: (M3WAN - Core)?		
A-	9+2 microtubules	B-	9 triplets, from a centriole
C-	fine filaments plus a thick glycocalyx	D-	a hollow actin cylinder, no coat

8-	The main function of microvilli is to: (Important M3WAN - Core)?		
A-	increase the absorptive surface area	B-	seal the intercellular space
C-	provide motility	D-	anchor the cell to the basal lamina

9-	Caveolae are best described as: (Important M3WAN - Core)?		
A-	motile microtubule processes	B-	protective protein deposits
C-	attachment plaques	D-	invaginations from endo-/exocytosis

10-	The cuticle is best described as: (Important M3WAN - Core)?		
A-	lipid-rich, increasing absorption	B-	a protein substance protecting cells
C-	g. urinary passages	D-	formed of 9+2 microtubules
E-	exclusive to respiratory epithelium		

11-	Covering epithelium is named according to: (M3WAN - Core)?		
A-	germ layer origin only	B-	basement membrane presence only
C-	layer number, cell shape, apical specialization	D-	vascularity only

12-	Simple squamous epithelium is characterized by: (Important M3WAN - Core)?		
A-	thin fusiform cells	B-	g. Bowman's capsule
C-	tall columnar cells, digestive tract	D-	cuboidal cells, kidney tubules
E-	multiple layers, squamous surface		

13-	Regarding endothelium and mesothelium: (M3WAN - Core)?
A-	endothelium lines body cavities
B-	both are stratified squamous
C-	endothelium is simple columnar
D-	endothelium lines vessels; mesothelium lines cavities

14-	Simple cuboidal epithelium is found lining: (Important M3WAN - Core)?		
A-	the digestive tract, cardia to anus	B-	thyroid follicles and kidney tubules
C-	the urinary tract	D-	the trachea and bronchi

15-	Simple columnar epithelium is found lining: (Important M3WAN - Core)?		
A-	the trachea and bronchi	B-	Bowman's capsule
C-	the digestive tract, cardia to anus	D-	the urinary bladder

16-	The ciliated variant of simple columnar epithelium lines: (M3WAN - Core)?		
A-	the uterus, paranasal sinuses, spinal canal	B-	the trachea and bronchi
C-	the oral cavity and esophagus	D-	the skin epidermis

17-	Pseudostratified columnar epithelium is best described as: (Important M3WAN - Core)?
A-	all layers reaching the surface
B-	lining the skin epidermis
C-	a single layer of squamous cells
D-	all cells on the lamina, few reach the surface (e.g. trachea)

18-	Stratified squamous (non-keratinized) epithelium is found in: (Important M3WAN - Core)?		
A-	kidney tubules and thyroid	B-	the trachea and bronchi
C-	the oral cavity, esophagus, cornea, vagina	D-	the urinary tract

19-	Keratinized stratified squamous epithelium is NOT found in the: (Important M3WAN - Hard)?						
A-	epidermis	B-	cornea	C-	palm	D-	sole

20-	Stratified columnar epithelium is found in the: (M3WAN - Core)?		
A-	conjunctival fornix, urethra, pharynx	B-	digestive tract

C-	trachea and bronchi	D-	epidermis

21-	The ciliated variant of stratified columnar epithelium is found in the: (M3WAN - Core)?		
A-	digestive tract	B-	epidermis
C-	urinary bladder	D-	soft palate, larynx, fetal esophagus

22-	Regarding the structure of transitional epithelium: (Important M3WAN - Core)?
A-	layers stay constant regardless of distension
B-	superficial cells are always squamous
C-	many layers when empty, fewer when distended
D-	it is a single cuboidal layer

23-	Transitional epithelium is found lining the: (Important M3WAN - Core)?		
A-	digestive tract	B-	urinary tract, calyx to urethra
C-	trachea and bronchi	D-	oral cavity

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	D	A	C	B	D	C	A	D	B
11	12	13	14	15	16	17	18	19	20
C	A	D	B	C	A	D	C	B	A
21	22	23	24	25	26	27	28	29	30
D	C	B	-	-	-	-	-	-	-

إخواني ! تأهبوا ليومٍ ترادف فيه العَبَرات، وتعظم الحَسَرات، فَيعضُّ الظالم على يديه ويقول : يا حسرتا !

يوم يقول لك: أين من أرضيتَ عنك بغضبي عليك ؟

يا ابن آدم أين من كنتَ تَزيَّنتَ له وبالقبيح بارزتني ؟

ما هذا التذلل بين يديّ وقد كنت جبّاراً عنيداً ؟ ! . . طالما ذُكّرْتَ بموقفك هذا فتناسيتَ، وطالما بُصّرْتَ بأمرك

هذا فتعاميتَ، ولم تزدد إلاّ فراراً !..

يا حسرةَ العاصين، يا ذُلّ مقام المتجبرين، واخيبة المضطرين، واخسارة المُسرِفِيْنْ.

ابن الجوزي | المنثور

Lecture 13: Secretory epithelium & glands

Pre Exams & Formatives

1-	**Compound tubule alveolar gland is? (MID 30, FINAL 30)**				
A-	Salivary gland	B-	dudenal gland	C-	Sweat gland
D-	Pancreas	E-	pituitary gland		

2-	**Steroid forming cells are rich in? (MID 31)**				
A-	R'E'R	B-	Lipid droplets	C-	S'E'R
D-	both:aandb	E-	both:b&c		

3-	**Which of the following is a type of secretion in which some cytoplasm is lost with the product & is found in the mammary glands? (FINAL 31)**				
A-	Holocrine	B-	Merocrine	C-	Apocrine
D-	Mucous	E-	None of these		

4-	**What is a gland called if the secretory portion is tube shaped ending in a flask shaped area at the terminus? (FINAL 31)**				
A-	simple gland	B-	compound gland	C-	tubular
D-	alveolar	E-	tubule alveolar		

5-	**Adrenal gland is? (FINAL 32)**				
A-	unicellular glands	B-	exocrine gland	C-	rich in blood supply
D-	endocrine gland	E-	both:c&d		

6-	**An example of apocrine glands is: (2ND TRIAL 32)?**						
A-	mammary gland	B-	sebaceous glands	C-	goblet cells	D-	salivary glands

7-	**Sebaceous glands are? (FINAL 33)**		
A-	Apocrine	B-	Merocrine
C-	Simple branched acinar	D-	Compound tubulo acinar

8-	**One is true about apocrine? (FINAL 34)**		
A-	an example is mammary gland	B-	an example is sebaceous gland
C-	eccrine sweat gland		

9-	**Mode of secretion of Sebaceous gland is? (2ND TRIAL 35)**				
A-	Apocrine	B-	Holocrine	C-	Merocrine

10-	The Pancreas is classified as? (FINAL SNU 1)				
A-	Mixed gland	B-	Endocrine gland	C-	Exocrine gland

11-	What is the type of Mammary gland?(FINAL 37)				
A-	Holocrine	B-	Merocrine	C-	Apocrine

12-	Type of exocrine gland of pancreas is: (Formative 36)?		
A-	compound acinar & serous acini	B-	serous acini
C-	compound acinar	D-	non of the above

13-	A characteristic of endocrine glands is that they: (2ND TRIAL 37)		
A-	Secrete their hormones directly into the blood	B-	Have extensive duct systems
C-	Secrete their products onto epithelial surfaces	D-	Are completely avascular

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
A	E	C	E	E	A	C	A	B	A
11	12	13	14	15	16	17	18	19	20
C	C	A	-	-	-	-	-	-	-

Lecture Qs

1-	Secretory material from adjacent cells is accumulated in channels called: (Lecture Qs)						
A-	caveoles	B-	tight junction	C-	gap junction	D-	canaliculi

2-	What is a gland called if the secretory portion is flask shaped? (Lecture Qs)						
A-	Compound gland	B-	Tubular	C-	Alveolar	D-	Tubuloalveolar

3-	Goblet cells of intestinal epithelium are examples of: (Lecture Qs)		
A-	unicellular glands	B-	simple columnar epithelium
C-	stratified epithelium	D-	pseudostratified epithelium

4-	Simple tubular gland is: (Lecture Qs)						
A-	Salivary gland	B-	Brunners gland	C-	Sweat gland	D-	Intestinal gland

5-	Holocrine glands: (Lecture Qs)
A-	some cells are destroyed, to form the secretion
B-	no change occur in the secretory cells

C-	the apex of the cell is destroyed during secretion
D-	mamary gland

6-	Which of the following is a multicellular gland? (Lecture Qs)						
A-	Salivary gland	B-	Goblet cell	C-	Basal cell	D-	Basket cell

7-	Exocrine glands are defined as compound glands when the excretory ducts are: (Lecture Qs)		
A-	Branched	B-	Unbranched
C-	Draining mucous and serous acini	D-	tall

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
D	C	A	D	A	A	A	-	-	-

M3WAN Bank

1-	Glands arise embryologically by: (M3WAN - Core)?		
A-	mesenchymal differentiation only	B-	neural crest migration
C-	CT cell migration into epithelium	D-	epithelial proliferation plus CT invasion

2-	Exocrine glands are best defined as glands that: (M3WAN - Core)?		
A-	are ductless, blood-transported	B-	retain a surface connection, with ducts
C-	combine ductless/ducted equally	D-	occur exclusively in the digestive tract

3-	Endocrine glands are best defined as: (M3WAN - Core)?		
A-	glands with a secretory portion and ducts	B-	glands composed of a single cell
C-	ductless glands, blood-transported secretion	D-	glands classified by secretory unit shape

4-	A classic mixed gland example is the: (Important M3WAN - Core)?		
A-	pancreas (exocrine plus endocrine islets)	B-	parotid, purely exocrine
C-	pituitary, purely endocrine	D-	thyroid, purely exocrine

5-	A unicellular gland is best exemplified by the: (Important M3WAN - Core)?		
A-	numerous cells of an acinus	B-	salivary glands
C-	goblet cell of the intestine	D-	ductless gland

6-	A multicellular gland is best exemplified by the: (M3WAN - Core)?		
A-	single cell	B-	goblet cell exclusively

C-	endocrine gland only	D-	salivary glands

7-	**Regarding gland classification by secretion nature: (M3WAN - Core)?**
A-	parotid is serous, esophageal is mucous, submandibular is seromucous
B-	parotid is mucous, esophageal is serous
C-	sebaceous is a mucous gland
D-	sweat gland is seromucous

8-	**Glands "of special nature" include: (M3WAN - Core)?**
A-	only sweat glands
B-	mammary, ceruminous, sebaceous, lacrimal, sweat
C-	exclusively endocrine glands
D-	glands secreting exclusively mucus

9-	**In the merocrine mode of secretion: (M3WAN - Core)?**
A-	the entire cell disintegrates
B-	exocytosis occurs, without membrane loss (eccrine sweat)
C-	apical cytoplasm is pinched off
D-	the secretory portion is a single cell

10-	**The apocrine mode of secretion e. g. the mammary gland: (Important M3WAN - Core)?**
A-	pinches off lipid droplet plus cytoplasm
B-	releases product by simple exocytosis
C-	involves breakdown of the entire cell
D-	occurs by diffusion across the membrane

11-	**The holocrine mode of secretion e.g. the sebaceous gland: (Important M3WAN - Core)?**
A-	exocytosis, no membrane loss
B-	apical cytoplasm pinched off
C-	simple diffusion
D-	entire cell breaks down, lipid-rich outpour

12-	**A simple tubular gland is best exemplified by: (M3WAN - Core)?**
A-	a coiled tubule, unbranched duct
B-	a bulbous portion, unbranched duct
C-	intestinal mucosa glands: straight tubule, no duct
D-	a bifurcated portion, short duct

13-	**A simple coiled tubular gland is exemplified by: (M3WAN - Core)?**		
A-	sweat glands: coiled tubule, unbranched duct	B-	a straight tubule, no duct
C-	acini clustered around a duct	D-	tubuloalveoli attached to a duct

14-	**A simple branched tubular gland is exemplified by: (M3WAN - Core)?**
A-	a single unbranched acinus
B-	sweat glands
C-	oral cavity/esophageal glands: bifurcated, short duct
D-	acini clustered around one duct

15-	**A simple acinar (alveolar) gland is exemplified by: (M3WAN - Core)?**
A-	a straight tubule, no duct
B-	penile urethra glands: bulbous portion, unbranched duct
C-	acini around a branching duct
D-	tubuloalveoli, branching duct

16-	**A simple branched acinar gland is exemplified by: (Important M3WAN - Core)?**
A-	a single unbranched tubule
B-	a coiled tubule, unbranched duct
C-	tubuloalveoli attached to a single duct
D-	sebaceous glands: acini clustered around a duct

17-	**A simple branched tubulo-acinar gland is exemplified by: (M3WAN - Core)?**
A-	a single straight tubule, no duct
B-	an unbranched acinus, opens on surface
C-	minor salivary glands: tubuloalveoli attached to a duct
D-	the pancreas, with repeatedly branching duct

18-	**A compound tubular gland is exemplified by gastric and bulbourethral glands, characterized by: (M3WAN - Core)?**		
A-	a single unbranched tubule, no duct	B-	an acinus with a branching duct
C-	tubuloalveoli, unbranched duct	D-	a repeatedly branching duct system

19-	**A compound acinar gland is exemplified by the: (Important M3WAN - Core)?**		
A-	major salivary glands	B-	pancreas
C-	mammary gland	D-	sweat glands

20-	A compound tubulo-acinar gland is exemplified by the: (Important M3WAN - Core)?		
A-	major salivary glands	B-	pancreas exclusively
C-	sebaceous glands	D-	simple unbranched duct

21-	A compound saccular gland e.g. mammary/prostate, is characterized by: (M3WAN - Core)?
A-	gastric-like secretory units
B-	large units, several ducts opening separately
C-	minor salivary-like units
D-	a single main duct

22-	Endocrine hormones are: (M3WAN - Core)?		
A-	blood-carried, with sustained effects	B-	released directly onto an epithelial surface
C-	acting only on adjacent cells	D-	requiring a duct system

23-	Endocrine glands are characteristically: (M3WAN - Core)?		
A-	poorly vascular	B-	reliant on diffusion alone
C-	served by non-fenestrated capillaries	D-	richly vascular, with fenestrated capillaries

24-	Regarding endocrine cell ultrastructure: (Important M3WAN - Hard)?
A-	polypeptide cells have SER/lipid; steroid cells have RER/granules
B-	both types are identical
C-	polypeptide cells: RER/granules at vascular pole; steroid cells: SER/lipid, large Golgi
D-	neither cell type has a Golgi apparatus

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	B	C	A	C	D	A	B	B	A
11	12	13	14	15	16	17	18	19	20
D	C	A	C	B	D	C	D	B	A
21	22	23	24	25	26	27	28	29	30
B	A	D	C	-	-	-	-	-	-

عن كُرْزِ بنِ وَبَرَةَ، قال: بَلَغَني أنَّ كَعْبًا قال:

«إنَّ الملائكةَ يَنظُرونَ من السَّماءِ إلى الَّذينَ يُصلُّونَ باللَّيلِ، كما تَنظُرونَ أنتُم إلى نجومِ السَّماءِ»

ابنُ الجوزي | قِيامُ اللَّيل.

Lecture 14: Connective tissue: Cells

Pre Exams & Formatives

1-	All of these cells are free connective tissue cells EXCEPT? (MID 30)				
A-	Macrophages	B-	plasma cells	C-	microphages
D-	mast cells	E-	Erythroblast		

2-	All of these cells are freeT. cells EXCEPT? (FINAL 30)				
A-	plasma cells	B-	fat cells	C-	mast cells
D-	macrophages	E-	leucocytes		

3-	Which cell produces heparin? (FINAL 30)				
A-	Fibroblasts	B-	Lymphocytes	C-	Mast cells
D-	macrophages	E-	plasma cells		

4-	Plasma cell? (FINAL 30)				
A-	Antigen presenting cell	B-	Immunity	C-	Growth
D-	Differentiation	E-	remodeling		

5-	Macrophage? (FINAL 30)				
A-	Antigen presenting cell	B-	Immunity	C-	Growth
D-	Differentiation	E-	remodeling		

6-	Cell that is most often found along capillaries and resembles fibroblast is? (MID 31)				
A-	plasma cell	B-	lymphocyte	C-	macrophage
D-	pericyte	E-	mesenchymal cell		

7-	Mast cells contain? (MID 31)
A-	mitochondria arranged in a spoke-wheel (cart-wheel) pattern
B-	deeply stained basophilic granules
C-	large Golgi
D-	many primary lysosomes
E-	both:b&d

8-	Cells that remove damaged cells or pathogens from connective tissue are?(FINAL 31)				
A-	Fibroblasts	B-	Adipocytes	C-	Melanocytes
D-	Macrophages	E-	Mast cells		

9-	Which connective tissue cell is a tissue macrophage? (FINAL 31)				
A-	Fibroblast	B-	Myofibroblast	C-	Histiocyte
D-	Plasma cell	E-	Mast cell		

10-	Which of the following describes a neutrophil? (FINAL 31)
A-	abundant, agranular, especially effective against cancer cells
B-	abundant, granular, especially effective against bacteria
C-	rare, agranular, releases antimicrobial defensins
D-	rare, granular, contains multiple granules packed with histamine
E-	abundant, granular contains specific granules likely to be primary lysosomes

11-	A patient has been experiencing severe, anaphylactic shock symptoms that are reduced when she takes an antihistamine. Before the treatment, this patient was likely to have had increased activity of which leukocyte? (FINAL 31)				
A-	basophils	B-	neutrophils	C-	monocytes
D-	natural killer cells	E-	T'lymphocytes		

12-	Which cell has large blue granules, often obscuring the nucleus?(FINAL 31)								
A-	neutrophil	B-	lymphocyte	C-	monocytes	D-	eosinophil	E-	basophils

13-	How many percent of all leucocytes in healthy individuals are lymphocytes?(FINAL 31)								
A-	0.5%	B-	30%	C-	5%	D-	3%	E-	75%

14-	Blood leucocytes are ? (FINAL 32)
A-	not true cells
B-	Their increase in number is called Leucopenia
C-	capable of amoeboid movement
D-	Their total number varies from 4-11 million/ml
E-	None of the above

15-	Cell associated with parasitic infestation: ? (FINAL 32)								
A-	neutrophil	B-	eosinophil	C-	basophil	D-	monocyte	E-	lymphocyte

16-	All of these cells are connective tissue cells EXCEPT: ? (FINAL 32)								
A-	Pericytes	B-	fibroblasts	C-	plasma cells	D-	fat cells	E-	Goblet cells

17-	Antibodies are secreted by? (2ND TRIAL 32, Formative 36)		
A-	mast cells	B-	plasma cells

C-	fat cells	D-	Mesenchymal cells

18-	**Blood platelets: (2ND TRIAL 32)?**
A-	Release growth factors which delay healing wounds
B-	Have central rounded nucleus
C-	Granulomere represent chromtin
D-	Hyalomere is the peripheral part

19-	**The C'T' cell which contains many lysosomes is: (2ND TRIAL 32)?**						
A-	mast cell	B-	fibroblast	C-	plasma cell	D-	macrophage

20-	**Eosinophils associated with? (FINAL 33)**		
A-	Parasitic infestation	B-	Acute bacterial infection
C-	Antibody production	D-	None of the above

21-	**Fibroblasts secrete? (FINAL 33)**		
A-	Collagen fibers only	B-	Antibodies
C-	Extracellular protein substances	D-	Heparin

22-	**Monocytes? (FINAL 33)**
A-	The largest cell in blood film
B-	Have multilobed nucleus
C-	Have granules contain mediators of anaphylactic shock
D-	None of the above

23-	**Origin of macrophage is? (FINAL 33)**		
A-	Monocyte	B-	Bone marrow
C-	Mesenchymal stem cell	D-	Non of the above

24-	**About basophil? (FINAL 34)**		
A-	secrete mediators cause anaphylactic shock	B-	rounded nucleus
C-	has only azurophilic granules	D-	largest cell in blood film

25-	**All are true about pericytes except? (FINAL 34)**
A-	associated with the endothelium of blood vessels
B-	mesenchymal in origin
C-	precursor of plasma cell

26-	Which of these releases cytokines? (FINAL 34)						
A-	fibroblast	B-	mesenchymal cell	C-	plasma cell	D-	macrophage

27-	Mast cell is? (2ND TRIAL 35, Formative 36, Formative 37)		
A-	Branched and phagocytic	B-	Can differentiate into other
C-	T-cells	D-	Secrete heparin and histamine
E-	Rich in melanin pigment		

28-	Foreign body giant cells are formed by: (2ND TRIAL 37)		
A-	Fusion of macrophages	B-	Fusion of plasma cells
C-	Division of mast cells	D-	Maturation of lymphocytes

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
E	B	C	B	A	D	E	D	C	B
11	12	13	14	15	16	17	18	19	20
A	E	B	C	B	E	B	A	D	A
21	22	23	24	25	26	27	28	29	30
C	A	A	A	C	D	D	-	-	-

Lecture Qs

1-	Electron dense granules are present in: (Lecture Qs)		
A-	mast cells	B-	plasma cells
C-	fat cells	D-	reticulin (reticular) cells

2-	All of these cells are connective tissue cells EXCEPT: (Lecture Qs)		
A-	macrophages	B-	fibroblasts
C-	plasma cells	D-	red blood corpuscles

3-	All of these cells are free C'T. cells EXCEPT: (Lecture Qs)				
A-	plasma cells	B-	fibroblasts	C-	mast cells
D-	macrophages	E-	Leucocytes		

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	D	B	-	-	-	-	-	-	-

M3WAN Bank

1-	Regarding fixed and free CT cells: (M3WAN - Core)?
A-	fixed cells are short-lived, blood-derived
B-	both types are equally long-lived
C-	free cells include fibroblasts/adipocytes
D-	fixed cells are long-lived; free cells are short-lived, blood-derived

2-	The fixed cells of connective tissue include: (Important M3WAN - Core)?		
A-	mast, plasma, macrophage, leukocyte	B-	fibroblast, fat cell, mesenchymal cell, pericyte
C-	fibroblast and macrophage only	D-	lymphocyte and neutrophil only

3-	The free cells of connective tissue include: (Important M3WAN - Core)?
A-	mast, plasma, macrophage, giant cell, leukocyte
B-	fibroblast and fat cell only
C-	pericyte and mesenchymal cell only
D-	osteoblast and osteoclast only

4-	Mesenchymal cells are best described as: (M3WAN - Core)?		
A-	multinucleated, bone marrow only	B-	spindle-shaped, acidophilic, pyknotic nucleus
C-	branched, pale basophilic, origin of CT cells	D-	non-branched, cartilage only

5-	Regarding fibroblasts and fibrocytes: (M3WAN - Core)?
A-	fibroblasts arise from macrophages
B-	fibroblasts are terminally differentiated
C-	fibrocytes are the young, active form
D-	fibroblasts arise from mesenchyme; fibrocytes are old

6-	The basophilic cytoplasm of active fibroblasts is due to: (Important M3WAN - Core)?		
A-	ribosomes and rough ER	B-	abundant lysosomes
C-	abundant smooth ER	D-	being acidophilic, not basophilic

7-	All are fibroblast functions EXCEPT: (Important M3WAN - Hard)?		
A-	tropocollagen secretion	B-	elastin/microfibril secretion
C-	antibody secretion	D-	wound repair

8-	The adipocyte is structurally characterized by: (M3WAN - Core)?		
A-	a branched cell, basophilic cytoplasm	B-	a spherical cell, flattened nucleus, signet-ring

C-	a multinucleated, ruffled-border cell	D-	RER-filled cytoplasm

9-	**Pericytes are best characterized as cells that: (Important M3WAN - Core)?**
A-	are found only in lymphoid organs
B-	are incapable of division
C-	are precursors of plasma cells only
D-	lie on capillaries, source new fibroblasts/SM cells

10-	**Macrophages of connective tissue arise from: (Important M3WAN - Core)?**		
A-	osteoprogenitor cells	B-	mesenchymal cells directly
C-	monocytes of the blood	D-	lymphocytes

11-	**Special forms of macrophages include: (M3WAN - Core)?**		
A-	Kupffer cells, microglia, osteoclasts	B-	plasma cells and mast cells
C-	fibrocytes in the CNS	D-	pericytes in the liver

12-	**Macrophages are identified by light microscopy through: (M3WAN - Core)?**		
A-	a clock-face nucleus	B-	a kidney-shaped nucleus, Trypan blue uptake
C-	basophilic granules	D-	silver staining only

13-	**By electron microscopy, macrophages show: (M3WAN - Core)?**
A-	smooth ER, no lysosomes
B-	a negative Golgi, RER only
C-	spoke-wheel mitochondria
D-	phagosomes, lysosomes, RER, Golgi, mitochondria

14-	**The functions of macrophages include: (Important M3WAN - Core)?**
A-	phagocytosis only
B-	phagocytosis, cytokine release, antigen presentation
C-	antibody secretion only
D-	histamine/heparin secretion

15-	**Foreign body giant cells are best described as: (M3WAN - Core)?**		
A-	derived directly from lymphocytes	B-	formed by fusion of fibroblasts
C-	formed by fusion of macrophages/monocytes	D-	arising from mast cell degranulation

16-	**Leukocytes present in loose CT: (M3WAN - Core)?**		
A-	migrate from capillaries/venules	B-	are exclusively erythrocytes

C-	are exclusively platelets	D-	are permanently resident, not blood-derived

17-	Plasma cells arise in the connective tissue from: (Important M3WAN - Core)?		
A-	monocytes	B-	mesenchymal cells directly
C-	fibroblasts	D-	Blymphocytes

18-	By light microscopy, the plasma cell shows: (Important M3WAN - Core)?
A-	a small, spindle-shaped body
B-	multiple nuclei, ruffled border
C-	an eccentric nucleus, basophilic cytoplasm, negative Golgi
D-	histamine/heparin granules

19-	By electron microscopy, the plasma cell shows: (M3WAN - Core)?		
A-	smooth ER, sparse ribosomes	B-	RER cisternae, clock-face nucleus
C-	phagosomes, multivesicular bodies	D-	a single large lipid droplet

20-	The main function of the plasma cell is: (Important M3WAN - Core)?		
A-	continuous immunoglobulin secretion	B-	histamine/heparin secretion
C-	phagocytosis	D-	tropocollagen secretion

21-	Mast cells originate from: (M3WAN - Core)?		
A-	monocytes	B-	Blymphocytes
C-	mesenchymal cells directly	D-	bone marrow stem cells

22-	By light microscopy, the mast cell is: (Important M3WAN - Core)?		
A-	the largest CT cell, basophilic granules	B-	a small cell, clock-face nucleus
C-	a cell with signet-ring appearance	D-	identified by Trypan blue uptake

23-	By electron microscopy, the mast cell shows: (M3WAN - Core)?		
A-	abundant RER, large Golgi	B-	ER, small Golgi, few mitochondria
C-	phagosomes, dense bodies	D-	a single large lipid droplet

24-	The main function of the mast cell is: (Important M3WAN - Core)?		
A-	immunoglobulin secretion	B-	phagocytosis, antigen presentation
C-	heparin and histamine secretion	D-	tropocollagen/elastin secretion

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	B	A	C	D	A	C	B	D	C
11	12	13	14	15	16	17	18	19	20
A	B	D	B	C	A	D	C	B	A
21	22	23	24	25	26	27	28	29	30
D	A	B	C	-	-	-	-	-	-



عن ابنِ عبَّاسٍ؛ أنَّ رسولَ اللهِ ﷺ كانَ إذا قامَ إلى الصَّلاةِ في جوفِ الليلِ يقولُ:

«اللَّهُمَّ لَكَ الحَمْدُ أَنْتَ نُورُ السَّمَوَاتِ وَالأَرْضِ، وَلَكَ الحَمْدُ أَنْتَ قَيِّامُ السَّمَوَاتِ وَالأَرْضِ، وَلَكَ الحَمْدُ أَنْتَ رَبُّ السَّمَوَاتِ وَالأَرْضِ وَمَنْ فِيهِنَّ، أَنْتَ الحَقّ، وَقَوْلُكَ الحَقّ، وَوَعْدُكَ الحَقّ، وَلِقَاؤُكَ حَقّ، وَالجَنَّةُ حَقّ، وَالنَّارُ حَقّ، وَالسَّاعَةُ حَقّ، اللَّهُمَّ لَكَ أَسْلَمْتُ، وَبِكَ آمَنْتُ، وَعَلَيْكَ تَوَكَّلْتُ، وَإِلَيْكَ أَنَبْتُ، وَبِكَ خَاصَمْتُ، وَإِلَيْكَ حَاكَمْتُ، فَاغْفِرْ لِي مَا قَدَّمْتُ وَأَخَّرْتُ، وَأَسْرَرْتُ وَأَعْلَنْتُ، أَنْتَ الَّذِي لَا إِلَهَ إِلَّا أَنْتَ»

ابنُ الجوزي | قيامُ اللَّيل.

Lecture 15: Connective tissue: Fibers

Pre Exams & Formatives

1-	Collagen fibers have the following characters EXCEPT? (MID 31)		
A-	they are composed of tropocollagen	B-	their main origin is fibroblast
C-	they are acidophilic	D-	they are branched
E-	they show periodicity		

2-	Reticular fibers are very thin collagen fibers? (MID 31)								
A-	Type I	B-	Type II	C-	Type III	D-	Type IV	E-	Type V

3-	What stain would be best to demonstrate the elastic fibers in elastic cartilage?(FINAL 31)		
A-	Mason trichrome stain	B-	Hematoxylin and Eosin stain
C-	Sudan stain	D-	Silver impregnation
E-	Orcein		

4-	Tendon has collagen type? (FINAL 34)						
A-	II	B-	I	C-	III	D-	IV

5-	Which of the following would be best suited to visualize reticular fibers: (Formative 36)?		
A-	Sudan stain	B-	Hematoxylin and eosin stain
C-	Masson's trichrome stain	D-	Silver impregnation

6-	Tendon is a connective tissue formed from (Formative 36)?		
A-	collagen fibers type I & fibroblast	B-	macrophage
C-	fibroblast	D-	collagen fibers type II

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
D	C	E	B	D	A	-	-	-	-

Lecture Qs

1-	Type of protein fiber that is most common. It is long, straight, and flexible and does not have branches. (Lecture Qs)				
A-	Reticular	B-	Collagen	C-	Fibrocartilage
D-	Elastic	E-	Lacunae		

2-	Which of the following would be best suited to visualize reticular fibers? (Lecture Qs)		
A-	Wright's stain	B-	Hematoxylin and eosin stain
C-	Sudan stain	D-	Silver stain
E-	Orcein		

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
B	D	-	-	-	-	-	-	-	-

M3WAN Bank

1-	Connective tissue proper is best characterized by: (M3WAN - Core)?		
A-	closely packed cells	B-	avascularity
C-	widely separated cells, abundant matrix	D-	exclusively immune defense

2-	Connective tissue is classified into: (M3WAN - Core)?
A-	embryonic CT, CT proper, CT with special properties
B-	loose and dense CT only
C-	cartilage and bone only
D-	blood and lymphoid tissue only

3-	The three fiber types of connective tissue are: (M3WAN - Core)?		
A-	keratin, elastin, actin	B-	micro/intermediate filaments, microtubules
C-	reticulin, tropomyosin, myosin	D-	collagenous, elastic, reticular

4-	All are correct characteristics of collagen fibers EXCEPT: (Important M3WAN - Hard)?		
A-	composed of tropocollagen	B-	the fibers branch
C-	fibroblast origin	D-	showing periodicity

5-	By light microscopy, collagen fibers appear as: (M3WAN - Core)?		
A-	invisible, requiring silver stain	B-	weakly acidophilic, orcein-stained
C-	long, wavy pink bundles, non-branching fibers	D-	deeply basophilic, thin, branching

6-	By electron microscopy, collagen shows periodicity of: (Important M3WAN - Core)?					
A-	none	B-	25 nm	C-	100 nm	D-	67 nm

7-	The hierarchical assembly of collagen is: (M3WAN - Core)?		
A-	fibers → fibrils → tropocollagen → bundles	B-	tropocollagen → fibrils → fibers → bundles

C-	bundles → fibers → tropocollagen → fibrils	D-	fibrils → bundles → tropocollagen → fibers

8-	**The main function of collagen fibers is: (M3WAN - Core)?**		
A-	strength	B-	elasticity
C-	lymphoid framework formation	D-	providing a diffusion medium

9-	**Type I collagen is found in: (Important M3WAN - Core)?**		
A-	hyaline/elastic cartilage	B-	reticular fibers
C-	the basal lamina	D-	dermis, bone, tendon, capsules

10-	**Type II collagen is found in: (M3WAN - Core)?**		
A-	the dermis and tendon	B-	reticular fibers
C-	hyaline/elastic cartilage	D-	the basal lamina

11-	**Type III collagen forms: (M3WAN - Core)?**		
A-	tendon and bone	B-	reticular fibers
C-	hyaline cartilage	D-	muscle external lamina

12-	**Type IV collagen is found in: (Important M3WAN - Core)?**						
A-	the basal lamina	B-	tendon	C-	reticular fibers	D-	hyaline cartilage

13-	**TypeV collagen is associated with: (M3WAN - Core)?**		
A-	the basal lamina	B-	reticular fibers
C-	hyaline cartilage	D-	muscle external lamina

14-	**By light microscopy, elastic fibers are: (M3WAN - Core)?**		
A-	thin, no striations, branching	B-	thick, non-branching, striated
C-	fine networks, invisible in H&E	D-	mainly composed of collagen

15-	**Regarding elastic fiber staining: (Important M3WAN - Core)?**		
A-	strongly acidophilic	B-	basophilic
C-	weakly acidophilic, orcein dark brown	D-	best seen by silver stain

16-	**By electron microscopy, the elastic fiber shows: (M3WAN - Core)?**		
A-	67 nm periodicity	B-	an elastin core, fibril sheath, no periodicity
C-	9+2 microtubules	D-	triplet microtubules

17-	**Elastic fibers are notably present in: (M3WAN - Core)?**		
A-	tendons exclusively	B-	the basal lamina exclusively

C-	reticular tissue exclusively	D-	the alveolar septa and aorta

18-	The main function of elastic fibers is: (M3WAN - Core)?		
A-	rigidity only	B-	lymphoid framework formation
C-	elasticity and strength	D-	serving as a collagen precursor

19-	Reticular fibers are best characterized as: (Important M3WAN - Core)?		
A-	fine networks, immature type III collagen	B-	thick, mature collagen bundles
C-	mainly composed of elastin	D-	visible black with H&E

20-	Ultrastructurally, reticular fibers are formed of: (M3WAN - Core)?		
A-	fibrils 67 nm, no periodicity	B-	fibrils 60 nm, collagen-type periodicity
C-	an elastin core, fibril sheath	D-	fibrils identical to elastic fibers

21-	By light microscopy, reticular fibers are: (M3WAN - Core)?		
A-	visible pink with H&E	B-	orcein dark brown
C-	invisible with all stains	D-	invisible in H&E, black with silver stain

22-	The main function of reticular fibers is: (M3WAN - Core)?		
A-	tensile strength like tendon	B-	elastic recoil
C-	providing a cellular framework	D-	forming the basal lamina

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
C	A	D	B	C	D	B	A	D	C
11	12	13	14	15	16	17	18	19	20
B	A	D	A	C	B	D	C	A	B
21	22	23	24	25	26	27	28	29	30
D	C	-	-	-	-	-	-	-	-

عجبًا لك! لا الدهر يعظك، ولا الحوادث تُنذرك!..

والساعات تُعدُّ عليك، والأفعال تُعدُّ منك.

وأحبّ أمريك إليك أعودهما بالضرر عليك!.

ابن الجوزي | المدهش.

Lecture 16: C.T. proper: Ground substance & Types

Pre Exams & Formatives

1-	**Cornea of eye is?(FINAL 30)**		
A-	Dense regular white fibrous C'T	B-	Dense irregular white fibrous C'T
C-	Dense regular elastic C'T	D-	Dense irregular elastic C'T
E-	reticular C'T		

2-	**The following regarding glycoprotein is correct?(FINAL 30)**
A-	stains black with silver
B-	have a great elastic recoil
C-	contain myosin
D-	formed of thin core of protein bound to GAGs
E-	formed of bulk protein core bound to carbohydrates

3-	**The stroma (framework) of most lymphoid organs is composed of?(FINAL 31)**				
A-	Reticular connective tissue	B-	Fibrocartilage	C-	Dense connective tissue
D-	Areolar connective tissue	E-	Adipose connective tissue		

4-	**What type of tissue makes up the sclera of the eye?(FINAL 31)**		
A-	Mucous connective tissue	B-	Mesenchyme
C-	Loose irregular connective tissue	D-	Dense irregular connective tissue
E-	Dense regular connective tissue		

5-	**Which of the following is a component of the ground substance?(FINAL 31)**				
A-	hyaluronic acid	B-	proteoglycans	C-	glycosaminoglycans
D-	chondroitin sulfate	E-	all of the above		

6-	**A beauty treatment for the reduction of wrinkles is the injection of hyaluronic acid into the wrinkle. What is hyaluronic acid?(FINAL 31)**				
A-	dermatan sulfate	B-	proteoglycan	C-	glycosaminoglycan
D-	chondroitin sulfate	E-	keratin sulfate		

7-	**What is areolar tissue?(FINAL 31)**		
A-	mucous connective tissue	B-	mesenchyme
C-	loose connective tissue	D-	dense irregular connective tissue
E-	dense regular connective tissue		

8-	Superficial fascia is very dense in the (2ND TRIAL 32)?						
A-	ear pinna	B-	eyelid	C-	sole of foot	D-	scrotum

9-	Dense irregular elastic CT is found in?(FINAL 34)				
A-	ligamentum nuchae	B-	sclera of eye	C-	aorta

10-	The main cyte in mucoid ? (FINAL 37)		
A-	Osteoblast	B-	Fibroblasts

11-	Dense regular elastic connective tissue is found in the: (2ND TRIAL 37)		
A-	Ligamentum nuchae	B-	Sclera of the eye
C-	Dermis of the skin	D-	Tendons

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
D	E	A	D	E	C	C	C	C	B
11	12	13	14	15	16	17	18	19	20
A	-	-	-	-	-	-	-	-	-

Lecture Qs

1-	Which statement is incorrect about tendon? (Lecture Qs)		
A-	Rich in elastic fibers	B-	Contains parallel bundles of collagen fibers
C-	Contains rows of fibroblasts	D-	Classified as dense regular C'T
E-	white in fresh condition		

2-	Hyalouronic acid is? (Lecture Qs)		
A-	Sulphated glycosaminoglycan	B-	Non sulphated glycosaminoglycan
C-	Glycoprotein	D-	Proteoglycans
E-	Soluble protein		

3-	Ligamentum nuchae is? (Lecture Qs)		
A-	Dense regular white fibrousC,T	B-	Dense irregular white fibrous C'T
C-	Dense regular elastic C'T	D-	Dense irregular elastic C' T
E-	None of the above		

4-	Which is not an example of supporting connective tissue? (Lecture Qs)				
A-	Hyaline cartilage	B-	Fibrocartilage	C-	Plasma
D-	Bone	E-	All of the above		

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	B	C	C	-	-	-	-	-	-

M3WAN Bank

1-	Loose (areolar) CT is present in: (M3WAN - Core)?		
A-	tendons exclusively	B-	sites of little stress: spaces, dermis, serosa
C-	cartilage exclusively	D-	bone marrow exclusively

2-	Loose CT is composed of: (M3WAN - Core)?		
A-	fibers and cells only	B-	amorphous matrix exclusively
C-	cells exclusively	D-	fibers, ground substance, cells

3-	The amorphous ground substance is: (M3WAN - Core)?		
A-	semi-fluid, with no particular form	B-	rigid and calcified
C-	parallel collagen bundles	D-	visible fibrils by light microscopy

4-	The consistency of ground substance varies: (M3WAN - Core)?
A-	always watery
B-	always hard
C-	watery in loose CT, firm in cartilage, hard in bone, liquid in blood
D-	always liquid

5-	A main function of ground substance is: (M3WAN - Core)?		
A-	mechanical support only	B-	nutrient/O2 diffusion medium
C-	immune defense only	D-	energy storage only

6-	The ground substance is composed of: (M3WAN - Core)?
A-	collagen and elastin only
B-	water exclusively
C-	cells exclusively
D-	GAGs, proteoglycans, glycoproteins, proteins, water

7-	Glycosaminoglycans (GAGs) are defined as: (M3WAN - Core)?		
A-	protein cores with carbohydrate side chains	B-	branched amino acid polymers
C-	unbranched, repeating disaccharide polymers	D-	peptides forming filaments

8-	Proteoglycans are defined as: (Important M3WAN - Core)?		
A-	protein cores with GAGs linked	B-	pure carbohydrate, no protein
C-	lipid-rich membrane molecules	D-	fibrous collagen proteins

9-	Adhesive glycoproteins function to: (M3WAN - Core)?		
A-	form the mitotic spindle	B-	attach ECM components together
C-	generate ATP	D-	transmit nerve impulses

10-	The soluble proteins of the matrix are: (M3WAN - Core)?		
A-	hydrolytic enzymes	B-	hormones
C-	structural collagen fibrils	D-	collagen precursors, leaked plasma proteins

11-	Regarding tissue fluid in loose C.T.: (M3WAN - Core)?
A-	it is retained during processing
B-	excess causes dehydration
C-	removed by processing, giving "areolar" empty spaces
D-	unrelated to edema

12-	Compared to loose C.T, dense C.T. has: (M3WAN - Core)?		
A-	fewer cells, more collagen/elastic fibers	B-	exclusively elastic fibers
C-	more cells than loose CT	D-	no fibroblasts

13-	Dense regular white C.T. e.g. tendon, shows: (Important M3WAN - Core)?		
A-	fibers in various directions	B-	collagen bundles, one plane/direction
C-	predominant elastic fibers	D-	fenestrated elastic membranes

14-	Dense irregular white C.T. e.g. sclera, shows: (Important M3WAN - Core)?		
A-	fibers regular, one plane	B-	predominant elastic fibers
C-	fenestrated elastic membranes	D-	collagen fibers in various directions

15-	Dense regular elastic C.T. e.g. ligamentum nuchae, shows: (M3WAN - Core)?
A-	predominant elastic fibers, regular plane
B-	predominant collagen, irregular
C-	fenestrated elastic membranes
D-	regular collagen

16-	Dense irregular elastic C.T. e.g. aorta media, shows: (Important M3WAN - Core)?
A-	regular collagen
B-	elastic fibers, regular, one plane
C-	elastic fibers, fenestrated membranes
D-	a reticular fiber framework

17-	Regarding adipose tissue types: (M3WAN - Core)?
A-	only unilocular exists
B-	unilocular: single droplet; multilocular: multiple + mitochondria
C-	only multilocular exists
D-	no blood supply

18-	Adipose tissue functions include: (M3WAN - Core)?		
A-	energy only	B-	insulation only
C-	shape only	D-	energy, shape, insulation, space filling

19-	Reticular connective tissue is: (Important M3WAN - Core)?
A-	the framework of tendons
B-	the framework of dense regular CT
C-	specialized loose CT, myeloid/lymphoid framework
D-	exclusive to the dermis

20-	Reticular cells are: (M3WAN - Core)?		
A-	specialized fibroblasts, sheathing fibers	B-	macrophages, phagocytosing fibers
C-	endothelial cells	D-	osteoblasts

21-	Mucoid CT is best described as: (Important M3WAN - Core)?
A-	dense collagen, found in tendons
B-	jelly-like, hyaluronic acid, umbilical cord/tooth pulp
C-	mainly adipocytes
D-	exclusive to bone marrow

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	D	A	C	B	D	C	A	B	D
11	12	13	14	15	16	17	18	19	20
C	A	B	D	A	C	B	D	C	A
21	22	23	24	25	26	27	28	29	30
B	-	-	-	-	-	-	-	-	-

Lecture 17: Cartilage

Pre Exams & Formatives

1-	**Fibro cartilage is present in these sites EXCEPT?(FINAL 30)**		
A-	lips of glenoid cavity	B-	Intervertebral discs
C-	symphysis pubis	D-	trachea
E-	semilunar cartilage of knee joint		

2-	**Which type of cartilage is found in the costal cartilage?(FINAL 30)**				
A-	hyaline cartilage	B-	elastic cartilage	C-	fibrocartilage
D-	both:a&b	E-	both:a&c		

3-	**Regarding chondroblast which of the following is true?(FINAL 30)**				
A-	protein secreting cell	B-	found inside cell nest	C-	found in the perichondrium
D-	both:a&c	E-	multinucleated		

4-	**What type of tissue makes up the "Adam's apple?(FINAL 31)**				
A-	Hyaline cartilage	B-	Fibrocartilage	C-	Elastic cartilage
D-	Fibrous connective tissue	E-	Both:a&b		

5-	**Perichondrium is present in?(FINAL 31)**				
A-	Articular cartilage	B-	Costal cartilage	C-	Elastic cartilage
D-	White fibrocartilage	E-	Both:b&c		

6-	**Which type of cartilage is characterized by the presence of thick bundles of collagen fibers?(FINAL 31)**				
A-	Hyaline cartilage	B-	Elastic cartilage	C-	Fibrocartilage
D-	Both:b&c	E-	Both:a&c		

7-	**Chondrocytes synthesize and secrete all of the following EXCEPT?(FINAL 31)**		
A-	Type II collagen	B-	Hyaluronic acid
C-	Type I collagen	D-	Sulphated glycosaminoglycans
E-	Both:b&c		

8-	**Tissue that is avascular and as a result is slow to heal is?(FINAL 31)**				
A-	Bone	B-	Cartilage	C-	Fibrous connective tissue
D-	Adipose connective tissue	E-	Loose connective tissue		

9-	**Perichondrium: (FINAL 32, FINAL SNU 1, FINAL 37)?**		
A-	Contains chondrocytes	B-	is absent in white fibrocartilage
C-	Has no role in cartilage growth	D-	Cover cartilage or bone
E-	none of the above		

10-	**cartilage matrix: ? (FINAL 32)**				
A-	is produced by fibroblast	B-	contains collagen type IV	C-	is avascular
D-	is acidophilic	E-	none of the above		

11-	**chondrocyte is: (2ND TRIAL 32)?**
A-	non dividable
B-	an epithelial cell
C-	contains basophilic cytoplasm
D-	branched cell which communicates with other cells through their cytoplasmic processes

12-	**elastic cartilage is: (2ND TRIAL 32)?**		
A-	present in aorta	B-	present in car pinna
C-	also called glassy cartilage	D-	with fibroblasts inside lacuna

13-	**Cartilage matrix?(FINAL 33)**		
A-	Is produced by chondroblast	B-	Contains collagen type IV
C-	Is penetrated by blood vessels	D-	None of the above

14-	**Chondrocyte?(FINAL 33)**
A-	Induces removal of cartilage matrix
B-	Cannot divide
C-	Has basophilic cytoplasm which become acidophilic and vacuolated in old state
D-	Is a branched cell which communicates with other cells through their cytoplasm processes

15-	**Which is true about cartilage matrix?(FINAL 34)**		
A-	penetrated by blood vessels	B-	calcified matrix
C-	contains collagen type II		

16-	**About chondroblast?(FINAL 34)**		
A-	non dividable	B-	found inside a lacunae
C-	acidophilic cytoplasm	D-	in the inner layer of perichondrium

17-	Type of collagen in hyaline cartilage? (2ND TRIAL 35)				
A-	Type I	B-	Type II	C-	Type III

18-	Chondroblast cells form: (FINAL SNU 1)?		
A-	inner layer of perichondrium	B-	outer layer of perichondrium

19-	The following is true about cartilage: (Formative 36)?		
A-	It has 4 types	B-	It has lymph vessels
C-	It is a specialized type of C'T	D-	It is very vascular

20-	perichodrium contains which of the following: (Formative 36)?						
A-	Blood vessels	B-	all of the above	C-	Fibroblasts	D-	Collagen fibers

21-	Types of cartilage are: (Formative 34)?						
A-	Hyaline	B-	Elastic	C-	Cancellous	D-	Both:A&B

22-	Connective tissue covering cartilage is called: (Formative 34)?						
A-	Periosteum	B-	Perichondrium	C-	Capsule	D-	All of the above

23-	Which type of collagen is primarily present in both hyaline and elastic cartilage? (2ND TRIAL 37)						
A-	Type I	B-	Type II	C-	Type III	D-	Type IV

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
D	A	D	A	E	C	E	B	B	C
11	12	13	14	15	16	17	18	19	20
C	B	A	C	C	D	B	A	C	B
21	22	23	24	25	26	27	28	29	30
D	B	B	-	-	-	-	-	-	-

M3WAN Bank

1-	Cartilage is best defined as: (M3WAN - Core)?		
A-	vascular, rigid CT	B-	flexible CT with no matrix
C-	calcified dense CT	D-	avascular CT, rigid but flexible, glassy matrix

2-	Cartilage histogenesis proceeds as: (M3WAN - Core)?		
A-	mesenchyme → chondrocyte → chondroblast	B-	mesenchyme → chondroblast → chondrocyte
C-	osteoblast → chondroblast	D-	fibroblast → chondrocyte

3-	Chondroblasts are located in the: (Important M3WAN - Core)?		
A-	inner (chondrogenic) layer of perichondrium	B-	outer fibrous layer of perichondrium
C-	center of the matrix	D-	bone marrow

4-	By EM, chondroblasts show features of: (M3WAN - Core)?
A-	phagocytic cells
B-	multinucleated cells
C-	protein-secreting cells: RER, Golgi, mitochondria
D-	cells devoid of organelles

5-	Chondroblasts function to secrete: (M3WAN - Core)?		
A-	type I collagen only	B-	hydroxyapatite
C-	keratin	D-	type II collagen and ECM

6-	Chondrocytes are found: (M3WAN - Core)?		
A-	in the perichondrium	B-	in the periosteum
C-	in the deeper part of cartilage	D-	outside the matrix

7-	Chondrocytes occupy spaces called: (M3WAN - Core)?		
A-	Howship's lacunae	B-	lacunae, surrounded by a capsule
C-	canaliculi	D-	Haversian canals

8-	A group of 2, 4, or 8 chondrocytes in one lacuna is called? (M3WAN - Core)				
A-	cell nest	B-	osteon	C-	capsule
D-	trabecula				

9-	All are chondrocyte functions EXCEPT: (Important M3WAN - Hard)?		
A-	synthesizing sulfated GAGs	B-	secreting type II collagen
C-	interstitial growth	D-	secreting type I collagen

10-	Hyaline cartilage has? (M3WAN - Core)		
A-	yellow, fibrous appearance	B-	pearly white, glassy appearance
C-	white, striated appearance	D-	red, vascular appearance

11-	Hyaline cartilage is found in the: (Important M3WAN - Core)?		
A-	ear pinna and epiglottis	B-	intervertebral discs
C-	articular surfaces and costal cartilage	D-	symphysis pubis

12-	Hyaline cartilage functions as: (M3WAN - Core)?		
A-	support and a model for bone development	B-	a hormone-producing tissue
C-	a site of hematopoiesis	D-	an immune organ

13-	The perichondrium is: (Important M3WAN - Core)?		
A-	absent from hyaline cartilage	B-	a loose, vascular CT
C-	present only in fibrocartilage	D-	dense irregular CT ensheathing cartilage

14-	The perichondrium's outer layer contains: (M3WAN - Core)?		
A-	fibroblasts, collagen, and blood vessels	B-	chondroblasts only
C-	osteoclasts	D-	hydroxyapatite

15-	The perichondrium's inner layer is: (Important M3WAN - Core)?		
A-	fibrous, avascular	B-	cellular (chondrogenic), with chondroblasts
C-	calcified	D-	rich in elastic fibers only

16-	Cartilage ground substance is composed of: (M3WAN - Core)?		
A-	collagen fibers only	B-	hydroxyapatite crystals
C-	tissue fluid and amorphous substance	D-	calcified matrix

17-	The sulfated GAG of cartilage matrix is: (M3WAN - Core)?		
A-	hyaluronic acid	B-	osteocalcin
C-	laminin	D-	chondroitin sulfate

18-	Cartilage matrix by H&E is: (M3WAN - Core)?						
A-	acidophilic	B-	unstained	C-	basophilic	D-	black

19-	Cartilage matrix metachromasia is due to: (M3WAN - Core)?		
A-	glycosaminoglycans	B-	collagen type I
C-	hydroxyapatite	D-	osteonectin

20-	The territorial (capsular) matrix around each chondrocyte is: (M3WAN - Core)?
A-	poor in GAGs
B-	rich in GAGs, more basophilic than the interterritorial matrix
C-	identical to the interterritorial matrix
D-	devoid of proteoglycans

21-	Collagen fibers of hyaline cartilage are invisible in H&E because: (Important M3WAN - Hard)?
A-	they are absent
B-	they are stained selectively
C-	they are digested during fixation
D-	they share the matrix's refractive index and are submicroscopic

22-	Elastic cartilage appears, in the fresh state: (M3WAN - Core)?						
A-	yellow	B-	white	C-	pink	D-	blue

23-	Elastic cartilage is found in the: (Important M3WAN - Core)?		
A-	symphysis pubis	B-	articular surfaces
C-	ear pinna and epiglottis	D-	intervertebral disc

24-	Elastic fibers of elastic cartilage are best shown with: (Important M3WAN - Core)?						
A-	H&E	B-	orcein stain	C-	PAS	D-	silver stain

25-	White fibrocartilage is characterized by: (Important M3WAN - Core)?
A-	a perichondrium with two layers
B-	yellow color, elastic fibers
C-	absence of chondrocytes
D-	rows of chondrocytes separated by type I collagen bundles

26-	Fibrocartilage, unlike hyaline and elastic cartilage: (Important M3WAN - Core)?		
A-	has a thick perichondrium	B-	is highly vascular
C-	lacks a perichondrium	D-	lacks chondrocytes

27-	Interstitial growth of cartilage occurs by: (Important M3WAN - Core)?		
A-	division of chondrocytes within the matrix	B-	perichondrial chondroblast activity
C-	osteoblast deposition	D-	vascular invasion

28-	Appositional growth of cartilage occurs from the: (Important M3WAN - Core)?		
A-	outer fibrous layer of perichondrium	B-	inner chondrogenic layer of perichondrium
C-	articular surface directly	D-	bone marrow

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	B	A	C	D	C	B	A	D	B
11	12	13	14	15	16	17	18	19	20
C	A	D	A	B	C	D	C	A	B
21	22	23	24	25	26	27	28	29	30
D	A	C	B	D	C	A	B	-	-

Lecture 18: Bone cells, bone matrix, periosteum & endosteum

Pre Exams & Formatives

1-	The outer surface of long bones is covered by? (MID 30, FINAL 30)				
A-	perichondrium	B-	endochondrium	C-	endosteum
D-	periosteum	E-	Bone marrow		

2-	Concerning osteoclast, it is? (MID 30, Formative 36, Formative 34)
A-	Large binucleated cells with basophilic vacuolated cytoplasm
B-	Part of the phagocytic system
C-	found in the center of the bone matrix
D-	Secrete bone matrix
E-	Only in compact bone

3-	The osteocyte processes are characterized by? (MID 30)		
A-	They are joined by gap junctions	B-	They contain a Golgi apparatus
C-	They contain microvilli	D-	Not connected with adjacent cells
E-	Secret bone matrix		

4-	Concerning osteoclast, it is?(FINAL 30)
A-	large binucleated cells with basophilic vacuolated cytoplasm
B-	monocyte in origin
C-	found in the center of the bone matrix
D-	secrete bone matrix
E-	present only in compact bone

5-	The osteocyte processes are characterized by?(FINAL 30)		
A-	they are present inside lacunae	B-	they contain a Golgi apparatus
C-	They contain microvilli	D-	not connected with adjacent cells
E-	secret bone matrix		

6-	Major inorganic constituents of bone matrix include?(FINAL 31)				
A-	Calcium phosphate	B-	Calcium oxalate	C-	Type I collagen
D-	Type III collagen	E-	Proteoglycans		

7-	Bone canaliculi contain?(FINAL 31)				
A-	Blood vessels	B-	Processes of osteocytes	C-	Osteoclasts

D-	Collagen fiber type I	E-	Both:a&c

8-	**Osteoclasts are derived from? (FINAL 31, FINAL 32)**				
A-	Osteogenic cells	B-	Osteoblast	C-	Osteocytes
D-	Monocytes	E-	Macrophage		

9-	**What cell is involved in laying down new bone?(FINAL 31)**								
A-	osteoclast	B-	osteon	C-	osteocyte	D-	osteoblast	E-	osteoid

10-	**Osteocytes arise most directly from ? (FINAL 32)**				
A-	chondrocytes	B-	osteoblasts	C-	osteoclasts
D-	monocytes	E-	osteoprogenitor cells		

11-	**In ordinary conditions the only bone cell that divide is ? (FINAL 32)**				
A-	The osteoprogenitor cell	B-	The osteocyte	C-	The osteoclast
D-	Both:b&c	E-	None of the above		

12-	**Concerning the periosteum; it (2ND TRIAL 32)?**		
A-	has layer of dense regular C'T	B-	Has no ability to form a new bone
C-	Covers the bone marrow canal	D-	Does not cover bone areas that share in joint

13-	**The osteoid is?(FINAL 33)**		
A-	Uncalcified bone matrix	B-	Formed by osteoprogenitor cell
C-	Contains type II collagen fibers	D-	Formed of calcified bone lamellae

14-	**Intracellular calcium deposition enzyme in osteoblast is stored in?(FINAL 33)**		
A-	Golgi apparatus	B-	Rough endoplasmic reticulum
C-	Lysosomes	D-	Matrix vesicle

15-	**Which of these is inside lacunae?(FINAL 34)**						
A-	osteocyte	B-	osteoblast	C-	osteoclast	D-	osteoid

16-	**One of the following has lysosomes?(FINAL 34)**
A-	Osteoclast

17-	**The outer fibrous layer of periosteum is?(FINAL 35)**						
A-	Dense regular	B-	Dense irregular	C-	thin regular	D-	thin irregular

18-	Hard and calcified? (2ND TRIAL 35)		
A-	Bone	B-	Cartilage

19-	Osteocytes? (2ND TRIAL 35)		
A-	Are derived from blood monocytes	B-	Live only for short time
C-	Possess long narrow cytoplasmic processes	D-	Give rise to osteoclasts

20-	Bone resorption by osteoclasts is due to: (2ND TRIAL 37)		
A-	Creation of an acidic medium	B-	Creation of an alkaline medium
C-	Secretion of collagen type I	D-	Deposition of hydroxyapatite crystals

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
D	B	A	B	A	A	B	D	D	B
11	12	13	14	15	16	17	18	19	20
A	D	A	D	A	A	A	A	C	A

M3WAN Bank

1-	Bone is best defined as: (M3WAN - Core)?		
A-	avascular dense CT	B-	flexible CT, uncalcified
C-	loose CT with cartilage cells	D-	calcified dense CT with four cell types

2-	Bone is covered by: (M3WAN - Core)?		
A-	perichondrium outside, endochondrium inside	B-	periosteum outside, endosteum inside
C-	endosteum outside, periosteum inside	D-	a capsule outside, matrix inside

3-	The organic bone matrix is composed of: (M3WAN - Core)?		
A-	type I collagen and ground substance	B-	hydroxyapatite and citrate
C-	type II collagen only	D-	calcium carbonate crystals

4-	The inorganic bone matrix is mainly: (M3WAN - Core)?						
A-	proteoglycans	B-	osteonectin	C-	hydroxyapatite	D-	type I collagen

5-	The four bone cell types are: (M3WAN - Core)?
A-	chondroblast, chondrocyte, osteoblast, osteocyte
B-	fibroblast, osteocyte, osteoclast, osteoblast
C-	mesenchymal cell, osteoblast, osteocyte, macrophage

D-	osteoprogenitor, osteoblast, osteocyte, osteoclast

6-	**Osteoprogenitor cells arise from: (M3WAN - Core)?**		
A-	osteoblasts	B-	monocytes
C-	undifferentiated mesenchymal cells	D-	chondrocytes

7-	**Osteoprogenitor cells are found: (M3WAN - Core)?**
A-	only in bone marrow
B-	in the periosteum, endosteum, and metaphyseal trabeculae
C-	exclusively in Howship's lacunae
D-	only in cartilage

8-	**By LM, osteoprogenitor cells are: (M3WAN - Core)?**
A-	oval, with a pale nucleus and scanty basophilic cytoplasm
B-	large, multinucleated, acidophilic
C-	flattened, with a ruffled border
D-	deeply basophilic, with a negative Golgi region

9-	**By EM, osteoprogenitor cells show: (M3WAN - Core)?**		
A-	abundant RER and secretory vesicles	B-	a ruffled border
C-	numerous lysosomes	D-	free ribosomes and a centriole

10-	**In ordinary conditions, the only dividing bone cell is the: (Important M3WAN - Core)?**		
A-	osteocyte	B-	osteoprogenitor cell
C-	osteoclast	D-	osteoblast

11-	**Osteoblasts arise from: (Important M3WAN - Core)?**		
A-	osteocytes	B-	monocytes
C-	osteoprogenitor cells	D-	chondroblasts

12-	**By LM, osteoblasts are: (Important M3WAN - Core)?**		
A-	basophilic cells with a negative Golgi region	B-	acidophilic, frothy, multinucleated
C-	pale, with scanty cytoplasm	D-	flattened, in lacunae

13-	**By EM, osteoblasts show: (M3WAN - Core)?**		
A-	a ruffled border	B-	free ribosomes and a centriole only
C-	numerous lysosomes and phagosomes	D-	numerous RER, Golgi, and procollagen vesicles

14-	Osteoblasts communicate with adjacent cells via: (Important M3WAN - Core)?		
A-	gap junctions	B-	tight junctions
C-	desmosomes	D-	no direct communication

15-	Osteoblast matrix vesicles contain: (Important M3WAN - Core)?		
A-	collagenase	B-	alkaline phosphatase
C-	acid phosphatase	D-	hyaluronidase

16-	An osteocyte arises when an osteoblast becomes: (Important M3WAN - Core)?		
A-	multinucleated	B-	fused with a monocyte
C-	surrounded by intercellular substance	D-	detached from the matrix

17-	Osteocytes occupy spaces called: (Important M3WAN - Core)?		
A-	Howship's lacunae	B-	cell nests
C-	Haversian canals	D-	lacunae, with processes in canaliculi

18-	Osteocyte processes communicate via: (Important M3WAN - Core)?						
A-	gap junctions	B-	tight junctions	C-	desmosomes	D-	no communication

19-	Compared with a young osteocyte, an older osteocyte shows: (M3WAN - Hard)?		
A-	more RER and Golgi	B-	fewer organelles, more glycogen
C-	a ruffled border	D-	multiple nuclei

20-	The main function of the osteocyte is: (Important M3WAN - Core)?		
A-	bone resorption	B-	collagen type II secretion
C-	maintenance of bone matrix	D-	fracture healing by division

21-	Osteoclasts arise by: (Important M3WAN - Core)?		
A-	differentiation of osteoprogenitor cells	B-	division of osteocytes
C-	differentiation of chondroblasts	D-	fusion of monocytes

22-	Osteoclasts occupy pits called: (Important M3WAN - Core)?						
A-	Howship's lacunae	B-	cell nests	C-	canaliculi	D-	osteons

23-	By LM, osteoclasts are: (Important M3WAN - Core)?		
A-	small, mononuclear, basophilic	B-	oval, pale-nucleated
C-	large, multinucleated (6-12 nuclei), acidophilic	D-	flattened, with a single nucleus

24-	The osteoclast border facing bone is: (Important M3WAN - Core)?		
A-	smooth and organelle-rich	B-	ruffled/striated, villi-like
C-	covered by a basement membrane	D-	indistinguishable from the rest of the cell

25-	Osteoclasts resorb bone by: (Important M3WAN - Core)?		
A-	secreting collagen type I	B-	forming matrix vesicles
C-	mitotic division	D-	acidic decalcification and lysosomal digestion

26-	Besides resorption, osteoclasts function in: (Important M3WAN - Core)?		
A-	collagen synthesis	B-	osteoid formation
C-	remodeling, repair, and calcium homeostasis	D-	hematopoiesis

27-	The periosteum is composed of: (Important M3WAN - Core)?		
A-	outer fibrous and inner osteogenic layers	B-	a single osteoprogenitor layer only
C-	calcified matrix only	D-	chondroblasts and chondrocytes

28-	The endosteum is composed of: (M3WAN - Core)?
A-	dense regular CT with fibroblasts
B-	a single layer of flattened osteoprogenitor cells
C-	multiple layers of osteoclasts
D-	hyaline cartilage

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	B	A	C	D	C	B	A	D	B
11	12	13	14	15	16	17	18	19	20
C	A	D	A	B	C	D	A	B	C
21	22	23	24	25	26	27	28	29	30
D	A	C	B	D	C	A	B	-	-

عن معاوية بن قُرَّة، قال:

دَخلتُ على الحسَنِ فقُلتُ: يا أبا سعيدٍ، أيُّ الأعمالِ أحبُّ إلى اللهِ عزَّ وجلَّ؟ قال:

«الصَّلاةُ في جوفِ اللَّيلِ والنَّاسُ نيامٌ.»

ابنُ الجوزي | قيامُ اللَّيل.

Lecture 19: Types of bone, Bone growth, remodeling & repair

Pre Exams & Formatives

1-	**Flat bone are formed by intramembranous ossification?(FINAL 30)**		
A-	True	B-	False

2-	**Osteoclast ? (FINAL 30)**				
A-	Antigen presenting cell	B-	Immunity	C-	Growth
D-	Differentiation	E-	remodeling		

3-	**Specific "structural units" of compact bone are?(FINAL 31)**				
A-	Interstitial lamellae	B-	Circumferential lamellae	C-	Osteons
D-	Volkmann's canals	E-	Haversian canal		

4-	**During endochondral ossification cartilage cells (chondrocytes)?(FINAL 31)**		
A-	Proliferate and form osteocytes	B-	Increase in size and form osteocytes
C-	Differentiate to osteogenic cells	D-	Die and leave empty spaces
E-	Both B and C		

5-	**Secondary center of ossification occurs in?(FINAL 31)**				
A-	Epiphysis	B-	Epiphyseal plate	C-	Diaphysis
D-	Vault of the skull	E-	Periosteum		

6-	**The osteon is ? (FINAL 32)**		
A-	uncalcified bone lamellae	B-	aligned parallel to Volkman's canal
C-	formed of concentrically calcified bone lamella	D-	appear only adjacent to bone marrow
E-	present in cancellous bone		

7-	**Which one of the following is a correct statement concerning Haversian systems? (2ND TRIAL 32)**		
A-	are found in fetal skeletons	B-	are found in spongy bone
C-	develop around Haversian's canals	D-	contain type III collagen fiber

8-	**Regarding the long bone all are true except (2ND TRIAL 32)?**		
A-	has compact outer layer	B-	has inner cancellous layer
C-	it forms blood	D-	has epiphyseal cartilage in the adult

9-	**Concerning the endochondral ossification of cartilage model?(FINAL 33)**
A-	Secondary center occurs in the diaphysis
B-	Primary center appears after birth
C-	Secondary ossification takes place in spiral manner
D-	It is responsible for the formation of flat bone

10-	**Which one of the following is NOT correct statement concerning compact bone?(FINAL 33)**
A-	Contain eccentric lamellae of bone
B-	Endosteum lined the bone marrow canal
C-	Outer circumferential lamellae under periosteum
D-	Contain type 1 collagen fiber

11-	**What is the way called when bone is directly formed without a cartilage template?(FINAL 34)**		
A-	intramembranous ossification	B-	interstitial growth
C-	endochondral ossification		

12-	**Growth of bone length is by?(FINAL 34)**		
A-	appositional growth	B-	interstitial growth
C-	intramembranous ossification	D-	endochondral

13-	**In which stage of endochondral bone growth does alkaline phosphatase exist? (FINAL SNU 1)**						
A-	Resting cartilage	B-	Proliferation	C-	Hypertrophy	D-	Calcification

14-	**Which from the following related to Compact bone ? (FINAL 37)**		
A-	Trabecula	B-	Haversian system

15-	**Spongy bone can be identified histologically by the presence of:: (Formative 36)?**		
A-	Volkmann's Canal	B-	Interstitial Lamellae
C-	Bony trabeculae and marrow cavity	D-	Haversian canal and concentric bony lamellae

Answers - Pre Exams & Formatives									
1	2	3	4	5	6	7	8	9	10
A	E	C	D	A	C	C	D	C	A
11	12	13	14	15	16	17	18	19	20
A	B	C	B	C	-	-	-	-	-

M3WAN Bank

1-	**Bone is organized into two histological forms: (M3WAN - Core)?**		
A-	osteoid and hydroxyapatite	B-	lamellar and woven only
C-	periosteal and endosteal	D-	compact and cancellous bone

2-	**Compact bone is found in the: (M3WAN - Core)?**		
A-	center of flat bones only	B-	shaft of long bones
C-	ends of long bones only	D-	bone marrow cavity only

3-	**Compact bone is covered by: (M3WAN - Core)?**
A-	periosteum outside, endosteum lining the marrow cavity
B-	endosteum outside, periosteum inside
C-	perichondrium on both surfaces
D-	cartilage on both surfaces

4-	**Compact bone is structurally organized into: (Important M3WAN - Core)?**		
A-	trabeculae	B-	Howship's lacunae only
C-	osteons (Haversian systems)	D-	a single cartilage plate

5-	**Each Haversian system consists of a Haversian canal surrounded by: (Important M3WAN - Core)?**		
A-	irregular trabeculae	B-	a cartilage capsule
C-	osteoclasts only	D-	concentric lamellae

6-	**Lamellae located between adjacent Haversian systems are called: (Important M3WAN - Core)?**		
A-	circumferential lamellae	B-	outer lamellae
C-	interstitial lamellae	D-	trabecular lamellae

7-	**Lamellae located just under the periosteum are called: (Important M3WAN - Core)?**		
A-	interstitial lamellae	B-	outer circumferential lamellae
C-	inner circumferential lamellae	D-	Haversian lamellae

8-	**Cancellous bone is found at the: (M3WAN - Core)?**		
A-	ends of long bones and center of flat bones	B-	shaft of long bones only
C-	outer part of short bones only	D-	articular cartilage

9-	**Cancellous bone is structurally composed of: (M3WAN - Core)?**		
A-	osteons with Haversian canals	B-	a single dense lamella

C-	interstitial lamellae only	D-	irregular trabeculae with marrow spaces

10-	Bone can be formed by two mechanisms: (Important M3WAN - Core)?
A-	endosteal and periosteal ossification
B-	intramembranous and endochondral ossification
C-	interstitial and appositional ossification
D-	primary and secondary lamellar ossification

11-	Intramembranous ossification typically forms: (Important M3WAN - Core)?				
A-	long bones	B-	short bones	C-	flat bones
D-	g. skull and clavicle	E-	articular cartilage		

12-	In intramembranous ossification, the bone formed is initially of the: (Important M3WAN - Core)?		
A-	spongy type	B-	compact type
C-	cartilaginous type	D-	lamellar compact type

13-	Endochondral ossification is principally responsible for forming: (Important M3WAN - Core)?		
A-	flat bones only	B-	the skull vault
C-	the clavicle	D-	short and long bones

14-	In endochondral ossification, the primary ossification center arises in the: (Important M3WAN - Core)?		
A-	shaft (diaphysis) of the cartilage model	B-	epiphysis
C-	epiphyseal plate	D-	perichondrium

15-	The first bone tissue formed in endochondral ossification surrounds the cartilage model as? (M3WAN - Core)		
A-	calcified cartilage core	B-	bone collar
C-	secondary ossification center	D-	cell nest

16-	Vascular invasion of calcified cartilage in endochondral ossification occurs through holes made by: (M3WAN - Core)?		
A-	chondroblasts	B-	osteoblasts
C-	osteoclasts in the bone collar	D-	fibroblasts

17-	Secondary centers of ossification appear: (Important M3WAN - Core)?		
A-	in the diaphysis, before birth	B-	in the perichondrium
C-	in the epiphyseal plate only	D-	at the epiphyses, after birth

18-	After ossification, hyaline cartilage persists at the: (M3WAN - Core)?		
A-	diaphysis only	B-	primary ossification center
C-	articular surface and epiphyseal plate	D-	bone collar

19-	Growth in length of long bones occurs mainly by: (Important M3WAN - Core)?		
A-	interstitial growth of epiphyseal cartilage	B-	appositional bone growth
C-	intramembranous ossification	D-	osteoclastic resorption

20-	The epiphyseal (growth) plate closes at approximately: (M3WAN - Core)?		
A-	25 years (males), 22 years (females)	B-	20 years (males), 17 years (females)
C-	15 years in both sexes	D-	30 years in both sexes

21-	The epiphyseal plate shows, in sequence, the zones of: (Important M3WAN - Core)?
A-	calcification, hypertrophy, proliferation, resting, ossification
B-	ossification, calcification, hypertrophy, proliferation, resting
C-	proliferation, resting, hypertrophy, ossification, calcification
D-	resting, proliferation, hypertrophy, calcification, ossification

22-	In the zone of proliferation, chondrocytes: (M3WAN - Core)?
A-	divide, forming rows parallel to the bone's long axis
B-	die, leaving empty spaces
C-	become rich in alkaline phosphatase
D-	are invaded by blood vessels

23-	In the zone of hypertrophy, chondrocytes: (Important M3WAN - Core)?
A-	die and degenerate
B-	are replaced by osteocytes
C-	enlarge and become rich in alkaline phosphatase
D-	divide repeatedly

24-	In the zone of calcification, chondrocytes: (M3WAN - Core)?
A-	proliferate actively
B-	die, as calcified matrix cuts off their nourishment
C-	form the bone collar
D-	form Howship's lacunae

25-	In the zone of ossification: (M3WAN - Core)?
A-	cartilage divides interstitially

B-	the bone collar first forms
C-	calcification begins
D-	vessels and mesenchymal cells invade, forming bone and marrow

26-	**Bone remodeling is best defined as: (Important M3WAN - Core)?**
A-	initial deposition of bone matrix only
B-	cartilage replacement by bone
C-	coupled osteoclastic resorption and osteoblastic formation
D-	growth of the epiphyseal plate

Answers - M3WAN Bank									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	**B**	**A**	**C**	**D**	**C**	**B**	**A**	**D**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	**A**	**D**	**A**	**B**	**C**	**D**	**C**	**A**	**B**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	**A**	**C**	**B**	**D**	**C**	**-**	**-**	**-**	**-**

تم بفضل الله ، هذا وما كان من توفيق فمن الله وحده وما كان من خطأ أو زلل فمن أنفسنا ومن الشيطان ، وفقنا الله وإياكم لخيري الدنيا والآخرة وأعاننا على طاعته .

نسألكم الدعاء لكل من سبقنا من إخواننا إلى الله أن يتغمدهم بواسع رحمته وأن يرفع قدرهم ويتجاوز عن سيئاتهم وأن يجمعنا بهم في جناته.